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(12) **United States Patent**
Lu(10) **Patent No.:** **US 12,398,387 B2**(45) **Date of Patent:** **Aug. 26, 2025**(54) **COMPOSITIONS AND METHODS FOR
PREPARING FACTOR Xa AND
DERIVATIVES**(71) Applicant: **Alexion Pharmaceuticals, Inc.**, Boston,
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MA (US)(*) Notice: Subject to any disclaimer, the term of this
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U.S.C. 154(b) by 761 days.(21) Appl. No.: **17/629,999**(22) PCT Filed: **Aug. 5, 2020**(86) PCT No.: **PCT/US2020/045039**

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17, 2020, provisional application No. 62/884,652,
filed on Aug. 8, 2019.(51) **Int. Cl.**
C12N 9/64 (2006.01)(52) **U.S. Cl.**
CPC **C12N 9/6432** (2013.01); **C07K 2319/31**
(2013.01); **C07K 2319/50** (2013.01)(58) **Field of Classification Search**CPC **C12N 9/6432**; **C07K 2319/31**; **C07K**
2319/50

See application file for complete search history.

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& Hampton LLP(57) **ABSTRACT**The present disclosure relates to protein sequences which
can be used to generate factor Xa proteins and derivatives
thereof. The protein sequences include a factor Xa light
chain portion, a heavy chain catalytic domain portion, and
an activation peptide C-terminal to the heavy chain catalytic
domain portion.**14 Claims, 6 Drawing Sheets****Specification includes a Sequence Listing.**

FIG. 1

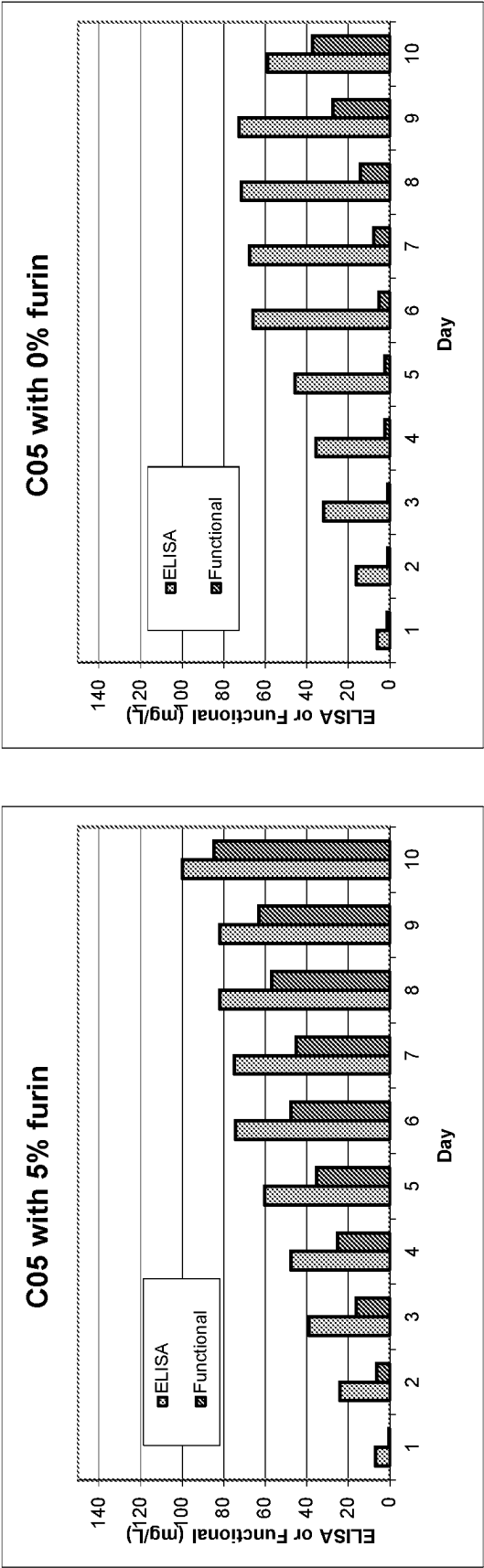


FIG. 2A

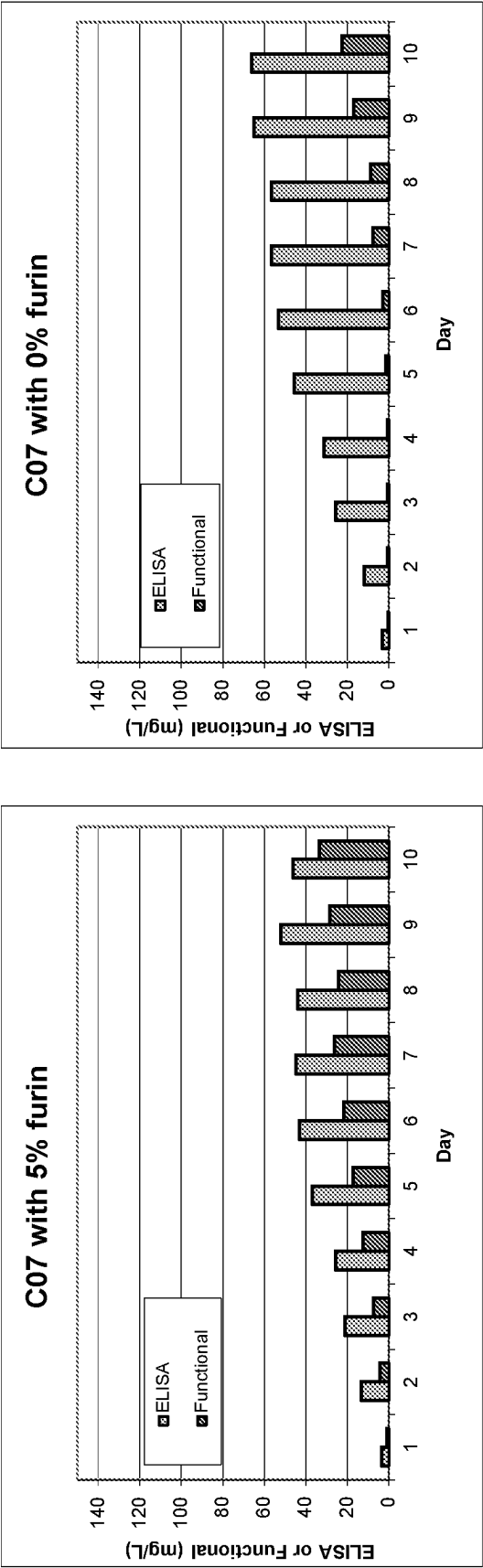


FIG. 2B

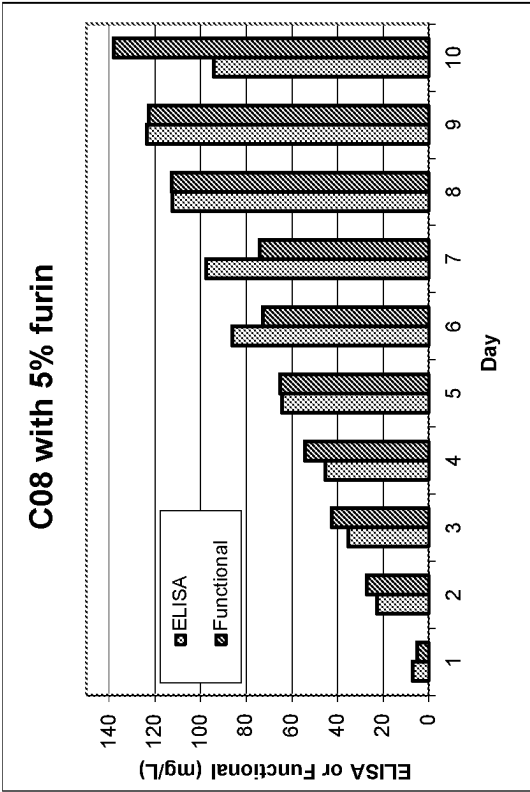
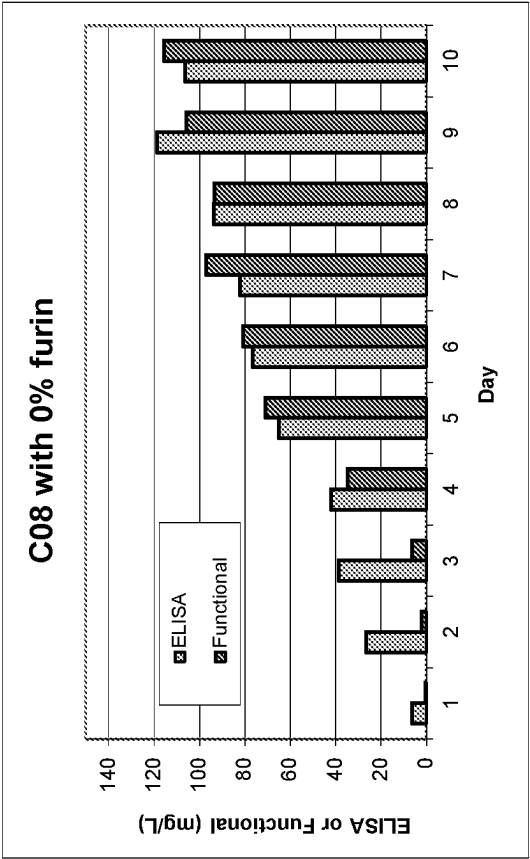


FIG. 2C

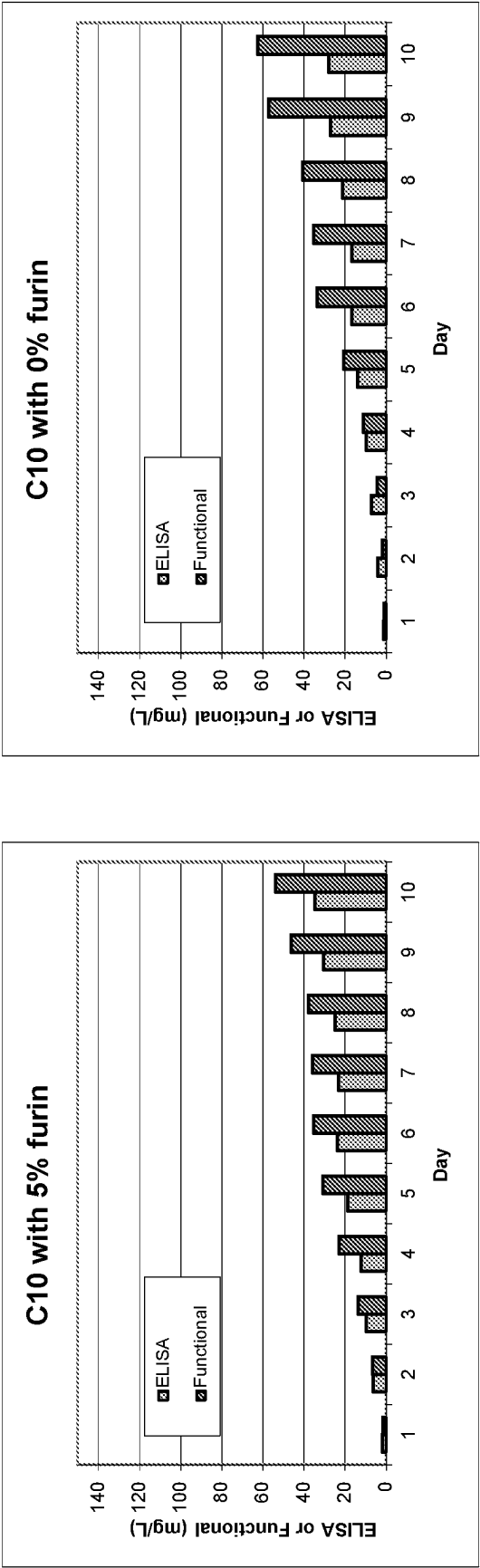


FIG. 2D

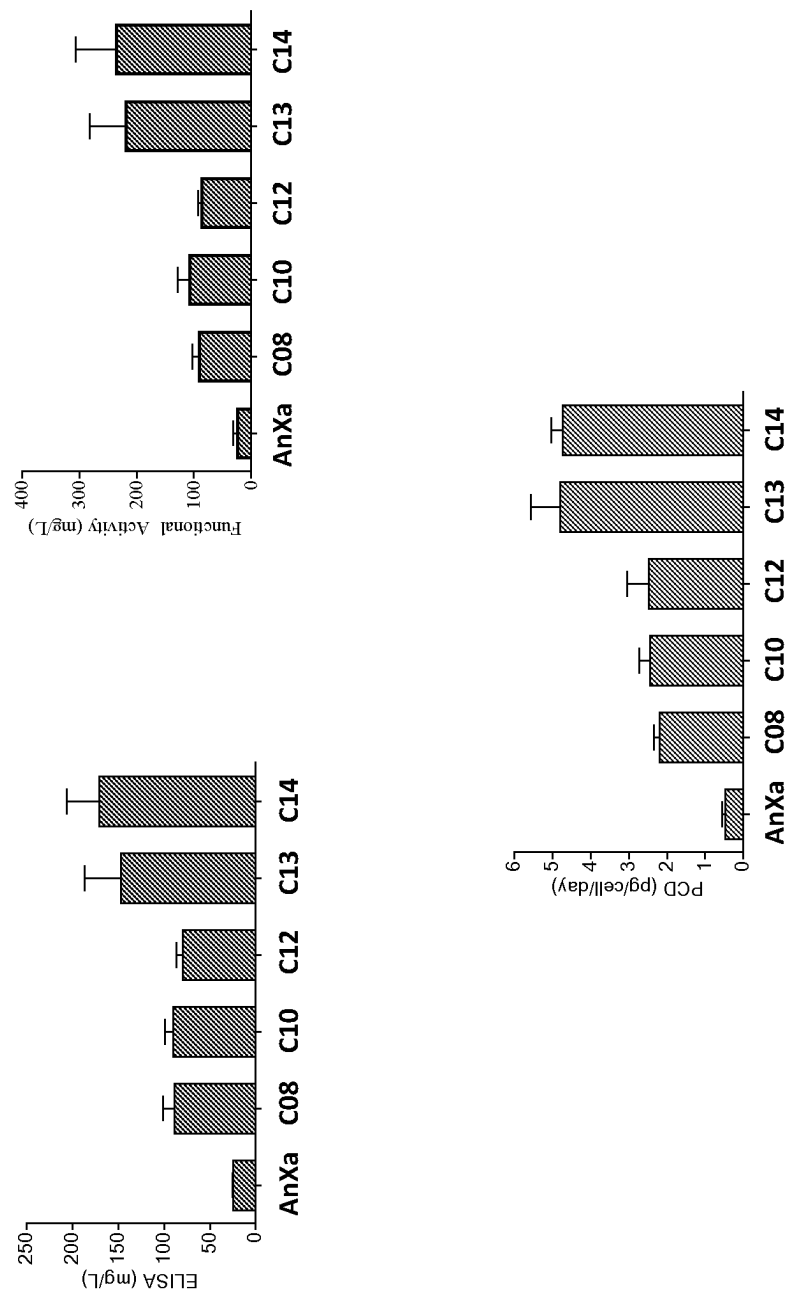


FIG. 3

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COMPOSITIONS AND METHODS FOR PREPARING FACTOR Xa AND DERIVATIVES

CROSS REFERENCE TO RELATED APPLICATIONS

This application is the United States National Phase Application of PCT/US2020/045039, filed Aug. 5, 2020, and claims the benefit under 35 U.S.C. § 119 (e) of U.S. Provisional Application Ser. No. 62/884,652, filed Aug. 8, 2019, and 62/990,885, filed Mar. 17, 2020, the content of each of which is incorporated by reference in its entirety into the present disclosure.

SEQUENCE LISTING

The instant application contains a Sequence Listing which has been submitted electronically in ASCII format and is hereby incorporated by reference in its entirety. Said ASCII copy, created on Sep. 11, 2020, is named 37JE-267815-WO_SL.txt and is 138,454 bytes in size.

BACKGROUND

Recombinant factor Xa (fXa) and its derivatives, such as Andexanet alfa, can be made from host mammalian cell lines. Andexanet alfa (or simply Andexanet) is a drug that has been approved in the U.S. and Europe for patients treated with rivaroxaban or apixaban, when reversal of anticoagulation is needed due to life-threatening or uncontrolled bleeding. Rivaroxaban and apixaban are factor Xa inhibitors, a group of anticoagulant (anti-blood clotting) drugs that also include betrixaban and edoxaban, as well as low molecular weight heparins (LMWHs). Andexanet is a modified recombinant derivative of factor Xa (fXa). It acts as a decoy molecule, and binds to the inhibitor and relieves its inhibition of fXa, thus restores normal coagulation activity of fXa.

Factor Xa, as well as Andexanet, have two chains linked by a disulfide bond between the two chains. Recombinant native fXa is usually made first with expression of an inactive precursor factor X (fX), followed by a second step of activating the expressed fX to fXa by physiologic enzymes (e.g., FVIIa/TF, FIXa/FVIIIa) or by non-physiologic activators (e.g., RVV-X). The difference between fX and fXa is the removal of a 52 amino acid residues of the activation peptide (AP) at the N-terminus of the fX heavy chain. The activation step is necessary for converting the inactive fX to the native fXa, because the typical production cell line, e.g., CHO cell, is unable to process and remove the AP.

In contrast, andexanet is directly expressed as a fully processed and functional molecule that can be purified directly from the harvested cell culture fluid. This is accomplished by replacing the AP sequence in the native fX with a —RKR— tripeptide to form a —RKRKR— (SEQ ID NO: 7) linker between the heavy- and light-chains, which can be processed by the CHO cells.

SUMMARY

The present disclosure provides composition and methods for preparing two-chain, activated, factor Xa proteins or the derivatives thereof. It is discovered that when an activation peptide (AP) is fused to the C-terminal end of the heavy chain of the factor Xa protein or derivative, the resulting protein can be more efficiently expressed, and the attachment of the activation peptide (AP) to the heavy chain does not affect the activity of the protein. By contrast, adding the activation peptide to the other parts of the protein is either

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not useful in enhancing expression (e.g. when attached to the C-terminus of the light chain) or even presents challenges for manufacturing (e.g. when attached to the N-terminus of the heavy chain). Further, contrary to the conventional wisdom (see, e.g., Branchini et al, *J Thromb Haemost* 2015; 13:1468-74, and Ferrarese et al., *Thrombosis Res.* 2019, 173:4-11) that the removal of the C-terminal 20 amino acid residues of FX heavy chain (the beta peptide) does not negatively impact the protein expression of factor X, it is discovered herein that such a large truncation is not desired for efficient expression of fXa and derivatives. More surprisingly, the impact of C-terminal truncation on protein expressions can be rescued by fusing the AP to the C-terminal of FXa and derivatives thereof.

In one embodiment, the present disclosure provides a protein comprising the amino acid sequence of the formula (I) of:

LC-L1-HC-L2-AP (I)

wherein:

LC comprises the amino acid sequence of SEQ ID NO:13 or a peptide having at least 85% sequence identity to SEQ ID NO:13,

L1 is a peptide linker comprising a protease recognition site,

HC comprises the amino acid sequence of SEQ ID NO:11 or a peptide having at least 85% sequence identity to SEQ ID NO:11,

L2 is absent or is a peptide linker that cannot be processed by the protease, and

AP comprises an activation peptide,

wherein the protein is capable of, when the L1 is processed by the protease, producing a two-chain polypeptide comprising the LC and HC on separate chains connected by a disulfide bond, wherein the two-chain polypeptide is capable of binding to a factor Xa inhibitor.

In another embodiment, provided is a protein comprising the amino acid sequence of the formula (II) of:

HSA-L2-LC-L1-HC (II)

wherein:

HSA is a human serum albumin (HSA) or a variant having at least 85% sequence identity to the HSA;

L1 is a peptide linker comprising a protease recognition site;

L2 is absent or is a peptide linker that cannot be processed by the protease;

LC comprises the amino acid sequence of SEQ ID NO:13 or a peptide having at least 85% sequence identity to SEQ ID NO:13; and

HC comprises the amino acid sequence of SEQ ID NO:11 or a peptide having at least 85% sequence identity to SEQ ID NO:11,

wherein the protein is capable to, when the L1 is processed by the protease, produce a two-chain polypeptide comprising the LC and HC on separate chains, wherein the two-chain polypeptide is capable of binding to a factor Xa inhibitor.

In another embodiment, provided is a protein comprising the amino acid sequence of the formula (II) of:

LC-L1-HC-L2-HSA (III)

wherein:

HSA is a human serum albumin (HSA) or a variant having at least 85% sequence identity to the HSA;

L1 is a peptide linker comprising a protease recognition site;

L2 is absent or is a peptide linker that cannot be processed by the protease;

LC comprises the amino acid sequence of SEQ ID NO:13 or a peptide having at least 85% sequence identity to SEQ ID NO:13; and

HC comprises the amino acid sequence of SEQ ID NO:11 or a peptide having at least 85% sequence identity to SEQ ID NO:11,

wherein the protein is capable to, when the L1 is processed by the protease, produce a two-chain polypeptide comprising the LC and HC on separate chains, wherein the two-chain polypeptide is capable of binding to a factor Xa inhibitor.

In some embodiments, the LC does not include amino acid residues 1-45 of SEQ ID NO: 2. In some embodiments, L2 is absent.

Also provided, in some embodiments, is a two-chain polypeptide comprising a light chain that comprises the amino acid sequence of SEQ ID NO:13 or a peptide having at least 85% sequence identity to SEQ ID NO:13, and a heavy chain that comprises a first fragment comprising the amino acid sequence of SEQ ID NO:11 or a peptide having at least 85% sequence identity to SEQ ID NO:11, and a second fragment that is a the C-terminal side of the first fragment and comprises an activation peptide, wherein the two-chain polypeptide is capable of binding to a factor Xa inhibitor.

Also provided is a two-chain polypeptide comprising a light chain that comprises a first fragment comprising a human serum albumin (HSA) or a variant having at least 85% sequence identity to the HSA, and a second fragment comprising the amino acid sequence of SEQ ID NO: 13 or a peptide having at least 85% sequence identity to SEQ ID NO:13, and a heavy chain that comprises the amino acid sequence of SEQ ID NO:11 or a peptide having at least 85% sequence identity to SEQ ID NO:11, wherein the two-chain polypeptide is capable of binding to a factor Xa inhibitor, and wherein the light chain does not include amino acid residues 1-45 of SEQ ID NO:2.

Polynucleotides, cells transfected with the polynucleotides, and methods are also provided, in some embodiments, to prepare the two-chain polypeptides.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 illustrates the structures of constructs C01-C14.

FIG. 2A-2D show the expression and activity testing results for C05 (2A), C07 (2B), C08 (2C) and C10 (2D).

FIG. 3 shows the expression and activity testing results for C12-C14, using andexanet precursor (AnXa), C08 and C10 as references.

DETAILED DESCRIPTION

I. Definitions

All numerical designations, e.g., pH, temperature, time, concentration, and molecular weight, including ranges, are approximations which are varied (+) or (−) by increments of 0.1. It is to be understood, although not always explicitly stated that all numerical designations are preceded by the term “about”. It also is to be understood, although not always explicitly stated, that the reagents described herein are merely exemplary and that equivalents of such are known in the art.

The term “protein” and “polypeptide” are used interchangeably and in their broadest sense to refer to a compound of two or more subunit amino acids, amino acid analogs or peptidomimetics. The subunits may be linked by peptide bonds. In another embodiment, the subunit may be linked by other bonds, e.g., ester, ether, etc. A protein or peptide must contain at least two amino acids and no limitation is placed on the maximum number of amino acids

which may comprise a protein's or peptide's sequence. As used herein the term “amino acid” refers to either natural and/or unnatural or synthetic amino acids, including glycine and both the D and L optical isomers, amino acid analogs and peptidomimetics. Single letter and three letter abbreviations of the naturally occurring amino acids are listed below. A peptide of three or more amino acids is commonly called an oligopeptide if the peptide chain is short. If the peptide chain is long, the peptide is commonly called a polypeptide or a protein.

“Factor Xa” or “fXa” or “fXa protein” refers to a serine protease in the blood coagulation pathway, which is produced from the inactive factor X (fX). Factor Xa is activated by either factor IXa with its cofactor, factor VIIIa, in a complex known as intrinsic Xase, or factor VIIa with its cofactor, tissue factor, in a complex known as extrinsic Xase. fXa forms a membrane-bound prothrombinase complex with factor Va and is the active component in the prothrombinase complex that catalyzes the conversion of prothrombin to thrombin. Thrombin is the enzyme that catalyzes the conversion of fibrinogen to fibrin, which ultimately leads to blood clot formation. Thus, the biological activity of fXa is sometimes referred to as “procoagulant activity” herein.

Factor Xa is a two chain molecule linked by one disulfide bond between the two chains. The light chain (LC) has 139 amino acid (amino acids 1 through 139 of SEQ ID NO:2) residues and contains the γ -carboxyglutamic acid (Gla)-rich domain (amino acids 1-45 of SEQ ID NO:2), including a short aromatic stack (AS) (amino acids 40-45 of SEQ ID NO:2), followed by two epidermal growth factor (EGF)-like domains (EGF1: amino acids 46-84, EGF2: amino acids 85-128 of SEQ ID NO:2).

The heavy chain (HC), prior to activation, has 306 amino acids and contains a 52 amino acids activation peptide (AP: amino acids 143-194 of SEQ ID NO:2) followed by the catalytic domain (amino acids 195-448 of SEQ ID NO:2). The catalytic triad equivalents to H57-D102-S195 in chymotrypsin numbering are located at His236, Asp282, and Ser379 in the fX sequence (SEQ ID NO:2) (amino acids 236, 282 and 379 of SEQ ID NO:2). The heavy chain contains the serine protease, trypsin-like active site and the N-terminal activation peptide which is glycosylated. The heavy chain has at least three forms, α , β , and γ , which differ due to the cleavage of a C-terminal peptide in the heavy chain.

The nucleotide sequence coding human factor X (“fX”) can be found in GenBank, “NM_000504”. The corresponding amino acid sequence and domain structure of fX are described in Leytus et al, *Biochemistry*, 1986, 25:5098-5102. The domain structure of mature fX is also described in Venkateswarlu, D. et al, *Biophysical Journal*, 2002, 82:1190-1206. Upon catalytic cleavage of the first 52 residues of the heavy chain, fX is activated to fXa (SEQ ID NO: 3). fXa contains a light chain with post-translation of glutamic acid residues to gamma-carboxyglutamic acid) and a heavy chain. The first 45 amino acid residues of the light chain is called the Gla domain because it contains 11 post-translationally modified γ -carboxyglutamic acid residues (Gla). It also contains a short (6 amino acid residues) aromatic stack sequence. Chymotrypsin digestion selectively removes the 1-44 residues resulting in Gla-domainless fXa. The serine protease catalytic domain of fXa locates at the C-terminal heavy chain. The heavy chain of fXa is highly homologous to other serine proteases such as thrombin, trypsin, and activated protein C.

TABLE 1

Polypeptide Sequence of Human Factor X (SEQ ID NO: 1)	
1	MGRPLHLVLL SASLAGLLLL GESLFIRREQ ANNILARVTR ANSFLEEMKK GHLERECMEE
61	TCSYEEAREV FEDSDKTNEF WNKYKGDQDC ETSPCQNQGK CKDGLGEYTC TCLEGFEGKN
121	CELFTRLKCS LDNGDCDQFC HEEQNSVVCs CARGYTLADN GKACIPTGPY PCGKQTLERR
181	KRSVAQATSS SGEAPDSITW KPYDAADLDP TENPFDLLDF NQTQPERGDN NLTRIVGGQE
241	CKDGECPWQA LLINEENEGF CGGTILSEFY ILTAACHLYQ AKRFKVRVGD RNTEQEEGGE
301	AVHEVEVVIK HNRFTKETYD FDI AVLRLKT PITFRMNVAP ACLPERDWAE STLMTQKTGI
361	VSGFGRTHEK GRQSTRLKML EVPNVDNRNSC KLSSSFIITQ NMFACGYDTK QEDACQGDSC
421	GPHVTRFKDT YFVTGIVSWG EGCARKGKYG IYTKVTAFLK WIDRSMKTRG LPKAKSHAPE
481	VITSSPLK

TABLE 2

Polypeptide Sequence of Mature Human Factor X (SEQ ID NO: 2)	
1	ANSFLEEMKK GHLERECMEE TCSYEEAREV FEDSDKTNEF WNKYKGDQDC ETSPCQNQGK
61	CKDGLGEYTC TCLEGFEGKN CELFTRLKCS LDNGDCDQFC HEEQNSVVCs CARGYTLADN
121	GKACIPTGPY PCGKQTLERR KRSVAQATSS SGEAPDSITW KPYDAADLDP TENPFDLLDF
181	NQTQPERGDN NLTRIVGGQE CKDGECPWQA LLINEENEGF CGGTILSEFY ILTAACHLYQ
241	AKRFKVRVGD RNTEQEEGGE AVHEVEVVIK HNRFTKETYD FDI AVLRLKT PITFRMNVAP
301	ACLPERDWAE STLMTQKTGI VSGFGRTHEK GRQSTRLKML EVPNVDNRNSC KLSSSFIITQ
361	NMFACGYDTK QEDACQGDSC GPHVTRFKDT YFVTGIVSWG EGCARKGKYG IYTKVTAFLK
421	WIDRSMKTRG LPKAKSHAPE VITSSPLK

TABLE 3

Polypeptide Sequence of Factor Xa (SEQ ID NO: 3)	
1	ANSFLEEMKK GHLERECMEE TCSYEEAREV FEDSDKTNEF WNKYKGDQDC ETSPCQNQGK
61	CKDGLGEYTC TCLEGFEGKN CELFTRLKCS LDNGDCDQFC HEEQNSVVCs CARGYTLADN
121	GKACIPTGPY PCGKQTLER
181	IVGGQE CKDGECPWQA LLINEENEGF CGGTILSEFY ILTAACHLYQ
241	AKRFKVRVGD RNTEQEEGGE AVHEVEVVIK HNRFTKETYD FDI AVLRLKT PITFRMNVAP
301	ACLPERDWAE STLMTQKTGI VSGFGRTHEK GRQSTRLKML EVPNVDNRNSC KLSSSFIITQ
361	NMFACGYDTK QEDACQGDSC GPHVTRFKDT YFVTGIVSWG EGCARKGKYG IYTKVTAFLK
421	WIDRSMKTRG LPKAKSHAPE VITSSPLK

“Native fXa” or “wild-type fXa” refers to the fXa naturally present in plasma or being isolated in its original, unmodified form, which processes the biological activity of activating prothrombin therefore promoting formation of blood clot. The term includes naturally occurring polypeptides isolated from tissue samples as well as recombinantly produced fXa. “Active fXa” refers to fXa having the biological activity of activating prothrombin. “Active fXa” may be a native fXa or modified fXa that retains procoagulant activity.

“fXa Derivatives” or “modified fXa” or “derivatives of a factor Xa protein” refers to fXa proteins that have been modified but can still bind, either directly or indirectly, to a factor Xa inhibitor.

The derivatives may have modified active sites and/or a modified Gla domain. Additional modifications are also contemplated. It is contemplated that such modifications may be made in one or more of the following ways: deletion of one or more of the amino acid from the sequence,

substitution of one or more amino acid residues with one or more different amino acid residues, and/or manipulation of one or more amino acid side chains or its "C" or "N" terminals.

The term "active site" refers to the part of an enzyme or antibody where a chemical reaction occurs. A "modified active site" is an active site that has been modified structurally to provide the active site with increased or decreased chemical reactivity or specificity. It is contemplated that the active site includes not only the actual site but also a domain containing the active site. Examples of active sites include, but are not limited to, the catalytic domain of human factor X comprising the 235-488 amino acid residues, and the catalytic domain of human factor Xa comprising the 195-488 amino acid residues. The catalytic triad equivalent to H57-D102-S195 in chymotrypsin numbering are located at His236, Asp282, and Ser379. Examples of modified active site include, but are not limited to, the catalytic triad residues individually or in combination. One modification relates to a fXa derivative having a modified active site Ser379Ala. Additional examples, include modifications to the catalytic domain of human factor Xa comprising 195-448 amino acid residues with at least one amino acid substitution at position Arg306, Glu310, Arg347, Lys351, Lys414, or Arg424.

The term "factor Xa inhibitors" or "inhibitors of factor Xa" refer to compounds that can inhibit, either directly or indirectly, the coagulation factor Xa's activity of catalyzing conversion of prothrombin to thrombin in vitro and/or in vivo. Examples of known fXa inhibitors include, without limitation, edoxaban, fondaparinux, idraparinux, biotinylated idraparinux, enoxaparin, fragmin, NAP-5, rNAPc2, tissue factor pathway inhibitor, DX-9065a (as described in, e.g., Herbert, J. M., et al. *J Pharmacol Exp Ther*. 1996 276 (3): 1030-8), YM-60828 (as described in, e.g., Taniuchi, Y., et al. *Thromb Haemost.* 1998 79 (3): 543-8), YM-150 (as described in, e.g., Eriksson, B. I. et. al. *Blood* 2005; 106 (11), Abstract 1865), apixaban, rivaroxaban, PD-348292 (as described in, e.g., Pipeline Insight: Antithrombotics-Reducing the Untreated Prophylaxis Market, 2007), otamixaban, razaxaban (DPC906), BAY 59-7939 (as described in, e.g., Turpie, A. G., et al. *J. Thromb. Haemost.* 2005, 3 (11): 2479-86), edoxaban (as described in, e.g., Hylek EM, Curr Opin Invest Drugs 2007 8 (9): 778-783), LY517717 (as described in, e.g., Agnelli, G., et al. *J. Thromb. Haemost.* 2007 5 (4): 746-53), GSK913893, betrixaban and derivatives thereof. Low molecular weight heparin ("LMWH") is also considered a factor Xa inhibitor.

In one embodiment, the derivative of the invention binds, either directly or indirectly to a factor Xa inhibitor. The terms "binding," "binds," "recognition," or "recognize" as used herein are meant to include interactions between molecules that may be detected using, for example, a hybridization assay. The terms are also meant to include "binding" interactions between molecules. Interactions may be, for example, protein-protein, protein-nucleic acid, protein-small molecule or small molecule-nucleic acid in nature. Binding may be "direct" or "indirect". "Direct" binding comprises direct physical contact between molecules. "Indirect" binding between molecules comprises the molecules having direct physical contact with one or more intermediate molecules simultaneously. For example, it is contemplated that derivatives of the invention indirectly bind and substantially neutralize low molecular weight heparin and other indirect inhibitors of factor Xa. This binding can result in the formation of a "complex" comprising the interacting molecules. A "complex" refers to the binding of two or more molecules held together by covalent or non-covalent bonds, interactions or forces.

"Neutralize," "reverse" or "counteract" the activity of an inhibitor of fXa or similar phrases refer to inhibit or block the factor Xa inhibitory or anticoagulant function of a fXa

inhibitor. Such phrases refer to partial inhibition or blocking of the function, as well as to inhibiting or blocking most or all of fXa inhibitor activity, in vitro and/or in vivo. These terms also refer to corrections of at least about 20% of fXa inhibitor dependent pharmacodynamic or surrogate markers. Examples of markers include, but are not limited to INR, PR, aPTT, ACT, anti-fXa units, thrombin generation (Technothrombin TGA), thromboelastography, CAT (calibrated automated thrombogram) and the like.

A "composition" is intended to mean a combination of active agent and another compound or composition, inert (for example, a detectable agent or label) or active, such as an adjuvant.

A "pharmaceutical composition" is intended to include the combination of an active agent with a carrier, inert or active, making the composition suitable for diagnostic or therapeutic use in vitro, in vivo or ex vivo.

As used herein, the term "equivalent thereof" when referring to a reference protein, polypeptide or nucleic acid, intends those having minimal homology while still maintaining desired functionality. It is contemplated that any modified protein mentioned herein also includes equivalents thereof. For example, the homology can be, at least 75% homology and alternatively, at least 80%, or alternatively at least 85%, or alternatively at least 90%, or alternatively at least 95%, or alternatively 98% percent homology and exhibit substantially equivalent biological activity to the reference polypeptide or protein. A polynucleotide or polynucleotide region (or a polypeptide or polypeptide region) has a certain percentage (for example, 80%, 85%, 90%, or 95%) of "sequence identity" to another sequence means that, when aligned, that percentage of bases (or amino acids) are the same in comparing the two sequences. It should be noted that when only the heavy chain of fXa (or a related serine protease) is used, the overall homology might be lower than 75%, such as, for example, 65% or 50% however, the desired functionality remains.

A polynucleotide or polynucleotide region (or a polypeptide or polypeptide region) has a certain percentage (for example, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 98% or 99%) of "sequence identity" to another sequence means that, when aligned, that percentage of bases (or amino acids) are the same in comparing the two sequences.

II. Preparation of Factor Xa and Derivatives

Andexanet alfa, or simply andexanet, is a modified factor Xa polypeptide that has been approved in the U.S. and Europe for patients treated with rivaroxaban or apixaban, when reversal of anticoagulation is needed due to life-threatening or uncontrolled bleeding. Also referred to as the r-Antidote, the structure and activities of andexanet are described in U.S. Pat. No. 8,153,590.

Andexanet is a processed two-chain polypeptide processing product of SEQ ID NO:4, after cleavage of the —RKRRKR— (SEQ ID NO:7) linker. Andexanet is represented by SEQ ID NO: 5, which includes a light chain (SEQ ID NO:6) and a heavy chain (SEQ ID NO:8). The light chain and the heavy chain are connected with a single disulfide bond between Cysteine 98 (Cys98) of the light chain and Cysteine 108 (Cys108) of the heavy chain. Like the wild-type fXa, in certain production batches, Andexanet undergoes post-translational modifications resulting in glycosylation at certain amino acid residues, e.g., Ser56, Ser72, Ser76 and Thr82 of the light chain and Thr249 of the heavy chain, and a modified residue, (3R)-3-hydroxyAsp at Asp29 of the light chain. Further, in addition to the inter-chain disulfide bond, there can be intra-chain disulfide bonds formed between Cysteines 16 and 27, 21 and 36, 38 and 47, 55 and 66, 62 and 75, and 77 and 90 of the light chain, and between Cysteines 7 and 12, 27 and 43, 156 and 170, and 181 and 209 of the heavy chain.

TABLE 4

Amino acid sequence of andexanet precursor prior to removal of -RKRRKR-linker (SEQ ID NO: 4)	
Light Chain Fragment (SEQ ID NO: 6)	
1 ANSFL	F WNKYKGDQC ETSPCQNQGK
61 CKDGLGEYTC TCLEGFEGKN CELFTRKLCS LDNGDCDQFC HEEQNSVVC S CARGYTLADN	
121 GKACIPTGPY PCGKQTLER	
Linker (SEQ ID NO: 7)	
RKRRKR	
Heavy Chain Fragment (SEQ ID NO: 8)	
181	IVGGQE CKDGECPWQA LLINEENEGF CGGTILSEFY ILTAACHLYQ
241	AKRPFKVRVGD RNTQEEGGE AVHEVEVVIK HNRFTKETD FDI AVLRLKT PITFRMNVAP
301	ACLPERDWAE STLTMTQGTGI VSGFGRTHEK GRQSTRKLML EVPNVDNRNSC KLSSSFIIITQ
361	NMFCAGYDTK QEDACQGDAG GPHVTRFKDT YFVTGIVSWG EGCARKGKYG IYTKVTAFLK
421	WIDRSMKTRG LPKAKSHAPE VITSSPLK

Amino acid residue number is based on mature inactive fX (SEQ ID NO: 2)

TABLE 5

Amino acid sequence of andexanet (SEQ ID NO: 5)	
Light Chain (SEQ ID NO: 6)	
1 ANSFL	F WNKYKGDQC ETSPCQNQGK
61 CKDGLGEYTC TCLEGFEGKN CELFTRKLCS LDNGDCDQFC HEEQNSVVC S CARGYTLADN	
121 GKACIPTGPY PCGKQTLER	
Heavy Chain (SEQ ID NO: 8)	
181	IVGGQE CKDGECPWQA LLINEENEGF CGGTILSEFY ILTAACHLYQ
241	AKRPFKVRVGD RNTQEEGGE AVHEVEVVIK HNRFTKETD FDI AVLRLKT PITFRMNVAP
301	ACLPERDWAE STLTMTQGTGI VSGFGRTHEK GRQSTRKLML EVPNVDNRNSC KLSSSFIIITQ
361	NMFCAGYDTK QEDACQGDAG GPHVTRFKDT YFVTGIVSWG EGCARKGKYG IYTKVTAFLK
421	WIDRSMKTRG LPKAKSHAPE VITSSPLK

Amino acid residue number is based on mature inactive fX (SEQ ID NO: 2)

The precursor of Andexanet, SEQ ID NO:4, contains three mutations relative to the wild type fXa. The first mutation is the deletion of 6-39 aa in the Gla-domain of fX. The second mutation is replacing the activation peptide sequence 143-194 aa with —RKR—. This produces a —RKRRKR— (SEQ ID NO:7) linker connecting the light chain and the heavy chain. Upon secretion, this linker is cleaved resulting in a two-chain polypeptide (Andexanet). The third mutation is mutation of active site residue S379 to an Ala residue.

Due to these structural changes, Andexanet does not compete with fXa in assembling into the prothrombinase complex and has reduced or no catalytic activities. Accordingly, Andexanet can sequester circulating fXa inhibitors without interfering with the native coagulation mechanisms. Similar antidotes have also been disclosed, including those that further have deletions in the EGF-1 or EGF-2 domain.

SEQ ID NO:4 is the precursor protein that is expressed in host cells to produce Andexanet, which can be digested by endogenous or a super-transfected furin protein that targets the —RKRRKR— (SEQ ID NO:7) linker. It was believed that the activation peptide (AP) from the wild-type fX is not necessary as it has been replaced by the artificial —RKRRKR— (SEQ ID NO:7) linker.

It was discovered herein, however, that inclusion of the AP can increase the expression of the processed two-chain product. Nevertheless, inclusion of the AP in the precursor proteins, such as at the N-terminus of the heavy chain, presents a challenge for manufacturing. Protein manufacturing is typically done with CHO cells, which do not possess enzymatic activity to process factor X naturally for the cleavage site -LTR— between AP and the N-terminus of the heavy chain (SEQ ID NO: 2). Therefore, the fXa derivatives include certain linkers that can be cleaved by endogenous furin in CHO cells or co-transfected with a furin

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protease which recognizes the —RKR— and —RKRRKR— (SEQ ID NO:7) linkers.

In this context, it was discovered that, if the AP is attached to the protein through furin-recognizable linkers (e.g., C04-C06), furin could process it too early, defeating the purpose of increasing expression. Additionally, the processed two-chain molecule might be inactive if the furin-recognizable linker was not processed properly. On the other hand, if the AP is fused to the light chain in a non-cleavable manner (e.g., C07), it did not have the ability to increase protein expression. Only when the AP was fused to the C-terminal end of the heavy chain, in a non-cleavable manner, did it have the best effect in increasing protein expression and functional activity.

Another discovery was that deletion of the 13 or even 15 C-terminal amino acid residues (Delta HC-13 and Delta HC-15, respectively) of the heavy chain did not have significant impact on the expression or activity of the protein. By contrast, deletion of 20 C-terminal amino acid residues from the C-terminus of HC (Delta HC-20) significantly reduced the expression of the protein. This was unexpected in view of the conventional wisdom as demonstrated in, e.g., Branchini et al., *Journal of Thrombosis and Haemostasis*, 13:1468-1474 (2015), that truncation of native FX up to 20 amino acids residuals did not significantly affect factor X expression. More surprisingly, the impact of C-terminal truncation on protein expressions could be rescued by fusing AP to the C-terminal of FXa and derivatives.

In accordance with one embodiment of the present disclosure, therefore, provided is a precursor protein that includes a sequence of formula (I):

LC-L1-HC-L2-AP

(I).

Here, LC denotes a protein fragment that includes the amino acid sequence of SEQ ID NO: 13 or a biological equivalents, such as a peptide having at least 85% sequence identity to SEQ ID NO:13. HC denotes a protein fragment that includes the amino acid sequence of SEQ ID NO: 11 or a biological equivalents, such as a peptide having at least 85% sequence identity to SEQ ID NO:11. In some embodiments, L2 includes at least the first 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 amino acid residues of the beta peptide (RGLPKAK-SHAPEVITSSPLK (SEQ ID NO:38)).

L1 denotes a protein linker that includes a protease recognition site. The protease, in some embodiments, may be furin. L2, on the other hand, can be null (in other words, optional) or a peptide linker. When L2 is the linker, however, L2 cannot be processed by the protease. Therefore, when the protein is incubated with the protease, the protease can digest the protein to make a two-chain polypeptide, one of which includes LC, and the other includes HC-L2-AP. In some embodiments, the produced two-chain polypeptide is capable of binding to a factor Xa inhibitor.

Interestingly, in a first set of experiments, the expression-enhancing effect of AP was not observed with human serum albumin (HSA), a protein commonly used to enhance expression or stability of expressed proteins. See C10 in Example 2. However, in a further experiment (Example 3), two precursor constructs with the HSA fused to the N-terminus of the light chain, C13 and C14 exhibited greatly improved expression and activities. Accordingly, it is contemplated that the full deletion of the Gla domain (C10 had partial deletion of the Gla domain) enhanced the effect of HSA. In other words, the expression-enhancing effect of HSA was more pronounced when the HSA was fused to the EGF1 domain directly.

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In accordance with another embodiment of the present disclosure, therefore, provided is a precursor protein that includes a sequence of formula (II):

HSA-L2-LC-L1-HC

(II).

In another embodiment, also provided is a precursor protein that includes a sequence of formula (III):

LC-L1-HC-L2-HSA

(III).

Here, HSA denotes a human serum albumin (HSA) or a variant having at least 85% sequence identity to the HSA; L1 is a peptide linker comprising a protease recognition site; L2 is absent or is a peptide linker that cannot be processed by the protease; LC comprises the amino acid sequence of SEQ ID NO:13 or a peptide having at least 85% sequence identity to SEQ ID NO: 13; and HC comprises the amino acid sequence of SEQ ID NO:11 or a peptide having at least 85% sequence identity to SEQ ID NO:11. The protein of formula (II), when the L1 is processed by the protease, produce a two-chain polypeptide comprising the LC and HC on separate chains, wherein the two-chain polypeptide is capable of binding to a factor Xa inhibitor. In some embodiments, the LC does not include amino acid residues 1-45 of SEQ ID NO:2. In some embodiments, L2 is absent.

In some embodiments, the protein of formula (I) can further include a HSA fused to the N-terminal end of the light chain, or the C-terminal end of the heavy chain. In some embodiments, the protein of formula (II) can further include an AP fused to the C-terminal end of the heavy chain. In some embodiments, the protein of formula (III) can further include an AP fused to the N-terminal end of the light chain.

In some embodiments, any protein of the present disclosure can be fused to an Fc fragment of an immunoglobulin. The fragment crystallizable region (Fc region) is the tail region of an antibody that interacts with cell surface receptors called Fc receptors and some proteins of the complement system. In IgG, IgA and IgD antibody isotypes, the Fc region is composed of two identical protein fragments, derived from the second and third constant domains of the antibody's two heavy chains.

In some embodiments, the Fc fragment used is an IgG, such as IgG1, IgG2 or IgG4, Fc fragment. In some embodiments, the Fc fragment further includes, besides the CH2 and CH3 regions, the CH1 region. An example Fc fragment sequence is provided in Table 24, SEQ ID NO: 46. In some embodiments, the fusion protein includes a signal peptide, such as SEQ ID NO: 34-36 or 45.

In some embodiments, only one chain of the Fc fragment is fused to the fXa derivative. In some embodiments, both chains of the Fc fragment are fused to the fXa derivative.

The term “furin” or “paired basic amino acid cleaving enzyme,” as used herein, refers to a protein having an amino acid sequence substantially identical to any of the representative Furin sequences of GenBank Accession Nos. NP_002560 (human), NP_001074923 (mouse) or NP_062204 (rat). Suitable cDNA encoding furin are provided at GenBank Accession Nos. NM_002569 (human), NM_001081454 (mouse) or NM_019331 (rat). In a particular aspect, furin refers to a human furin.

Human serum albumin (HSA) is the serum albumin found in human blood. HSA constitutes about half of serum protein. It is produced in the liver and is soluble in water. Albumin transports hormones, fatty acids, and other compounds, buffers pH, and maintains oncotic pressure, among other functions. Albumin is synthesized in the liver as preproalbumin, which has an N-terminal peptide that is

removed before the nascent protein is released from the rough endoplasmic reticulum. The product, proalbumin, is in turn cleaved in the Golgi vesicles to produce the secreted albumin. The HSA can have the native sequence or with a single Cys34Ser mutation (SEQ ID NO:15) to remove the free cysteine in native HSA, as tested herein.

AP denotes a protein fragment that includes an activation peptide. An "activation peptide" is peptide that is attached, either covalently or non-covalently, to a protein and maintains that protein inactive until the activation peptide is removed. In some embodiments, the activation peptide includes four or more glycosylation sites. It is known that amino acid residues such as Asp, Ser, Tyr, and Thr are suitable glycosylation sites. In some embodiments, the AP is from 10 to 100 amino acid residues long (or alternatively from 10 to 80, from 15 to 70, from 20 to 60, or from 10 to 50 amino acid residues long) and includes 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20 glycosylation sites.

Activation peptides exist in wild-type proteins, such as the precursors of factor IX (fIX), factor X (fX), factor XIII (fXIII), factor II (prothrombin) and protein C. Their respective sequences are listed in Table 6.

TABLE 6

Example activation peptides
Activation peptide of factor X (SEQ ID NO: 12)
SVQAATSSSG EAPDSITWKP YDAADLDPTF NPFDDLDFNQ
TQPERGDNNL TR
Activation peptide of factor IX (SEQ ID NO: 31)
AETVFPDQVDY VNSTEAEITL DNITQSTQSF NDFTR
Activation peptide of factor XIII (SEQ ID NO: 32)
MSETSRATFG GRRVPPNNS NAAEDDLPTV ELQGVVPR
Activation peptide of Protein C (SEQ ID NO: 33)
DTEDQEDQVD PR
Activation peptide of factor II, AP1 (SEQ ID NO: 39)
ANTFLEEVRK GNLERECVEE TCSYEEAFEA LESSTATDVF
WAKYTACETA RTPRDKLAAC LEGNCAEGLG TNYRGHVNIT
RSGLQCQLWR SRYPHKPEIN STHPGADLQ ENFCRNPDS
TTGPWCYTTD PTVRRQECIS PVCQGDQVTV AMTPR
Activation peptide of factor II, AP2 (SEQ ID NO: 40)
SEGSSVNLSP PLEQCVPDRG QQYQGR LAVT THGLPCLAWA
SAQAKALSKH QDFNSAVQLV ENFCRNPDDG EEGVWCYVAG
KPGDFGYCDL NYCEEAVEEE TGDGLDESD RAIEGRTATS
EYQTFNPR

In some embodiments, the activation peptide includes the amino acid sequence of SEQ ID NO: 12, 31, 32, 33, 39 or 40, or include an amino acid sequence having at least 85% sequence identity to SEQ ID NO:12, 31, 32, 33, 39 or 40. In some embodiments, the activation peptide includes an amino acid sequence having at least 85%, or alternatively at

least 70%, 75%, 80%, 90%, 95%, 98% or 99%, sequence identity to SEQ ID NO:12. In some embodiments, the activation peptide includes an amino acid sequence derived from SEQ ID NO:12 with one, two or three amino acid addition, deletions and/or substitutions. In some embodiments, the activation peptide includes an amino acid sequence having at least 85%, or alternatively at least 70%, 75%, 80%, 90%, 95%, 98% or 99%, sequence identity to SEQ ID NO:31. In some embodiments, the activation peptide includes an amino acid sequence derived from SEQ ID NO:31 with one, two or three amino acid addition, deletions and/or substitutions. In some embodiments, the activation peptide includes an amino acid sequence having at least 85%, or alternatively at least 70%, 75%, 80%, 90%, 95%, 98% or 99%, sequence identity to SEQ ID NO:32. In some embodiments, the activation peptide includes an amino acid sequence derived from SEQ ID NO:32 with one, two or three amino acid addition, deletions and/or substitutions. In some embodiments, the activation peptide includes an amino acid sequence having at least 85%, or alternatively at least 70%, 75%, 80%, 90%, 95%, 98% or 99%, sequence identity to SEQ ID NO:33. In some embodiments, the activation peptide includes an amino acid sequence derived from SEQ ID NO:33 with one, two or three amino acid addition, deletions and/or substitutions. In some embodiments, the activation peptide includes an amino acid sequence having at least 85%, or alternatively at least 70%, 75%, 80%, 90%, 95%, 98% or 99%, sequence identity to SEQ ID NO:39. In some embodiments, the activation peptide includes an amino acid sequence derived from SEQ ID NO:39 with one, two or three amino acid addition, deletions and/or substitutions. In some embodiments, the activation peptide includes an amino acid sequence having at least 85%, or alternatively at least 70%, 75%, 80%, 90%, 95%, 98% or 99%, sequence identity to SEQ ID NO:40. In some embodiments, the activation peptide includes an amino acid sequence derived from SEQ ID NO:40 with one, two or three amino acid addition, deletions and/or substitutions. In some embodiments, the activation peptide is from 10 to 100 amino acid residues long (or alternatively from 10 to 80, from 15 to 70, from 20 to 60, or from 10 to 50 amino acid residues long) and includes 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20 glycosylation sites.

In some embodiments, the protease is furin, but other proteases are also within the scope of the present disclosure, for example, other proprotein convertases such as PC5 or coagulation enzymes such as FXa and thrombin. Examples of peptide linkers suitable for furin include —RKR— and —RKRKR— (SEQ ID NO:7).

As provided, in some embodiments, L2 is absent (null). In some embodiments, L2 is peptide linker of 1-50 (or 1-40, 1-30, 1-25, 1-20, 5-40, 10-30, 15-35) amino acid residues in length. The peptide linker is preferably flexible, such as those having at least 30%, 40%, 50%, 60%, 70%, 80% or 90% Gly and/or Ser.

The produced two-chain polypeptide, in some embodiment, is suitable to be used as an antidote to factor fXa inhibitor-based anticoagulant treatments. In some embodiments, the two-chain polypeptide cannot compete with fXa in assembling into the prothrombinase complex, has reduced capacity to assemble into a prothrombin complex, or cannot assemble into a prothrombin complex. Assembly into the prothrombin complex requires a functional Gla domain. In some embodiments, the LC does not include a substantial portion of the Gla domain, such as does not include amino acids 6-39 of SEQ ID NO:2. In some embodiments, the LC does not include amino acids 1-45 of SEQ ID NO:2.

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In some embodiments, the LC includes the amino acid sequence of SEQ ID NO:6 or includes an amino acid sequence having at least 85%, or alternatively at least 70%, 75%, 80%, 90%, 95%, 98% or 99%, sequence identity to SEQ ID NO:6.

In some embodiments, the LC includes the full light chain sequence of the wild-type fXa (i.e., amino acid residues 1-139 of SEQ ID NO:3), or an amino acid sequence having at least 85%, or alternatively at least 70%, 75%, 80%, 90%, 95%, 98% or 99%, sequence identity to the full length light chain sequence of the wild-type fXa.

The produced two-chain polypeptide, in some embodiments, has reduced catalytic activity as compared to a wild-type human factor Xa. This can be achieved, for instance, by mutation of one or more of the active sites in the heavy chain, e.g., His236, Asp282, and Ser379 (amino acids 236, 282 and 379 of SEQ ID NO:2).

In some embodiments, the HC includes at least the amino acid sequence of SEQ ID NO: 10 or an amino acid sequence having at least 85%, or alternatively at least 70%, 75%, 80%, 90%, 95%, 98% or 99%, sequence identity to SEQ ID NO:10. In some embodiments, the HC includes at least the amino acid sequence of SEQ ID NO:11 or an amino acid sequence having at least 85%, or alternatively at least 70%, 75%, 80%, 90%, 95%, 98% or 99%, sequence identity to SEQ ID NO:11. In some embodiments, the HC includes the amino acid sequence of SEQ ID NO: 7 or an amino acid sequence having at least 85%, or alternatively at least 70%, 75%, 80%, 90%, 95%, 98% or 99%, sequence identity to SEQ ID NO:7.

In some embodiments, the HC includes the full heavy chain sequence of the wild-type fXa (i.e., amino acid residues 140-393 of SEQ ID NO:3), or an amino acid sequence having at least 85%, or alternatively at least 70%, 75%, 80%, 90%, 95%, 98% or 99%, sequence identity to the full length heavy chain sequence of the wild-type fXa.

In some embodiments, the protein (or the HC) does not include the 13 C-terminal amino acid residues (436-448 of SEQ ID NO:2) of the heavy chain. In some embodiments, the protein (or the HC) does not include the 15 C-terminal amino acid residues (434-448 of SEQ ID NO:2) of the heavy chain.

Non-limiting examples of the protein sequences include SEQ ID NO:25 and 26.

In some embodiments, a signal peptide (or in combination with pro-peptide) is included at the N-terminal end of the protein. Example signal/pro-peptides are listed in Table 7.

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TABLE 7

Signal or signal/pro-peptides	
Human fX signal peptide (SEQ ID NO: 34)	
MGRPLHLVLL	SASLAGLLLL GESLFIRREQ ANNILARVTR
Human prothrombin signal peptide (SEQ ID NO: 35)	
MAHVRGLQLP	GCLALAALCS LVHS
Human transferrin signal peptide (SEQ ID NO: 36)	
MRLAVGALLV	CAVLGLCLA
Prothrombin signal/pro-peptide (SEQ ID NO: 37)	
MAHVRGLQLP	GCLALAALCS LVHSQHVFLA PQQARSLLQR VRR
Ig kappa LC signal peptide (SEQ ID NO: 45)	
METDTLLLV	LLLWVPGSTG

The disclosed precursor protein can have the same amino acid residues as the wild-type fX, except for the indicated deletions. In a preferred embodiment, certain amino acid substitutions are introduced, such as the Ser379Ala substitution as shown in SEQ ID NO:4, leading to production of an effective antidote to fXa inhibitors. In some embodiments, the precursor protein is capable of, when the L1 and L2 are digested, producing a two-chain polypeptide capable of binding to a factor Xa inhibitor. When amino acid substitutions are introduced, in some embodiments, the digested two-chain polypeptide cannot compete with fXa in assembling into the prothrombinase complex, has reduced capacity to assemble into a prothrombin complex, or cannot assemble into a prothrombinase complex, and/or has reduced catalytic activity as compared to a wild-type human factor Xa.

Also provided, in some embodiments, are two-chain polypeptides that can be obtained by processing the precursor protein of the present disclosure. In one embodiment, provided is a two-chain polypeptide comprising a light chain that comprises the amino acid sequence of SEQ ID NO: 13 or a peptide having at least 85% sequence identity to SEQ ID NO:13, and a heavy chain that comprises a first fragment comprising the amino acid sequence of SEQ ID NO:11 or a peptide having at least 85% sequence identity to SEQ ID NO:11, and a second fragment that is at the C-terminal side of the first fragment and comprises an activation peptide, wherein the two-chain polypeptide is capable of binding to a factor Xa inhibitor. Examples of such two-chain polypeptides are provided in Tables 8 and 9 below.

TABLE 8

Amino acid sequence of activated product of construct C08 (SEQ ID NO: 29)	
Light Chain (SEQ ID NO: 13)	
1	DGDQC ETSPCQNQGK
61	CKDGLGEYTC TCLEGFEGKN CELFTRKLCS LDNGDCDQFC HEEQNSVVCs CARGYTLADN
121	GKACIPTGPY PCGKQTLER

TABLE 8-continued

Amino acid sequence of activated product of construct C08 (SEQ ID NO: 29)	
Heavy Chain (SEQ ID NO: 16) HC Portion (SEQ ID NO: 11)	
181	IVGGQE CKDGECPWQA LLINEENEGF CGGTILSEFY ILTAACHLYQ
241	AKRFKVRVGD RNTEQEEGGE AVHEVEVVIK HNRFTKETD FDI AVLRLKT PITFRMNVAP
301	ACLPERDWAE STLMTQKTGI VSGFGRTHEK GRQSTRLKML EVPNVDNRNSC KLSSSFIIITQ
361	NMFCAGYDTK QEDACQGDAG GPHVTRFKDT YFVTGIVSWG EGCARKGKYG IYTKVTAFLK
421	WIDRSMKTRG LPK
AP Portion (SEQ ID NO: 12)	
121	SVAQATSS SGEAPDSITW KPYDAADLDP TENPFDLLDF
181	NQTQPERGDN NLTR

Amino acid residue number is based on mature inactive FX (SEQ ID NO: 2)

TABLE 9

Amino acid sequence of activated product of construct C09 (SEQ ID NO: 30)	
Light Chain (SEQ ID NO: 13)	
1	DGDQC ETSPCQNGK
61	CKDGLGEYTC TCLEGFEGKN CELFTRKLCS LDNGDCDQFC HEEQNSVVC S CARGYTLADN
121	GKACIPTGPY PCGKQTLER
Heavy Chain (SEQ ID NO: 17) HC Portion (SEQ ID NO: 11)	
181	IVGGQE CKDGECPWQA LLINEENEGF CGGTILSEFY ILTAACHLYQ
241	AKRFKVRVGD RNTEQEEGGE AVHEVEVVIK HNRFTKETD FDI AVLRLKT PITFRMNVAP
301	ACLPERDWAE STLMTQKTGI VSGFGRTHEK GRQSTRLKML EVPNVDNRNSC KLSSSFIIITQ
361	NMFCAGYDTK QEDACQGDAG GPHVTRFKDT YFVTGIVSWG EGCARKGKYG IYTKVTAFLK
421	WIDRSMKTRG LPK
Linker (SEQ ID NO: 14)	
SSGSGSGSGG SGGSGSGSGS GSGSGSGSG S	
AP Portion (SEQ ID NO: 12)	
121	SVAQATSS SGEAPDSITW KPYDAADLDP TENPFDLLDF
181	NQTQPERGDN NLTR

Amino acid residue number is based on mature inactive FX (SEQ ID NO: 2)

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Polynucleotides and Host Cells

This disclosure also provides polynucleotides encoding the disclosed proteins, and host cells containing one or more of the polypeptides. In one aspect, the polypeptides are expressed and present on the cell surface (extracellularly). Suitable cells containing the inventive polypeptides include prokaryotic and eukaryotic cells, which include, but are not limited to bacterial cells, yeast cells, insect cells, animal cells, mammalian cells, murine cells, rat cells, sheep cells, simian cells and human cells. Examples of bacterial cells include *Escherichia coli*, *Salmonella enterica* and *Strepto-*

coccus gordonii. The cells can be purchased from a commercial vendor such as the American Type Culture Collection (ATCC, Rockville Maryland, USA) or cultured from an isolate using methods known in the art. Examples of suitable eukaryotic cells include, but are not limited to 293T HEK cells, as well as the hamster cell line CHO, BHK-21; the murine cell lines designated NIH3T3, NSO, C127, the simian cell lines COS, Vero; and the human cell lines HeLa, PER.C6 (commercially available from Crucell) U-937 and Hep G2. A non-limiting example of insect cells include *Spodoptera frugiperda*. Examples of yeast useful for expres-

sion include, but are not limited to *Saccharomyces*, *Schizosaccharomyces*, *Hansenula*, *Candida*, *Torulopsis*, *Yarrowia*, or *Pichia*. See e.g., U.S. Pat. Nos. 4,812,405; 4,818,700; 4,929,555; 5,736,383; 5,955,349; 5,888,768 and 6,258,559.

In some embodiments, the host cell is further transfected with a polynucleotide that encodes a furin protein.

In addition to species specificity, the cells can be of any particular tissue type such as neuronal or alternatively a somatic or embryonic stem cell such as a stem cell that can or cannot differentiate into a neuronal cell, e.g., embryonic stem cell, adipose stem cell, neuronal stem cell and hematopoietic stem cell. The stem cell can be of human or animal origin, such as mammalian.

EXAMPLES

The invention is further understood by reference to the following examples, which are intended to be purely exemplary of the invention. The present invention is not limited in scope by the exemplified embodiments, which are intended as illustrations of single aspects of the invention only. Any methods that are functionally equivalent are within the scope of the invention. Various modifications of the invention in addition to those described herein will become apparent to those skilled in the art from the foregoing description and accompanying figures. Such modifications fall within the scope of the appended claims.

Example 1. Preparation of Expression Vectors

This example generated polynucleotide constructs coding for precursor proteins that can be used to produce factor Xa derivatives, including Andexanet, which can be used to neutralize factor Xa inhibitors. The precursor protein sequences are listed in Tables 10-20 and illustrated in FIG. 1, referred to as C01-C11.

Compared to the precursor (SEQ ID NO:4) of Andexanet, C01 (SEQ ID NO:18) contained a deletion of 20 C-terminal amino acid residues of the heavy chain. Similarly, C02 (SEQ ID NO:19) and C03 (SEQ ID NO:20) contained a deletion of the 13 and 15 C-terminal amino acid residues, respectively.

On the basis of C03, the factor Xa activation peptide (AP) was added back in precursor C04 (SEQ ID NO:21). Different from the wild-type fX, a —RKRRKR— (SEQ ID NO:7) linker was placed between the AP and the heavy chain to facilitate processing by furin (there was also a —RKR— linker between the light chain and the AP, as in the wild type fX). Another version of this precursor, C05 (SEQ ID NO:22), which did not have the C-terminal truncation but removed the N-terminal 11 amino acids of the light chain, was also prepared. Yet another precursor, C06 (SEQ ID NO:23) contained both the N-terminal 11-amino acid truncation and the C-terminal 15-amino acid truncation, was also prepared. In C07 (SEQ ID NO:24), the natural —RKR— linker was removed between the light chain and the AP.

In precursors C08 (SEQ ID NO:25) and C09 (SEQ ID NO:26), the AP was placed at the C-terminal side of the heavy chain. The AP was immediately fused to the heavy chain in C08 and was fused through an artificial linker (—KSS(GSS)_gGSS— (SEQ ID NO:14)) in C09.

In precursors C10 (SEQ ID NO:27) and C11 (SEQ ID NO:28), a human serum albumin (HSA, SEQ ID NO:15) was fused to either the N-terminus or the C-terminus, where HSA can have the native sequence or with a single Cys34Ser mutation (SEQ ID NO:15) to remove the free cysteine in native HSA. In C10, the heavy chain was intact and in C11, the heavy chain had a 15-amino acid truncation at the C-terminus.

TABLE 10

Amino acid sequence of construct C01 (SEQ ID NO: 18)	
Light Chain Fragment (SEQ ID NO: 6)	
1 ANSFL	F WNKYKDGQDC ETSPCQNQ GK
61 CKDGLGEYTC	TCLEGFEGKN CELFTRKLC S LDNGDCDQFC HEEQNSVVCS CARGYTLADN
121 GKACIPTGPY	PCGKQTLER
Linker (SEQ ID NO: 7)	
RKRRKR	
Heavy Chain Fragment truncated-20 (SEQ ID NO: 9)	
181	IVGGQE CKDGECPWQA LLINEENEGF CGGTILSEFY ILTAAHCLYQ
241 AKRFKVRVGD	RNTEQEEGGE AVHEVEVVIK HNRFTKET YD FDI AVLRLKT PITFRMNVAP
301 ACLPERDWAE	STLMTQKTGI VSGFGRTHEK GRQSTR LKML EVPYVDRNSC KLSSSFIITQ
361 NMF CAGYDTK	QEDACQGDAG GPHVTRFKDT YFVTGIVSWG EGCARKGKYG IYTKVTAFLK
421 WIDRSMKT	

Amino acid residue number is based on mature inactive fX (SEQ ID NO: 2)

TABLE 11

Amino acid sequence of construct C02 (SEQ ID NO: 19)	
Light Chain Fragment (SEQ ID NO: 6)	
1 ANSFL	F WNKYKGDQC ETSPCQNQGK
61 CKDGLGEYTC TCLEGFEGKN CELFTRKLCS LDNGDCDQFC HEEQNSVVC	S CARGYTLADN
121 GKACIPTGPY PCGKQTLER	
Linker (SEQ ID NO: 7)	
RKRRKR	
Heavy Chain Fragment truncated-13 (SEQ ID NO: 10)	
181	IVGGQE CKDGECPWQA LLINEENEGF CGGTILSEFY ILTAACHLYQ
241	AKRFKVRVGD RNTEQEEGGE AVHEVEVVIK HNRFTKETD FDI AVLRLKT PITFRMNVAP
301	ACLPERDWAE STLMTQKTGI VSGFGRTHEK GRQSTRLKML EPPYVDRNSC KLSSSPIITQ
361	NMFCAGYDTK QEDACQGDAG GPHVTRFKDT YFVTGIVSWG EGCARKGKYG IYTKVTAFLK
421	WIDRSMKTRG LPKAK

Amino acid residue number is based on mature inactive fx (SEQ ID NO: 2)

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TABLE 12

Amino acid sequence of construct C03 (SEQ ID NO: 20)	
Light Chain Fragment (SEQ ID NO: 6)	
1 ANSFL	F WNKYKGDQC ETSPCQNQGK
61 CKDGLGEYTC TCLEGFEGKN CELFTRKLCS LDNGDCDQFC HEEQNSVVC	S CARGYTLADN
121 GKACIPTGPY PCGKQTLER	
Linker (SEQ ID NO: 7)	
RKRRKR	
Heavy Chain Fragment truncated-15 (SEQ ID NO: 11)	
181	IVGGQE CKDGECPWQA LLINEENEGF CGGTILSEFY ILTAACHLYQ
241	AKRFKVRVGD RNTEQEEGGE AVHEVEVVIK HNRFTKETD FDI AVLRLKT PITFRMNVAP
301	ACLPERDWAE STLMTQKTGI VSGFGRTHEK GRQSTRLKML EPPYVDRNSC KLSSSPIITQ
361	NMFCAGYDTK QEDACQGDAG GPHVTRFKDT YFVTGIVSWG EGCARKGKYG IYTKVTAFLK
421	WIDRSMKTRG LPK

Amino acid residue number is based on mature inactive fx (SEQ ID NO: 2)

TABLE 13

Amino acid sequence of construct C04 (SEQ ID NO: 21)	
Light Chain Fragment (SEQ ID NO: 6)	
1 ANSFL	F WNKYKGDQC ETSPCQNQGK
61 CKDGLGEYTC TCLEGFEGKN CELFTRKLCS LDNGDCDQFC HEEQNSVVC	S CARGYTLADN
121 GKACIPTGPY PCGKQTLER	

TABLE 13-continued

Amino acid sequence of construct C04 (SEQ ID NO: 21)	
Linker	
121	R KR
Activation Peptide (SEQ ID NO: 12)	
121	SVAQATSS SGEAPDSITW KPYDAADLDP TENPFDLLDF
181	NQTQPERGDN NLTR
Linker (SEQ ID NO: 7)	
RKRRKR	
Heavy Chain Fragment truncated-15 (SEQ ID NO: 11)	
181	IVGGQE CKDGECPWQA LLINEENEGF CGGTILSEFY ILTAAHCLYQ
241	AKRFKVRVGD RNTEQEEGGE AVHEVEVVIK HNRFTKETVD FDI AVLRLKT PITFRMNVAP
301	ACLPERDWAE STLMTQKTGI VSGFGRTHEK GRQSTRLKML EVPNVDNRNSC KLSSSPIITQ
361	NMFCAGYDTK QEDACQGDAG GPHVTRFKDT YFVTGIVSWG EGCARKGKYG IYTKVTAFLK
421	WIDRSMKTRG LPK
Amino acid residue number is based on mature inactive fX (SEQ ID NO: 2)	

TABLE 14

Amino acid sequence of construct C05 (SEQ ID NO: 22)	
Light Chain Fragment truncated (SEQ ID NO: 13)	
1	DGDQC ETSPCQNQGK
61	CKDGLGEYTC TCLEGFEGKN CELFTRKLCS LDNGDCDQFC HEEQNSVVCs CARGYTLADN
121	GKACIPTGPY PCGKQTLER
Linker	
121	R KR
Activation Peptide (SEQ ID NO: 12)	
121	SVAQATSS SGEAPDSITW KPYDAADLDP TENPFDLLDF
181	NQTQPERGDN NLTR
Linker (SEQ ID NO: 7)	
RKRRKR	
Heavy Chain Fragment (SEQ ID NO: 8)	
181	IVGGQE CKDGECPWQA LLINEENEGF CGGTILSEFY ILTAAHCLYQ
241	AKRFKVRVGD RNTEQEEGGE AVHEVEVVIK HNRFTKETVD FDI AVLRLKT PITFRMNVAP
301	ACLPERDWAE STLMTQKTGI VSGFGRTHEK GRQSTRLKML EVPNVDNRNSC KLSSSPIITQ
361	NMFCAGYDTK QEDACQGDAG GPHVTRFKDT YFVTGIVSWG EGCARKGKYG IYTKVTAFLK
421	WIDRSMKTRG LPKAKSHAPE VITSSPLK
Amino acid residue number is based on mature inactive fX (SEQ ID NO: 2)	

TABLE 15

Amino acid sequence of construct C06 (SEQ ID NO: 23)	
Light Chain Fragment truncated(SEQ ID NO: 13)	
1	DGDQC ETSPCQNQGK
61	CKDGLGEYTC TCLEGFEGKN CELFTRKLCS LDNGDCDQFC HEEQNSVVCs CARGYTLADN
121	GKACIPTGPY PCGKQTLER
Linker	
121	R KR
Activation Peptide (SEQ ID NO: 12)	
121	SVAQATSS SGEAPDSITW KPYDAADLDP TENPFDLLDF
181	NQTQPERGDN NLTR
Linker (SEQ ID NO: 7)	
	RKRRKR
Heavy Chain Fragment truncated-15 (SEQ ID NO: 11)	
181	IVGGQE CKDGECPWQA LLINEENEGF CGGTILSEFY ILTAAHCLYQ
241	AKRFKVRVGD RNTEQEEGGE AVHEVEVVIK HNRFTKETD FDI AVLRLKT PITFRMNVAP
301	ACLPERDWAE STLMTQKTGI VSGFGRTHEK GRQSTRKLML EYPYVDRNSC KLSSSFIITQ
361	NMFCAGYDTK QEDACQGDAG GPHVTRFKDT YFVTGIVSWG EGCARKGKYG IYTKVTAFLK
421	WIDRSMKTRG LPK

Amino acid residue number is based on mature inactive fX (SEQ ID NO: 2)

TABLE 16

Amino acid sequence of construct C07 (SEQ ID NO: 24)	
Light Chain Fragment (SEQ ID NO: 13)	
1	DGDQC ETSPCQNQGK
61	CKDGLGEYTC TCLEGFEGKN CELFTRKLCS LDNGDCDQFC HEEQNSVVCs CARGYTLADN
121	GKACIPTGPY PCGKQTLER
Activation Peptide (SEQ ID NO: 12)	
121	SVAQATSS SGEAPDSITW KPYDAADLDP TENPFDLLDF
181	NQTQPERGDN NLTR
Linker (SEQ ID NO: 7)	
	RKRRKR
Heavy Chain Fragment truncated-15 (SEQ ID NO: 11)	
181	IVGGQE CKDGECPWQA LLINEENEGF CGGTILSEFY ILTAAHCLYQ
241	AKRFKVRVGD RNTEQEEGGE AVHEVEVVIK HNRFTKETD FDI AVLRLKT PITFRMNVAP
301	ACLPERDWAE STLMTQKTGI VSGFGRTHEK GRQSTRKLML EYPYVDRNSC KLSSSFIITQ
361	NMFCAGYDTK QEDACQGDAG GPHVTRFKDT YFVTGIVSWG EGCARKGKYG IYTKVTAFLK
421	WIDRSMKTRG LPK

Amino acid residue number is based on mature inactive fX (SEQ ID NO: 2)

TABLE 17

Amino acid sequence of construct C08 (SEQ ID NO: 25)	
Light Chain Fragment (SEQ ID NO: 13)	
1	DGDQC ETSPCQNQGK
61	CKDGLGEYTC TCLEGFEGKN CELFTRKLCS LDNGDCDQFC HEEQNSVVCs CARGYTLADN
121	GKACIPTGPY PCGKQTLER
Linker (SEQ ID NO: 7)	
RKRKRK	
Heavy Chain Fragment truncated-15 (SEQ ID NO: 11)	
181	IVGGQE CKDGECPWQA LLINEENEGF CGGTILSEFY ILTAAHCLYQ
241	AKRFKVRVGD RNTQEEGGE AVHEVEVVIK HNRFTKETD FDI AVLRLKT PITFRMNAP
301	ACLPERDWA E STLMTQKTGI VSGFGRTHEK GRQSTRKML EVPYVDRNSC KLSSSFIITQ
361	NMFCAGYDTK QEDACQDAG GPHVTRFKDT YFVTGIVSWG EGCARKGKYG IYTKVTAPLK
421	WIDRSMKTRG LPK
Activation Peptide (SEQ ID NO: 12)	
121	SVAQATSS SGEAPDSITW KPYDAADLDP TENPFDLLDF
181	NQTQPERGDN NLTR

Amino acid residue number is based on mature inactive fx (SEQ ID NO: 2)

TABLE 18

Amino acid sequence of construct C09 (SEQ ID NO: 26)	
Light Chain Fragment truncated (SEQ ID NO: 13)	
1	DGDQC ETSPCQNQGK
61	CKDGLGEYTC TCLEGFEGKN CELFTRKLCS LDNGDCDQFC HEEQNSVVCs CARGYTLADN
121	GKACIPTGPY PCGKQTLER
Linker (SEQ ID NO: 7)	
RKRKRK	
Heavy Chain Fragment truncated-15 (SEQ ID NO: 11)	
181	IVGGQE CKDGECPWQA LLINEENEGF CGGTILSEFY ILTAAHCLYQ
241	AKRFKVRVGD RNTQEEGGE AVHEVEVVIK HNRFTKETD FDI AVLRLKT PITFRMNAP
301	ACLPERDWA E STLMTQKTGI VSGFGRTHEK GRQSTRKML EVPYVDRNSC KLSSSFIITQ
361	NMFCAGYDTK QEDACQDAG GPHVTRFKDT YFVTGIVSWG EGCARKGKYG IYTKVTAPLK
421	WIDRSMKTRG LPK
Linker (SEQ ID NO: 14)	
SSGGSGGSGG SGGSGGSGGS GSGSGGSGSG S	
Activation Peptide (SEQ ID NO: 12)	
121	SVAQATSS SGEAPDSITW KPYDAADLDP TENPFDLLDF
181	NQTQPERGDN NLTR

Amino acid residue number is based on mature inactive fx (SEQ ID NO: 2)

TABLE 19

Amino acid sequence of construct C10 (SEQ ID NO: 27)	
HSA (Human serum albumin, SEQ ID NO: 15)	
DAHKSEVAHR FDKLGEENFK ALVLIAFAQY LQQSPFEDHV KLVNEVTEFA KTCVADESAE	
NCDKSLHTLF GDKLCTVATL RETYGEMADC CAKQEPERNE CFLQHKDDNP NLPRLVRPEV	
DVMCTAFHDN EETFLKKYLY EIARRHPYFY APELLFFAKR YKAAFTECCQ AADKAACLLP	
KLDELDEGK ASSAKQRLKC ASLQKFGERA FKAWAVARLS QRFPKAEFAE VSKLVTDLTK	
VHTECCHGDL LECADDRADL AKYICENQDS ISSKLKECCE KPILLEKSHCI AEVENDEMPA	
DLPSLAADFV ESKDVCKNYA EAKDVFLGMF LYEYARRHPD YSVVLLRLA KTYETTLKEC	
CAAADPHECY AKVFDEFKPL VEEPQNLIKQ NCELFEQLGE YKFQNALLVR YTKKVPQVST	
PTLVEVSRNL GKGSKCKKH PEAKRMPCAE DYLSVVLNQL CVLHEKTPVS DRVTKCCTES	
LVNRRPCFSA LEVDETYVPK EFNAETFTFH ADICTLSEKE RQIKKQTALV ELVKHKPKAT	
KEQLKAVMDD FAAFVEKCCK ADDKETCFAE EGKKLVAASQ AALGL	
Light Chain Fragment (SEQ ID NO: 6)	
1 ANSFL	F WNKYKGDQC ETSPCQNQGK
61 CKDGLGEYTC TCLEGFEGKN CELFTRKLCS LDNGDCDQFC HEEQNSVVCs CARGYTLADN	
121 GKACIPTGPY PCGKQTLER	
Linker (SEQ ID NO: 7)	
RKRRKR	
Heavy Chain Fragment (SEQ ID NO: 8)	
181	IVGGQE CKDGECPWQA LLINEENEGF CGGTILSEFY ILTAAHCLYQ
241	AKRFKVRVDG RNTQEEGGE AVHEVEVVIK HNRFTKETD FDI AVLRLKT PITFRMNVAP
301	ACLPERDWA E STLMTQKTGI VSGFGRTHEK GRQSTRKLML EYPYVDRNSC KLSSSFIIQ
361	NMFCAGYDTK QEDACQGDAG GPHVTRFKDT YFVTGIVSWG EGCARKGKYG IYTKVTAFLK
421	WIDRSMKTRG LPKAKSHAPE VITSSPLK

Amino acid residue number is based on mature inactive FX (SEQ ID NO: 2)

TABLE 20

Amino acid sequence of construct C11 (SEQ ID NO: 28)	
Light Chain (SEQ ID NO: 6)	
1 ANSFL	F WNKYKGDQC ETSPCQNQGK
61 CKDGLGEYTC TCLEGFEGKN CELFTRKLCS LDNGDCDQFC HEEQNSVVCs CARGYTLADN	
121 GKACIPTGPY PCGKQTLER	
Linker (SEQ ID NO: 7)	
RKRRKR	
Heavy Chain Fragment truncated-15 (SEQ ID NO: 11)	
181	IVGGQE CKDGECPWQA LLINEENEGF CGGTILSEFY ILTAAHCLYQ
241	AKRFKVRVDG RNTQEEGGE AVHEVEVVIK HNRFTKETD FDI AVLRLKT PITFRMNVAP
301	ACLPERDWA E STLMTQKTGI VSGFGRTHEK GRQSTRKLML EYPYVDRNSC KLSSSFIIQ
361	NMFCAGYDTK QEDACQGDAG GPHVTRFKDT YFVTGIVSWG EGCARKGKYG IYTKVTAFLK
421	WIDRSMKTRG LPK

TABLE 20-continued

Amino acid sequence of construct C11 (SEQ ID NO: 28)	
HSA (Human serum albumin, SEQ ID NO: 15)	
DAHKSEVAHR FKDLGEENFK ALVLIQFAQY LQQSPFEDHV KLVNEVTEFA KTCVADESAE	
NCDKSLHTLF GDKLCTVATL RETYGEADC CAKQEPERNE CFLQHKDDNP NLPRLVRPEV	
DVMCTAFHDN EETFLKKYLY EIARRHPYFY APELLFFAKR YKAAFTECCQ AADKAACLLP	
KLDELDEGK ASSAKQRLKC ASLQKFGERA FKAWAVARLS QRFPKAEFAE VSKLVTDLTK	
VHTECCHGDL LECADDRADL AKYICENQDS ISSKLKECCE KPILLEKSHCI AEVENDEMPA	
DLPSLAADFV ESKDVCKNYA EAKDVFLGMF LYEYARRHPD YSVVLLRLA KTYETTLEKC	
CAAADPHECY AKVFDEFKPL VEEPQNLIKQ NCELFEQLGE YKFQNALLVY YTKKVPQVST	
PTLVEVSRNL GKGSGKCKH PEAKRMPCAE DYLSVVLNQL CVLHEKTPVS DRVTKCCTES	
LVNRRPCFSA LEVDETYVPK EFNAETFTFH ADICTLSEKE RQIKKQTLV ELVKHKPKAT	
KEQLKAVMDD FAFVKECKK ADDKETCFAE EGKLVAAASQ AALGL	

Amino acid residue number is based on mature inactive fX (SEQ ID NO: 2)

Polynucleotides coding for these precursor proteins (with suitable signal peptides) were synthesized de novo. Following verification of these polynucleotide sequences, they were ligated into an expression vector suitable for transfection to a mammalian host cell. An example expression vector was the AB1 vector. Another expression vector used was the pcDNA3.3 vector (Thermo Fisher Scientific).

The expression vectors were transfected into host cells CHO-DUXB11, CHO-S (Thermo Fisher Scientific), ExpiCHO-S (Thermo Fisher Scientific), DG44 (Thermo Fisher Scientific), or CHO-M. The transfection was carried out using the ExpiFectamine™ CHO transfection kit (Thermo Fisher Scientific) or electroporation with reagents and electroporator (Maxcyte) according to the manufacturer's recommendations.

The transfected CHO cells were cultured in shake flasks and monitored for protein expressions over time. The protein expressions were monitored transiently without prior use of antibiotics for stable cells selection. Protein expressions were also monitored using stable CHO cell pools. Stable pools were expanded and cell banks were prepared and stored in liquid nitrogen. Protein expressions with stable pools were cultured similarly as for transient expression by thawing a vial of the cell bank.

To improve the process of the linker connecting the heavy and light chains of the fXa derivatives, a second polynucleotide sequence encoding human furin protein was co-transfected in certain samples. The furin construct was introduced using a separate vector. For stable cell production, stable cell pools were first selected using pcDNA3.3 with the fXa derivatives and G-418 (0-1500 µg/mL) to generate stable pools (parental pools). The parental pools were further transfected with a second vector with furin (super-transfection). The super-transfected pools were further selected using antibiotics and prepare for cell banks.

The protein expression levels in cell culture were measured by ELISA (FX-EIA, Enzyme Research Laboratories) and characterized by western blots using monoclonal antibodies against FX/fXa heavy and light chains. Andexanet was used to construct the standard curve. Proteins in harvested cell culture fluids were purified by an affinity-based method using STI-resin to capture functional proteins.

Example 2. Testing of Expression Levels and Activities

This example tested the expression levels and activities of the fXa derivatives prepared in Example 1.

The expression and functional activities of Andexanet (AnXa) precursor (SEQ ID NO:4) and C01-C03 were tested in transient transfections (co-transfection with furin). Conditions included, chemical transfection with Expifectamine, followed by addition of enhancer and feed after 24 hrs, at 37° C., with 8% CO₂, 135 rpm. The results are shown in the table below.

Construct	Day post transfection	N number	ELISA (µg/mL)	Functional activity (µg/mL)
AnXa precursor	6	11	4.6 ± 1.6	3.8
C01	6	3	2.6 ± 0.8	BLQ*
C02	6	2	4.9 ± 2	5.9
C03	6	2	4.1 ± 0.9	3.1

*BLQ: below level of quantification

Stable pools of these constructs were generated. The total cell culture volume was 30 mL. The cells were passaged every 3 days up to full recovery of cells at >97% viability. Conditions for titer were as follows: 32° C., 5% CO₂, 135 rpm; addition of feed on Day 1, 4, 7, 10. The results are shown in the table below.

Construct	Day post transfection	N number	ELISA (µg/mL)	Functional activity (µg/mL)
AnXa precursor	10	1	22.3	18.5
C01	10	1	5.7	3.0
C02	10	1	38.5	26
C03	10	1	19.1	16.0

The following table shows the testing results when the cells were further transfected with furin.

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Construct	Day post transfection	N number	ELISA (μg/mL)	Functional activity (μg/mL)
AnXa precursor	10	1	27.3	13
C01	10	1	3.3	BLQ*
C02	10	1	12.2	12.2
C03	10	1	19.1	13.2

*BLQ: below level of quantification

The results showed that, in both transient and stable expression, C01 had low levels of expression, while C02 and C03 performed similarly to the Andexanet precursor. Therefore, truncation of 13 or 15 C-terminal residues from the heavy chain did not impact the protein expression or activity, but deletion of the 20-terminal residues had significant negative impact.

Compared to C01-C03, C04-C11 further included a factor X activation peptide (AP) or a human serum albumin (HSA). The AP is removed when the wild-type factor X is activated. In the production of Andexanet, the AP was not part of the construct. As shown in the results below, inclusion of the AP significantly increased protein expression. Further, when the AP was fused to the C-terminal end of the heavy chain (as opposed to the light chain), the construct yielded products that were most functionally active (C08 and C09). Fusion to the HSA was also helpful in increasing the expression and activity, but the impact was less pronounced as compared to AP.

Expression Level Measured by ELISA

	C04	C05	C06	C07	C08	C09	C10	C11
Day 1	4.1	6.2	1.0	3.1	3.0	2.5	1.4	1.3
Day 2	11.9	34.5	5.7	10.3	18.1	11.8	8.1	6.8

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-continued

	C04	C05	C06	C07	C08	C09	C10	C11
Day 3	38.1	49.3	11.7	20.4	43.3	31.1	12.5	11.7
Day 4	29.6	76.6	15.3	23.7	50.5	34.8	17.9	15.0
Day 5	34.7	117.4	18.3	40.5	59.1	48.4	24.3	17.6
Day 6	36.4	163.0	23.4	45.0	78.8	57.7	37.3	22.2
Day 7	36.1	172.8	19.8	49.6	83.9	61.0	36.6	22.4
Day 8	34.1	183.7	20.3	52.1	104.0	65.2	41.2	26.1
Day 9	32.9	205.1	21.0	63.8	111.2	76.9	49.9	31.2
Day 10	32.2	170.4	20.0	59.9	101.0	114.0	39.5	35.6

Functional Activities

	C04	C05	C06	C07	C08	C09	C10	C11
Day 1	0.2	0.1	0.0	0.22	0.2	0.2	0.3	0.3
Day 2	0.3	0.2	0.2	0.27	1.1	0.5	2.3	2.4
Day 3	0.8	0.5	0.4	0.32	5.0	2.6	6.4	8.3
Day 4	1.1	1.1	0.5	0.55	11.3	5.0	13.1	19.9
Day 5	2.6	2.9	0.7	0.63	32.8	10.7	27.0	32.7
Day 6	3.5	4.8	1.0	0.66	36.3	22.6	43.5	42.4
Day 7	6.9	10.3	1.5	1.61	43.9	32.3	33.0	43.3
Day 8	10.8	15.4	2.9	4.99	53.9	36.2	50.5	50.9
Day 9	11.1	26.1	6.0	10.06	74.0	—	58.4	56.0
Day 10	7.0	30.3	7.3	15.55	74.0	70.5	29.7	60.5

Based on the above results, C05, C07, C08 and C10 were further studied for their expression and functional activities. As shown in FIG. 2A-2D, co-transfection of the constructs with furin helped to increase functional protein titer in most cases. However, among C05, C07, C08 and C10, only C08 retained both the high expression level and functional activity.

Further, as shown in the table below, co-transfection of C08 with 5% furin yielded the highest Picogram/Cell/Day (PCD) ~1.9 in both calculations from ELISA and functional quantitative analyses.

Specific productivities based on ELISA and functional activity

Day	C05		C07		C08		C10	
	ELISA	Functional	ELISA	Functional	ELISA	Functional	ELISA	Functional
1	2.04	0.15	1.13	0.28	2.09	1.54	0.56	0.48
2	2.75	0.71	2.13	0.71	2.83	3.41	0.77	0.82
3	2.32	0.97	2.29	0.79	2.59	3.11	0.61	0.86
4	1.77	0.93	1.84	0.89	2.19	2.62	0.48	0.89
5	1.54	0.90	1.74	0.80	2.18	2.20	0.52	0.86
6	1.40	0.89	1.36	0.69	2.16	1.83	0.49	0.73
7	1.11	0.67	1.03	0.61	1.92	1.46	0.37	0.58
8	1.01	0.70	0.80	0.44	1.82	1.83	0.34	0.51
9	0.85	0.66	0.78	0.43	1.70	1.69	0.36	0.54
10	0.92	0.78	0.59	0.43	1.13	1.65	0.35	0.55
Average (Day 4-10)	1.23	0.79	1.16	0.61	1.87	1.90	0.42	0.67

Example 3. Preparation and Testing of Additional Constructs

Based on the testing results in Example 2, this Example prepared and tested a few additional constructs, the sequences of which are provided in the tables below.

TABLE 21

Amino acid sequence of construct C12 (SEQ ID NO: 41)	
Light Chain (SEQ ID NO: 6)	
1 ANSFL	F WNKYKGDQC ETSPCQNQGK
61 CKDGLGEYTC TCLEGFEGKN CELFTRKLCS LDNGDCDQFC HEEQNSVVCSCARGYTLADN	
121 GKACIPTGPY PCGKQTLER	
Linker	
121	R KR
Activation Peptide (SEQ ID NO: 12)	
121	SVAQATSS SGEAPDSITW KPYDAADLDP TENPFDLLDF
181	NQTQPERGDN NLTR
Linker (SEQ ID NO: 7)	
	RKRRKR
Heavy Chain Fragment (SEQ ID NO: 8)	
181	IVGGQE CKDGECPWQA LLINEENEGF CGGTILSEFY ILTAAHCLYQ
241	AKRFKVRVGD RNTQEEGGE AVHEVEVVIK HNRFTKETVD FDI AVLRLKT PITFRMNVAP
301	ACLPERDWAE STLMTQKTGI VSGFGRTHEK GRQSTRLKML EVPNVDNRSC KLSSEFIITQ
361	NMFCAGYDTK QEDACQGDAG GPHVTRFKDT YFVTGIVSWG EGCARKGKYG IYTKVTAFLK
421	WIDRSMKTRG LPKAKSHAPE VITSSPLK
Activation Peptide without N-terminal S (SEQ ID NO: 44)	
121	VAQATSS SGEAPDSITW KPYDAADLDP TENPFDLLDF
181	NQTQPERGDN NLTR

Amino acid residue number is based on mature inactive fX (SEQ ID NO: 2)

TABLE 22

Amino acid sequence of construct C13 (SEQ ID NO: 42)	
HSA (Human serum albumin, SEQ ID NO: 15)	
DAHKSEVAHR FKLGEENFK ALVLIAFAQY LQQSPFEDHV KLVNEVTEFA KTCVADESAA	
NCDKSLHTLF GDKLCTVATL RETYGEMADC CAKQEPERNE CFLQHKDDNP NLPRLVRPEV	
DVMCTAFHDN EETFLKKYLY EIARRHPYFY APELLFFAKR YKAAFTECCQ AADKAACLLP	
KLDELDRDEG ASSAKQLKCA SLQKFGERA FKAWAVARLS QRFPKAEFAE VSKLVTDLTG	
VHTECCHGDL LECADDRADL AKYICENQDS ISSKLKECCE KPILLEKSHCI AEVENDEMPA	
DLPSLAADFV ESKDVCKNYA EAKDVFLGMF LYEYARRHPD YSVVLLRLA KTYETTLEKC	
CAAADPHECY AKVFDEFKPL VEEPQNLIKQ NCELFEQLGE YKFNALLVR YTKKVPQVST	
PTLVEVSRNL GKVGSKCKKH PEAKRMPCAE DYLSVVLNQL CVLHEKTPVS DRVTKCCTES	
LVNRRPCFSA LEVDETYVPK EFNAETFTFH ADICTLSEKE RQIKKQATLV ELVKHKPKAT	
KEQLKAVMDD FAAFVEKCKC ADDKETCFAE EGKLVAAASQ AALGL	

TABLE 22-continued

Amino acid sequence of construct C13 (SEQ ID NO: 42)	
Light Chain Fragment truncated (SEQ ID NO: 13)	
1	DGDQC ETSPCQNQGK
61 CKDGLGEYTC TCLEGFEGKN CELFTRKLCS LDNGDCDQFC HEEQNSVVCs CARGYTLADN	
121 GKACIPTGPY PCGKQTLER	
Linker (SEQ ID NO: 7)	
RKRRKR	
Heavy Chain Fragment (SEQ ID NO: 8)	
181	IVGGQE CKDGECPWQA LLINEENEGF CGGTILSEFY ILTAAHCLYQ
241 AKRFKVRVGD RNTEQEEGGE AVHEVEVVIK HNRFTKETYD FDI AVLRLKT PITFRMN VAP	
301 ACLPERDWAE STLMTQKTGI VSGFGRTHEK GRQSTRLKML EVPYVDRNSC KLSSSFIITQ	
361 NMFAGYDTK QEDACQDAG GPHVTRFKDT YFVTGIVSWG EGCARKGKYG IYTKVTAFLK	
421 WIDRSMKTRG LPKAKSHAPE VITSSPLK	

Amino acid residue number is based on mature inactive fX (SEQ ID NO: 2)

25

TABLE 23

Amino acid sequence of construct C14 (SEQ ID NO: 43)	
HSA (Human serum albumin, SEQ ID NO: 15)	
DAHKSEVAHR FKDLGEENFK ALVLIAFAQY LQQSPFEDHV KLVNEVTEFA KTCVADES AE	
NCDKSLHTLF GDKLCTVATL RETYGEMADC CAKQEPERNE CFLQHKDDNP NLPRLVRPEV	
DVMCTAFHDN EETFLKKYLY EIARRHPYFY APELLFFAKR YKAAFTECCQ AADKAACLLP	
KLDEL RDEGK ASSAKQRLKC ASLQKFGERA FKAWAVARLS QRFPKAEFAE VSKLVTDLT K	
VHTECCHGDL LECADDRADL AKYICENQDS ISSKLKECCE KP LLEKSHCI AEVENDEMPA	
DLPSLAADFV ESKDVCKNYA EAKDVFLGMF LY EYARRHPD YSVVLLRLA KTYETTLEKC	
CAAADPHECY AKVFDEFKPL VEEPQNLIKQ NCELFEQLGE YKFQNAL LVR YTKKVPQVST	
PTLVEVSRNL GKVGSKCKH PEAKRMPCAE DYLSVVLNQL CVLHEKTPVS DRVTKCCTES	
LVNRRPCFSA LEVDETYVPK EFNAETFTFH ADICTLSEKE RQIKKQTALV ELVKHKPKAT	
KEQLKAVMDD FAAFVEKCCK ADDKETCF AE EGKKLVAASQ AALGL	
Light Chain Fragment truncated (SEQ ID NO: 13)	
1	DGDQC ETSPCQNQGK
61 CKDGLGEYTC TCLEGFEGKN CELFTRKLCS LDNGDCDQFC HEEQNSVVCs CARGYTLADN	
121 GKACIPTGPY PCGKQTLER	
Linker (SEQ ID NO: 7)	
RKRRKR	
Heavy Chain Fragment truncated-15 (SEQ ID NO: 11)	
181	IVGGQE CKDGECPWQA LLINEENEGF CGGTILSEFY ILTAAHCLYQ
241 AKRFKVRVGD RNTEQEEGGE AVHEVEVVIK HNRFTKETYD FDI AVLRLKT PITFRMN VAP	
301 ACLPERDWAE STLMTQKTGI VSGFGRTHEK GRQSTRLKML EVPYVDRNSC KLSSSFIITQ	

TABLE 23-continued

Amino acid sequence of construct C14 (SEQ ID NO: 43)	
361	NMFCAGYDTK QEDACQGDAG GPHVTRFKDT YFVTGIVSWG EGCARKGKYG IYTKVTAFLK
421	WIDRSMKTRG LPK
Activation Peptide (SEQ ID NO: 12)	
121	SVAQATSS SGEAPDSITW KPYDAADLDP TENPFDLLDF
181	NQTQPERGDN NLTR

Amino acid residue number is based on mature inactive fX (SEQ ID NO: 2)

The domain structures of these constructs are also illustrated in FIG. 1. C12, compared to C08, has an intact heavy chain and an additional AP inserted between the EGF2 and the heavy chain. C13 is similar to C10 but with an N-terminus truncated light chain. Finally, C14 included both a HSA at the N-terminus (like C14) and an AP at the C-terminus (like C08).

The expression levels and activities of these constructs, along with Andexanet precursor (AnXa), C10 and C12 as controls, were measured. The results are shown in the tables below and plotted in FIG. 3.

C13, which had a HSA fused directly to the EGF1 domain of the light chain, and C14, which further included an AP domain at the C-terminus of the heavy chain, exhibited the highest expression and activities. Interestingly, even though the only difference between C10 and C13 is that C10 further included certain amino acids from the Gla-domain (A1 to K11), C10's expression was markedly lower, suggesting that a direct fusion between the HSA and the EGF1 domain is beneficial. Moreover, the result of C12 suggests that adding an additional AP between the light chain and the heavy chain is not necessary.

ELISA (mg/L) (with each purified protein as standard)												
Day	AnXa		C08	C10		C12		C13			C14	
	n1	n2	n1	n1	n2	n1	n2	n1	n2	n3	n1	n2
Day 7	—	—	81.0	—	86.6	—	—	159.7	109.2	137.7	—	—
Day 8	—	—	83.1	85.8	93.9	77.9	87.7	175.5	103.3	173.5	—	—
Day 9	—	—	85.9	92.7	85.6	79.1	87.8	197.1	108.5	178.4	170.9	127.2
Day 10	25.5	25.3	107.4	108.4	82.2	71.8	77.1	195.0	135.0	161.5	213.9	173.5
Mean	25.4		89.4	90.7		80.2		152.9			171.4	
SD	0.1		12.2	8.8		6.3		33.5			35.4	

Functional Activity (mg/L) (with each purified protein as standard)												
	AnXa		C08	C10		C12		C13			C14	
Day	n1	n2	n1	n1	n2	n1	n2	n1	n2	n3	n1	n2
Day 8	—	27.0	—	116.0	102.7	86.6	94.7	—	287.4	173.5	260.9	158.0
Day 9	19.1	27.0	84.7	131.0	100.8	84.5	88.3	—	257.5	178.4	306.0	164.0
Day 10	20.9	34.3	99.4	126.3	80.7	—	—	181.9	308.3	161.5	318.0	216.5
Mean		25.6	92.1		109.6		88.5		221.2			237.2
SD		6.0	10.4		18.7		4.4		61.3			69.1

PCD (pg/cell/day)												
Day	AnXa		C08	C10		C12		C13			C14	
	n1	n2	n1	n1	n2	n1	n2	n1	n2	n3	n1	n2
Day 3	0.57	0.52	2.26	2.99	2.40	3.07	3.28	4.04	5.13	4.98	5.04	4.89
Day 4	0.57	0.40	2.02	2.70	2.24	2.40	2.72	4.60	4.68	4.98	4.61	4.10
Day 5	0.47	0.37	2.19	2.52	2.11	2.44	2.55	4.96	6.58	5.23	4.67	4.85
Day 7	0.53	0.42	2.34	2.38	2.33	1.62	1.87	3.62	4.94	3.99	5.03	4.74
Mean	0.48		2.20	2.46		2.49		4.81			4.74	
SD	0.08		0.13	0.28		0.56		0.75			0.30	

This example further demonstrates the benefit of fusing a HSA at the N-terminus of the light chain and fusing an AP domain at the C-terminus of the heavy chain in the constructs.

Example 4. Preparation and Testing of Fc Fusions

This example tested Fc fragments and fusions between the Fc fragments and a factor Xa derivative, such as Andexanet.

Two fusion constructs were prepared, as shown in Tables 25 and 26, which included the Fc fragment of Table 24, and two different signal peptides.

TABLE 24

Fc Fragment (SEQ ID NO: 46)	
P01857 IgG1 Human CH1-hinge-CH2-CH3	
5	ASTKGPSVFPP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS
	WNSGALTSGV HTFPAVLQSS GLYSLSSVVT VPSSSLGTQT
	YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPELLGG
10	PSVFLFPPKP KDTLMISRTP EVTCVVVDVS HEDPEVKFNW
	YVDGVEVHNA KTKPREEQYN STYRVVSVLT VLHQDWLNGK
	EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSRDE
15	LTKNQVSLTC LVKGFYPSDI AVEWESNGQP ENNYKTTTPV
	LDSDGSFFLY SKLTVDKSRW QQGNVFSCSV MHEALHNHYT
	QKSLSLSPGK

TABLE 25

Fc Fusion Construct F01 (SEQ ID NO: 47)	
Signal peptide of fXa (SEQ ID NO: 34)	
MGRPLHLVLL SASLAGLLLL GESLFIRREQ ANNILARVTR	
Light Chain Fragment (SEQ ID NO: 6)	
1	ANSFL F WNKYKGDQC ETSPCQNQGK
61	CKDGLGEYTC TCLEGFEGKN CELFTRKLCS LDNGDCDQFC HEEQNSVVCS CARGYTLADN
121	GKACIPTGPY PCGKQTLER
Linker (SEQ ID NO: 7)	
RKRRKR	
Heavy Chain Fragment (SEQ ID NO: 8)	
181	IVGGQE CKDGECPWQA LLINEENEGF CGGTILSEFY ILTAAHCLYQ
241	AKRFKVRVGD RNTEQEEGGE AVHEVEVVIK HNRFTKETD FDI AVLRLKT PITFRMNVAP
301	ACLP ERD WAE STLMTQKTGI VSGFGRTHEK GRQSTR LKML E VPYVDRNSC KLSSSFII TQ
361	NMF CAGYDTK QEDACQGDAG GPHVTRFKDT YFVTGIVSWG EGCARKGKYG IYTKVTAFLK
421	WIDRSMKTRG LPKAKSHAPE VITSSPLK
Fc fragment (SEQ ID NO: 46)	
ASTKGPSVFPP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV	
HTFPAVLQSS GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP	
KSCDKTHTCP PCPAPELLGG PSVFLFPPKP KDTLMISRTP EVTCVVVDVS	
HEDPEVKFNW YVDGVEVHNA KTKPREEQYN STYRVVSVLT VLHQDWLNGK	
EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSRDE LTKNQVSLTC	
LVKGFYPSDI AVEWESNGQP ENNYKTTTPV LDSDGSFFLY SKLTVDKSRW	
QQGNVFSCSV MHEALHNHYT QKSLSLSPGK	

Amino acid residue number is based on mature inactive fX (SEQ ID NO: 2)

TABLE 26

Fc Fusion Construct F02 (SEQ ID NO: 48)	
Signal peptide of fXa	
METDTLLLVV LLLWVPGSTG (SEQ ID NO: 45)	
Light Chain Fragment (SEQ ID NO: 6)	
1	ANSFL F WNKYKGDGQC ETSPCQNQGK
61	CKDGLGEYTC TCLEGFEGKN CELFTRKLCs LDNGDCDQFC HEEQNSVVCs CARGYTLADN
121	GKACIPTGPY PCGKQTLER
Linker (SEQ ID NO: 7)	
RKRRKR	
Heavy Chain Fragment (SEQ ID NO: 8)	
181	IVGGQE CKDGECPWQA LLINEENEGF CGGTILSEFY ILTAAHCLYQ
241	AKRPFKVRVGD RNTQELEGGE AVHEVEVVIK HNRFTKETVD FDI AVLRLKT PITFRMNVAP
301	ACLPERDWAE STLMTQKTGI VSGFGRTHEK GRQSTRKML EVPYVDRNSC KLSSEFIITQ
361	NMFCAGYDTK QEDACQGDAG GPHVTRFKDT YFVTGIVSWG EGCARKGKYG IYTKVTAFLK
421	WIDRSMKTRG LPKAKSHAPE VITSSPLK
Fc fragment (SEQ ID NO: 46)	
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV	
HTFPAVLQSS GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP	
KSCDKHTHCP PCPAPELLGG PSVFLFPPKP KDTLMISRTP EVTCVVVDVS	
HEDPEVKFNW YVDGVEVHNA KTKPREEQYN STYRVVSVLT VLGQDWLNGK	
EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSRDE LTKNQVSLTC	
LVKGFYPSDI AVEWESNGQP ENNYKTPPV LDSDGSFFLY SKLTVDKSRW	
QQGNVFSCSV MHEALHNHYT QKSLSLSPGK	

Amino acid residue number is based on mature inactive fX (SEQ ID NO: 2)

The expression of these constructs were tested in transient transfections. Conditions included, chemical transfection with Expifectamine, followed by addition of enhancer and feed after 24 hrs, at 37° C., with 8% CO₂, 135 rpm. As shown in the results below, fXa's signal peptide (construct F01) helped increase the expression of the fusion protein. Expression Level Measured by ELISA

	Fc Fragment		Fusion F01		Fusion F02	
Timepoint	Mean (µg/mL)	SD	Mean (µg/mL)	SD	Mean (µg/mL)	SD
Day 1	2.4	0.2	0.1	0.0	0.2	0.0
Day 2	5.6	0.9	1.0	0.1	1.1	0.2
Day 3	12.0	3.0	1.7	0.3	2.5	1.4
Day 4	14.4	1.9	3.9	0.0	2.3	0.4

-continued

	Fc Fragment		Fusion F01		Fusion F02	
Timepoint	Mean (µg/mL)	SD	Mean (µg/mL)	SD	Mean (µg/mL)	SD
Day 5	15.0	5.3	5.0	0.0	4.1	0.6
Day 6	15.8	3.9	6.7	0.3	4.7	0.8
Day 7	24.5	3.9	9.4	0.3	6.0	1.1

It is to be understood that while the invention has been described in conjunction with the above embodiments, that the foregoing description and examples are intended to illustrate and not limit the scope of the invention. Other aspects, advantages and modifications within the scope of the invention will be apparent to those skilled in the art to which the invention pertains.

SEQUENCE LISTING

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<212> TYPE: PRT

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<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 1

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 20          25          30
Asn Ile Leu Ala Arg Val Thr Arg Ala Asn Ser Phe Leu Glu Glu Met
 35          40          45
Lys Lys Gly His Leu Glu Arg Glu Cys Met Glu Glu Thr Cys Ser Tyr
 50          55          60
Glu Glu Ala Arg Glu Val Phe Glu Asp Ser Asp Lys Thr Asn Glu Phe
 65          70          75          80
Trp Asn Lys Tyr Lys Asp Gly Asp Gln Cys Glu Thr Ser Pro Cys Gln
 85          90          95
Asn Gln Gly Lys Cys Lys Asp Gly Leu Gly Glu Tyr Thr Cys Thr Cys
100          105          110
Leu Glu Gly Phe Glu Gly Lys Asn Cys Glu Leu Phe Thr Arg Lys Leu
115          120          125
Cys Ser Leu Asp Asn Gly Asp Cys Asp Gln Phe Cys His Glu Glu Gln
130          135          140
Asn Ser Val Val Cys Ser Cys Ala Arg Gly Tyr Thr Leu Ala Asp Asn
145          150          155          160
Gly Lys Ala Cys Ile Pro Thr Gly Pro Tyr Pro Cys Gly Lys Gln Thr
165          170          175
Leu Glu Arg Arg Lys Arg Ser Val Ala Gln Ala Thr Ser Ser Ser Gly
180          185          190
Glu Ala Pro Asp Ser Ile Thr Trp Lys Pro Tyr Asp Ala Ala Asp Leu
195          200          205
Asp Pro Thr Glu Asn Pro Phe Asp Leu Leu Asp Phe Asn Gln Thr Gln
210          215          220
Pro Glu Arg Gly Asp Asn Asn Leu Thr Arg Ile Val Gly Gly Gln Glu
225          230          235          240
Cys Lys Asp Gly Glu Cys Pro Trp Gln Ala Leu Leu Ile Asn Glu Glu
245          250          255
Asn Glu Gly Phe Cys Gly Gly Thr Ile Leu Ser Glu Phe Tyr Ile Leu
260          265          270
Thr Ala Ala His Cys Leu Tyr Gln Ala Lys Arg Phe Lys Val Arg Val
275          280          285
Gly Asp Arg Asn Thr Glu Gln Glu Glu Gly Gly Glu Ala Val His Glu
290          295          300
Val Glu Val Val Ile Lys His Asn Arg Phe Thr Lys Glu Thr Tyr Asp
305          310          315          320
Phe Asp Ile Ala Val Leu Arg Leu Lys Thr Pro Ile Thr Phe Arg Met
325          330          335
Asn Val Ala Pro Ala Cys Leu Pro Glu Arg Asp Trp Ala Glu Ser Thr
340          345          350
Leu Met Thr Gln Lys Thr Gly Ile Val Ser Gly Phe Gly Arg Thr His
355          360          365
Glu Lys Gly Arg Gln Ser Thr Arg Leu Lys Met Leu Glu Val Pro Tyr
370          375          380
Val Asp Arg Asn Ser Cys Lys Leu Ser Ser Ser Phe Ile Ile Thr Gln
385          390          395          400

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Asn Met Phe Cys Ala Gly Tyr Asp Thr Lys Gln Glu Asp Ala Cys Gln
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Gly Asp Ser Gly Gly Pro His Val Thr Arg Phe Lys Asp Thr Tyr Phe
      420                      425                      430

Val Thr Gly Ile Val Ser Trp Gly Glu Gly Cys Ala Arg Lys Gly Lys
      435                      440                      445

Tyr Gly Ile Tyr Thr Lys Val Thr Ala Phe Leu Lys Trp Ile Asp Arg
      450                      455                      460

Ser Met Lys Thr Arg Gly Leu Pro Lys Ala Lys Ser His Ala Pro Glu
      465                      470                      475                      480

Val Ile Thr Ser Ser Pro Leu Lys
      485

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<210> SEQ ID NO 2
<211> LENGTH: 448
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

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<400> SEQUENCE: 2

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Cys Met Glu Glu Thr Cys Ser Tyr Glu Glu Ala Arg Glu Val Phe Glu
 20          25          30

Asp Ser Asp Lys Thr Asn Glu Phe Trp Asn Lys Tyr Lys Asp Gly Asp
 35          40          45

Gln Cys Glu Thr Ser Pro Cys Gln Asn Gln Gly Lys Cys Lys Asp Gly
 50          55          60

Leu Gly Glu Tyr Thr Cys Thr Cys Leu Glu Gly Phe Glu Gly Lys Asn
 65          70          75          80

Cys Glu Leu Phe Thr Arg Lys Leu Cys Ser Leu Asp Asn Gly Asp Cys
 85          90          95

Asp Gln Phe Cys His Glu Glu Gln Asn Ser Val Val Cys Ser Cys Ala
100          105          110

Arg Gly Tyr Thr Leu Ala Asp Asn Gly Lys Ala Cys Ile Pro Thr Gly
115          120          125

Pro Tyr Pro Cys Gly Lys Gln Thr Leu Glu Arg Arg Lys Arg Ser Val
130          135          140

Ala Gln Ala Thr Ser Ser Ser Gly Glu Ala Pro Asp Ser Ile Thr Trp
145          150          155          160

Lys Pro Tyr Asp Ala Ala Asp Leu Asp Pro Thr Glu Asn Pro Phe Asp
165          170          175

Leu Leu Asp Phe Asn Gln Thr Gln Pro Glu Arg Gly Asp Asn Asn Leu
180          185          190

Thr Arg Ile Val Gly Gly Gln Glu Cys Lys Asp Gly Glu Cys Pro Trp
195          200          205

Gln Ala Leu Leu Ile Asn Glu Glu Asn Glu Gly Phe Cys Gly Gly Thr
210          215          220

Ile Leu Ser Glu Phe Tyr Ile Leu Thr Ala Ala His Cys Leu Tyr Gln
225          230          235          240

Ala Lys Arg Phe Lys Val Arg Val Gly Asp Arg Asn Thr Glu Gln Glu
245          250          255

Glu Gly Gly Glu Ala Val His Glu Val Glu Val Val Ile Lys His Asn
260          265          270

Arg Phe Thr Lys Glu Thr Tyr Asp Phe Asp Ile Ala Val Leu Arg Leu
275          280          285

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-continued

Lys Thr Pro Ile Thr Phe Arg Met Asn Val Ala Pro Ala Cys Leu Pro
290 295 300

Glu Arg Asp Trp Ala Glu Ser Thr Leu Met Thr Gln Lys Thr Gly Ile
305 310 315 320

Val Ser Gly Phe Gly Arg Thr His Glu Lys Gly Arg Gln Ser Thr Arg
325 330 335

Leu Lys Met Leu Glu Val Pro Tyr Val Asp Arg Asn Ser Cys Lys Leu
340 345 350

Ser Ser Ser Phe Ile Ile Thr Gln Asn Met Phe Cys Ala Gly Tyr Asp
355 360 365

Thr Lys Gln Glu Asp Ala Cys Gln Gly Asp Ser Gly Gly Pro His Val
370 375 380

Thr Arg Phe Lys Asp Thr Tyr Phe Val Thr Gly Ile Val Ser Trp Gly
385 390 395 400

Glu Gly Cys Ala Arg Lys Gly Lys Tyr Gly Ile Tyr Thr Lys Val Thr
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<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

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20 25 30

Asp Ser Asp Lys Thr Asn Glu Phe Trp Asn Lys Tyr Lys Asp Gly Asp
35 40 45

Gln Cys Glu Thr Ser Pro Cys Gln Asn Gln Gly Lys Cys Lys Asp Gly
50 55 60

Leu Gly Glu Tyr Thr Cys Thr Cys Leu Glu Gly Phe Glu Gly Lys Asn
65 70 75 80

Cys Glu Leu Phe Thr Arg Lys Leu Cys Ser Leu Asp Asn Gly Asp Cys
85 90 95

Asp Gln Phe Cys His Glu Glu Gln Asn Ser Val Val Cys Ser Cys Ala
100 105 110

Arg Gly Tyr Thr Leu Ala Asp Asn Gly Lys Ala Cys Ile Pro Thr Gly
115 120 125

Pro Tyr Pro Cys Gly Lys Gln Thr Leu Glu Arg Ile Val Gly Gly Gln
130 135 140

Glu Cys Lys Asp Gly Glu Cys Pro Trp Gln Ala Leu Leu Ile Asn Glu
145 150 155 160

Glu Asn Glu Gly Phe Cys Gly Gly Thr Ile Leu Ser Glu Phe Tyr Ile
165 170 175

Leu Thr Ala Ala His Cys Leu Tyr Gln Ala Lys Arg Phe Lys Val Arg
180 185 190

Val Gly Asp Arg Asn Thr Glu Gln Glu Glu Gly Gly Glu Ala Val His

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195					200					205					
Glu	Val	Glu	Val	Val	Ile	Lys	His	Asn	Arg	Phe	Thr	Lys	Glu	Thr	Tyr
210						215					220				
Asp	Phe	Asp	Ile	Ala	Val	Leu	Arg	Leu	Lys	Thr	Pro	Ile	Thr	Phe	Arg
225					230					235					240
Met	Asn	Val	Ala	Pro	Ala	Cys	Leu	Pro	Glu	Arg	Asp	Trp	Ala	Glu	Ser
				245						250				255	
Thr	Leu	Met	Thr	Gln	Lys	Thr	Gly	Ile	Val	Ser	Gly	Phe	Gly	Arg	Thr
		260					265						270		
His	Glu	Lys	Gly	Arg	Gln	Ser	Thr	Arg	Leu	Lys	Met	Leu	Glu	Val	Pro
		275					280					285			
Tyr	Val	Asp	Arg	Asn	Ser	Cys	Lys	Leu	Ser	Ser	Ser	Phe	Ile	Ile	Thr
	290					295					300				
Gln	Asn	Met	Phe	Cys	Ala	Gly	Tyr	Asp	Thr	Lys	Gln	Glu	Asp	Ala	Cys
305					310					315					320
Gln	Gly	Asp	Ser	Gly	Gly	Pro	His	Val	Thr	Arg	Phe	Lys	Asp	Thr	Tyr
			325						330					335	
Phe	Val	Thr	Gly	Ile	Val	Ser	Trp	Gly	Glu	Gly	Cys	Ala	Arg	Lys	Gly
		340						345					350		
Lys	Tyr	Gly	Ile	Tyr	Thr	Lys	Val	Thr	Ala	Phe	Leu	Lys	Trp	Ile	Asp
		355					360					365			
Arg	Ser	Met	Lys	Thr	Arg	Gly	Leu	Pro	Lys	Ala	Lys	Ser	His	Ala	Pro
	370					375					380				
Glu	Val	Ile	Thr	Ser	Ser	Pro	Leu	Lys							
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<212> TYPE: PRT

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<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

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Glu	Thr	Ser	Pro	Cys	Gln	Asn	Gln	Gly	Lys	Cys	Lys	Asp	Gly	Leu	Gly
		20						25					30		
Glu	Tyr	Thr	Cys	Thr	Cys	Leu	Glu	Gly	Phe	Glu	Gly	Lys	Asn	Cys	Glu
		35				40					45				
Leu	Phe	Thr	Arg	Lys	Leu	Cys	Ser	Leu	Asp	Asn	Gly	Asp	Cys	Asp	Gln
	50				55					60					
Phe	Cys	His	Glu	Glu	Gln	Asn	Ser	Val	Val	Cys	Ser	Cys	Ala	Arg	Gly
65					70					75				80	
Tyr	Thr	Leu	Ala	Asp	Asn	Gly	Lys	Ala	Cys	Ile	Pro	Thr	Gly	Pro	Tyr
			85					90						95	
Pro	Cys	Gly	Lys	Gln	Thr	Leu	Glu	Arg	Arg	Lys	Arg	Arg	Lys	Arg	Ile
			100					105					110		
Val	Gly	Gly	Gln	Glu	Cys	Lys	Asp	Gly	Glu	Cys	Pro	Trp	Gln	Ala	Leu
		115					120					125			
Leu	Ile	Asn	Glu	Glu	Asn	Glu	Gly	Phe	Cys	Gly	Gly	Thr	Ile	Leu	Ser
	130					135						140			
Glu	Phe	Tyr	Ile	Leu	Thr	Ala	Ala	His	Cys	Leu	Tyr	Gln	Ala	Lys	Arg
145				150						155					160

Phe	Lys	Val	Arg	Val	Gly	Asp	Arg	Asn	Thr	Glu	Gln	Glu	Gly	Gly	
				165					170					175	
Glu	Ala	Val	His	Glu	Val	Glu	Val	Val	Ile	Lys	His	Asn	Arg	Phe	Thr
			180					185					190		
Lys	Glu	Thr	Tyr	Asp	Phe	Asp	Ile	Ala	Val	Leu	Arg	Leu	Lys	Thr	Pro
		195					200					205			
Ile	Thr	Phe	Arg	Met	Asn	Val	Ala	Pro	Ala	Cys	Leu	Pro	Glu	Arg	Asp
		210				215					220				
Trp	Ala	Glu	Ser	Thr	Leu	Met	Thr	Gln	Lys	Thr	Gly	Ile	Val	Ser	Gly
225					230					235					240
Phe	Gly	Arg	Thr	His	Glu	Lys	Gly	Arg	Gln	Ser	Thr	Arg	Leu	Lys	Met
				245					250					255	
Leu	Glu	Val	Pro	Tyr	Val	Asp	Arg	Asn	Ser	Cys	Lys	Leu	Ser	Ser	Ser
			260					265					270		
Phe	Ile	Ile	Thr	Gln	Asn	Met	Phe	Cys	Ala	Gly	Tyr	Asp	Thr	Lys	Gln
		275					280					285			
Glu	Asp	Ala	Cys	Gln	Gly	Asp	Ala	Gly	Gly	Pro	His	Val	Thr	Arg	Phe
	290					295					300				
Lys	Asp	Thr	Tyr	Phe	Val	Thr	Gly	Ile	Val	Ser	Trp	Gly	Glu	Gly	Cys
305					310					315					320
Ala	Arg	Lys	Gly	Lys	Tyr	Gly	Ile	Tyr	Thr	Lys	Val	Thr	Ala	Phe	Leu
				325					330					335	
Lys	Trp	Ile	Asp	Arg	Ser	Met	Lys	Thr	Arg	Gly	Leu	Pro	Lys	Ala	Lys
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Glu	Thr	Ser	Pro	Cys	Gln	Asn	Gln	Gly	Lys	Cys	Lys	Asp	Gly	Leu	Gly
			20					25					30		
Glu	Tyr	Thr	Cys	Thr	Cys	Leu	Glu	Gly	Phe	Glu	Gly	Lys	Asn	Cys	Glu
		35					40					45			
Leu	Phe	Thr	Arg	Lys	Leu	Cys	Ser	Leu	Asp	Asn	Gly	Asp	Cys	Asp	Gln</

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Asp Arg Asn Thr Glu Gln Glu Gly Gly Glu Ala Val His Glu Val
 165 170 175
 Glu Val Val Ile Lys His Asn Arg Phe Thr Lys Glu Thr Tyr Asp Phe
 180 185 190
 Asp Ile Ala Val Leu Arg Leu Lys Thr Pro Ile Thr Phe Arg Met Asn
 195 200 205
 Val Ala Pro Ala Cys Leu Pro Glu Arg Asp Trp Ala Glu Ser Thr Leu
 210 215 220
 Met Thr Gln Lys Thr Gly Ile Val Ser Gly Phe Gly Arg Thr His Glu
 225 230 235 240
 Lys Gly Arg Gln Ser Thr Arg Leu Lys Met Leu Glu Val Pro Tyr Val
 245 250 255
 Asp Arg Asn Ser Cys Lys Leu Ser Ser Ser Phe Ile Ile Thr Gln Asn
 260 265 270
 Met Phe Cys Ala Gly Tyr Asp Thr Lys Gln Glu Asp Ala Cys Gln Gly
 275 280 285
 Asp Ala Gly Gly Pro His Val Thr Arg Phe Lys Asp Thr Tyr Phe Val
 290 295 300
 Thr Gly Ile Val Ser Trp Gly Glu Gly Cys Ala Arg Lys Gly Lys Tyr
 305 310 315 320
 Gly Ile Tyr Thr Lys Val Thr Ala Phe Leu Lys Trp Ile Asp Arg Ser
 325 330 335
 Met Lys Thr Arg Gly Leu Pro Lys Ala Lys Ser His Ala Pro Glu Val
 340 345 350
 Ile Thr Ser Ser Pro Leu Lys
 355

<210> SEQ ID NO 6
 <211> LENGTH: 105
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 6

Ala Asn Ser Phe Leu Phe Trp Asn Lys Tyr Lys Asp Gly Asp Gln Cys
 1 5 10 15
 Glu Thr Ser Pro Cys Gln Asn Gln Gly Lys Cys Lys Asp Gly Leu Gly
 20 25 30
 Glu Tyr Thr Cys Thr Cys Leu Glu Gly Phe Glu Gly Lys Asn Cys Glu
 35 40 45
 Leu Phe Thr Arg Lys Leu Cys Ser Leu Asp Asn Gly Asp Cys Asp Gln
 50 55 60
 Phe Cys His Glu Glu Gln Asn Ser Val Val Cys Ser Cys Ala Arg Gly
 65 70 75 80
 Tyr Thr Leu Ala Asp Asn Gly Lys Ala Cys Ile Pro Thr Gly Pro Tyr
 85 90 95
 Pro Cys Gly Lys Gln Thr Leu Glu Arg
 100 105

<210> SEQ ID NO 7
 <211> LENGTH: 6
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic

-continued

peptide

<400> SEQUENCE: 7

Arg Lys Arg Arg Lys Arg
1 5

<210> SEQ ID NO 8

<211> LENGTH: 254

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 8

Ile Val Gly Gly Gln Glu Cys Lys Asp Gly Glu Cys Pro Trp Gln Ala
1 5 10 15

Leu Leu Ile Asn Glu Glu Asn Glu Gly Phe Cys Gly Gly Thr Ile Leu
20 25 30

Ser Glu Phe Tyr Ile Leu Thr Ala Ala His Cys Leu Tyr Gln Ala Lys
35 40 45

Arg Phe Lys Val Arg Val Gly Asp Arg Asn Thr Glu Gln Glu Glu Gly
50 55 60

Gly Glu Ala Val His Glu Val Glu Val Val Ile Lys His Asn Arg Phe
65 70 75 80

Thr Lys Glu Thr Tyr Asp Phe Asp Ile Ala Val Leu Arg Leu Lys Thr
85 90 95

Pro Ile Thr Phe Arg Met Asn Val Ala Pro Ala Cys Leu Pro Glu Arg
100 105 110

Asp Trp Ala Glu Ser Thr Leu Met Thr Gln Lys Thr Gly Ile Val Ser
115 120 125

Gly Phe Gly Arg Thr His Glu Lys Gly Arg Gln Ser Thr Arg Leu Lys
130 135 140

Met Leu Glu Val Pro Tyr Val Asp Arg Asn Ser Cys Lys Leu Ser Ser
145 150 155 160

Ser Phe Ile Ile Thr Gln Asn Met Phe Cys Ala Gly Tyr Asp Thr Lys
165 170 175

Gln Glu Asp Ala Cys Gln Gly Asp Ala Gly Gly Pro His Val Thr Arg
180 185 190

Phe Lys Asp Thr Tyr Phe Val Thr Gly Ile Val Ser Trp Gly Glu Gly
195 200 205

Cys Ala Arg Lys Gly Lys Tyr Gly Ile Tyr Thr Lys Val Thr Ala Phe
210 215 220

Leu Lys Trp Ile Asp Arg Ser Met Lys Thr Arg Gly Leu Pro Lys Ala
225 230 235 240

Lys Ser His Ala Pro Glu Val Ile Thr Ser Ser Pro Leu Lys
245 250

<210> SEQ ID NO 9

<211> LENGTH: 234

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 9

Ile Val Gly Gly Gln Glu Cys Lys Asp Gly Glu Cys Pro Trp Gln Ala
1 5 10 15

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Leu Leu Ile Asn Glu Glu Asn Glu Gly Phe Cys Gly Gly Thr Ile Leu
 20 25 30
 Ser Glu Phe Tyr Ile Leu Thr Ala Ala His Cys Leu Tyr Gln Ala Lys
 35 40 45
 Arg Phe Lys Val Arg Val Gly Asp Arg Asn Thr Glu Gln Glu Glu Gly
 50 55 60
 Gly Glu Ala Val His Glu Val Glu Val Val Ile Lys His Asn Arg Phe
 65 70 75 80
 Thr Lys Glu Thr Tyr Asp Phe Asp Ile Ala Val Leu Arg Leu Lys Thr
 85 90 95
 Pro Ile Thr Phe Arg Met Asn Val Ala Pro Ala Cys Leu Pro Glu Arg
 100 105 110
 Asp Trp Ala Glu Ser Thr Leu Met Thr Gln Lys Thr Gly Ile Val Ser
 115 120 125
 Gly Phe Gly Arg Thr His Glu Lys Gly Arg Gln Ser Thr Arg Leu Lys
 130 135 140
 Met Leu Glu Val Pro Tyr Val Asp Arg Asn Ser Cys Lys Leu Ser Ser
 145 150 155 160
 Ser Phe Ile Ile Thr Gln Asn Met Phe Cys Ala Gly Tyr Asp Thr Lys
 165 170 175
 Gln Glu Asp Ala Cys Gln Gly Asp Ala Gly Gly Pro His Val Thr Arg
 180 185 190
 Phe Lys Asp Thr Tyr Phe Val Thr Gly Ile Val Ser Trp Gly Glu Gly
 195 200 205
 Cys Ala Arg Lys Gly Lys Tyr Gly Ile Tyr Thr Lys Val Thr Ala Phe
 210 215 220
 Leu Lys Trp Ile Asp Arg Ser Met Lys Thr
 225 230

<210> SEQ ID NO 10
 <211> LENGTH: 241
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 10

Ile Val Gly Gly Gln Glu Cys Lys Asp Gly Glu Cys Pro Trp Gln Ala
 1 5 10 15
 Leu Leu Ile Asn Glu Glu Asn Glu Gly Phe Cys Gly Gly Thr Ile Leu
 20 25 30
 Ser Glu Phe Tyr Ile Leu Thr Ala Ala His Cys Leu Tyr Gln Ala Lys
 35 40 45
 Arg Phe Lys Val Arg Val Gly Asp Arg Asn Thr Glu Gln Glu Glu Gly
 50 55 60
 Gly Glu Ala Val His Glu Val Glu Val Val Ile Lys His Asn Arg Phe
 65 70 75 80
 Thr Lys Glu Thr Tyr Asp Phe Asp Ile Ala Val Leu Arg Leu Lys Thr
 85 90 95
 Pro Ile Thr Phe Arg Met Asn Val Ala Pro Ala Cys Leu Pro Glu Arg
 100 105 110
 Asp Trp Ala Glu Ser Thr Leu Met Thr Gln Lys Thr Gly Ile Val Ser
 115 120 125
 Gly Phe Gly Arg Thr His Glu Lys Gly Arg Gln Ser Thr Arg Leu Lys

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130	135	140
Met Leu Glu Val Pro Tyr Val Asp Arg Asn Ser Cys Lys Leu Ser Ser		
145	150	155 160
Ser Phe Ile Ile Thr Gln Asn Met Phe Cys Ala Gly Tyr Asp Thr Lys		
	165	170 175
Gln Glu Asp Ala Cys Gln Gly Asp Ala Gly Gly Pro His Val Thr Arg		
	180	185 190
Phe Lys Asp Thr Tyr Phe Val Thr Gly Ile Val Ser Trp Gly Glu Gly		
	195	200 205
Cys Ala Arg Lys Gly Lys Tyr Gly Ile Tyr Thr Lys Val Thr Ala Phe		
	210	215 220
Leu Lys Trp Ile Asp Arg Ser Met Lys Thr Arg Gly Leu Pro Lys Ala		
225	230	235 240

Lys

<210> SEQ ID NO 11

<211> LENGTH: 239

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 11

Ile Val Gly Gly Gln Glu Cys Lys Asp Gly Glu Cys Pro Trp Gln Ala		
1	5	10 15
Leu Leu Ile Asn Glu Glu Asn Glu Gly Phe Cys Gly Gly Thr Ile Leu		
	20	25 30
Ser Glu Phe Tyr Ile Leu Thr Ala Ala His Cys Leu Tyr Gln Ala Lys		
	35	40 45
Arg Phe Lys Val Arg Val Gly Asp Arg Asn Thr Glu Gln Glu Glu Gly		
	50	55 60
Gly Glu Ala Val His Glu Val Glu Val Val Ile Lys His Asn Arg Phe		
	65	70 75 80
Thr Lys Glu Thr Tyr Asp Phe Asp Ile Ala Val Leu Arg Leu Lys Thr		
	85	90 95
Pro Ile Thr Phe Arg Met Asn Val Ala Pro Ala Cys Leu Pro Glu Arg		
	100	105 110
Asp Trp Ala Glu Ser Thr Leu Met Thr Gln Lys Thr Gly Ile Val Ser		
	115	120 125
Gly Phe Gly Arg Thr His Glu Lys Gly Arg Gln Ser Thr Arg Leu Lys		
	130	135 140
Met Leu Glu Val Pro Tyr Val Asp Arg Asn Ser Cys Lys Leu Ser Ser		
145	150	155 160
Ser Phe Ile Ile Thr Gln Asn Met Phe Cys Ala Gly Tyr Asp Thr Lys		
	165	170 175
Gln Glu Asp Ala Cys Gln Gly Asp Ala Gly Gly Pro His Val Thr Arg		
	180	185 190
Phe Lys Asp Thr Tyr Phe Val Thr Gly Ile Val Ser Trp Gly Glu Gly		
	195	200 205
Cys Ala Arg Lys Gly Lys Tyr Gly Ile Tyr Thr Lys Val Thr Ala Phe		
	210	215 220
Leu Lys Trp Ile Asp Arg Ser Met Lys Thr Arg Gly Leu Pro Lys		
225	230	235

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<210> SEQ ID NO 12
<211> LENGTH: 52
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

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<400> SEQUENCE: 12

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Ser Val Ala Gln Ala Thr Ser Ser Ser Gly Glu Ala Pro Asp Ser Ile
 1             5             10             15

Thr Trp Lys Pro Tyr Asp Ala Ala Asp Leu Asp Pro Thr Glu Asn Pro
      20             25             30

Phe Asp Leu Leu Asp Phe Asn Gln Thr Gln Pro Glu Arg Gly Asp Asn
      35             40             45

Asn Leu Thr Arg
      50

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<210> SEQ ID NO 13
<211> LENGTH: 94
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

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<400> SEQUENCE: 13

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```

Asp Gly Asp Gln Cys Glu Thr Ser Pro Cys Gln Asn Gln Gly Lys Cys
 1             5             10             15

Lys Asp Gly Leu Gly Glu Tyr Thr Cys Thr Cys Leu Glu Gly Phe Glu
      20             25             30

Gly Lys Asn Cys Glu Leu Phe Thr Arg Lys Leu Cys Ser Leu Asp Asn
      35             40             45

Gly Asp Cys Asp Gln Phe Cys His Glu Glu Gln Asn Ser Val Val Cys
      50             55             60

Ser Cys Ala Arg Gly Tyr Thr Leu Ala Asp Asn Gly Lys Ala Cys Ile
      65             70             75             80

Pro Thr Gly Pro Tyr Pro Cys Gly Lys Gln Thr Leu Glu Arg
      85             90

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<210> SEQ ID NO 14
<211> LENGTH: 31
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

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<400> SEQUENCE: 14

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Ser Ser Gly Gly Ser Gly Gly Ser Gly Gly Ser Gly Gly
 1             5             10             15

Ser Gly Gly Ser Gly Gly Ser Gly Gly Ser Gly Ser
      20             25             30

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<210> SEQ ID NO 15
<211> LENGTH: 585
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

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<400> SEQUENCE: 15

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Asp Ala His Lys Ser Glu Val Ala His Arg Phe Lys Asp Leu Gly Glu
 1             5             10             15

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Glu	Asn	Phe	Lys	Ala	Leu	Val	Leu	Ile	Ala	Phe	Ala	Gln	Tyr	Leu	Gln
			20				25						30		
Gln	Ser	Pro	Phe	Glu	Asp	His	Val	Lys	Leu	Val	Asn	Glu	Val	Thr	Glu
			35				40						45		
Phe	Ala	Lys	Thr	Cys	Val	Ala	Asp	Glu	Ser	Ala	Glu	Asn	Cys	Asp	Lys
			50				55						60		
Ser	Leu	His	Thr	Leu	Phe	Gly	Asp	Lys	Leu	Cys	Thr	Val	Ala	Thr	Leu
			65				70						75		
Arg	Glu	Thr	Tyr	Gly	Glu	Met	Ala	Asp	Cys	Cys	Ala	Lys	Gln	Glu	Pro
			85				90						95		
Glu	Arg	Asn	Glu	Cys	Phe	Leu	Gln	His	Lys	Asp	Asp	Asn	Pro	Asn	Leu
			100				105						110		
Pro	Arg	Leu	Val	Arg	Pro	Glu	Val	Asp	Val	Met	Cys	Thr	Ala	Phe	His
			115				120						125		
Asp	Asn	Glu	Glu	Thr	Phe	Leu	Lys	Lys	Tyr	Leu	Tyr	Glu	Ile	Ala	Arg
			130				135						140		
Arg	His	Pro	Tyr	Phe	Tyr	Ala	Pro	Glu	Leu	Leu	Phe	Phe	Ala	Lys	Arg
			145				150						155		
Tyr	Lys	Ala	Ala	Phe	Thr	Glu	Cys	Cys	Gln	Ala	Ala	Asp	Lys	Ala	Ala
			165				170						175		
Cys	Leu	Leu	Pro	Lys	Leu	Asp	Glu	Leu	Arg	Asp	Glu	Gly	Lys	Ala	Ser
			180				185						190		
Ser	Ala	Lys	Gln	Arg	Leu	Lys	Cys	Ala	Ser	Leu	Gln	Lys	Phe	Gly	Glu
			195				200						205		
Arg	Ala	Phe	Lys	Ala	Trp	Ala	Val	Ala	Arg	Leu	Ser	Gln	Arg	Phe	Pro
			210				215						220		
Lys	Ala	Glu	Phe	Ala	Glu	Val	Ser	Lys	Leu	Val	Thr	Asp	Leu	Thr	Lys
			225				230						235		
Val	His	Thr	Glu	Cys	Cys	His	Gly	Asp	Leu	Leu	Glu	Cys	Ala	Asp	Asp
			245				250						255		
Arg	Ala	Asp	Leu	Ala	Lys	Tyr	Ile	Cys	Glu	Asn	Gln	Asp	Ser	Ile	Ser
			260				265						270		
Ser	Lys	Leu	Lys	Glu	Cys	Cys	Glu	Lys	Pro	Leu	Leu	Glu	Lys	Ser	His
			275				280						285		
Cys	Ile	Ala	Glu	Val	Glu	Asn	Asp	Glu	Met	Pro	Ala	Asp	Leu	Pro	Ser
			290				295						300		
Leu	Ala	Ala	Asp	Phe	Val	Glu	Ser	Lys	Asp	Val	Cys	Lys	Asn	Tyr	Ala
			305				310						315		
Glu	Ala	Lys	Asp	Val	Phe	Leu	Gly	Met	Phe	Leu	Tyr	Glu	Tyr	Ala	Arg
			325				330						335		
Arg	His	Pro	Asp	Tyr	Ser	Val	Val	Leu	Leu	Leu	Arg	Leu	Ala	Lys	Thr
			340				345						350		
Tyr	Glu	Thr	Thr	Leu	Glu	Lys	Cys	Cys	Ala	Ala	Ala	Asp	Pro	His	Glu
			355				360						365		
Cys	Tyr	Ala	Lys	Val	Phe	Asp	Glu	Phe	Lys	Pro	Leu	Val	Glu	Glu	Pro
			370				375						380		
Gln	Asn	Leu	Ile	Lys	Gln	Asn	Cys	Glu	Leu	Phe	Glu	Gln	Leu	Gly	Glu
			385				390						395		
Tyr	Lys	Phe	Gln	Asn	Ala	Leu	Leu	Val	Arg	Tyr	Thr	Lys	Lys	Val	Pro
			405				410						415		
Gln	Val	Ser	Thr	Pro	Thr	Leu	Val	Glu	Val	Ser	Arg	Asn	Leu	Gly	Lys

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435	440	445
Ala Glu Asp Tyr Leu Ser	Val Val Leu Asn Gln	Leu Cys Val Leu His
450	455	460
Glu Lys Thr Pro Val Ser	Asp Arg Val Thr Lys	Cys Cys Thr Glu Ser
465	470	475
Leu Val Asn Arg Arg Pro	Cys Phe Ser Ala Leu	Glu Val Asp Glu Thr
485	490	495
Tyr Val Pro Lys Glu Phe	Asn Ala Glu Thr Phe	Thr Phe His Ala Asp
500	505	510
Ile Cys Thr Leu Ser Glu	Lys Glu Arg Gln Ile	Lys Lys Gln Thr Ala
515	520	525
Leu Val Glu Leu Val Lys	His Lys Pro Lys Ala	Thr Lys Glu Gln Leu
530	535	540
Lys Ala Val Met Asp Asp	Phe Ala Ala Phe Val	Glu Lys Cys Cys Lys
545	550	555
Ala Asp Asp Lys Glu Thr	Cys Phe Ala Glu Glu	Gly Lys Lys Leu Val
565	570	575
Ala Ala Ser Gln Ala Ala	Leu Gly Leu	
580	585	
<210> SEQ ID NO 16		
<211> LENGTH: 291		
<212> TYPE: PRT		
<213> ORGANISM: Artificial Sequence		
<220> FEATURE:		
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide		
<400> SEQUENCE: 16		
Ile Val Gly Gly Gln Glu Cys Lys Asp Gly Glu Cys Pro Trp Gln Ala		
1	5	10
Leu Leu Ile Asn Glu Glu Asn Glu Gly Phe Cys Gly Gly Thr Ile Leu		
20	25	30
Ser Glu Phe Tyr Ile Leu Thr Ala Ala His Cys Leu Tyr Gln Ala Lys		
35	40	45
Arg Phe Lys Val Arg Val Gly Asp Arg Asn Thr Glu Gln Glu Glu Gly		
50	55	60
Gly Glu Ala Val His Glu Val Glu Val Val Ile Lys His Asn Arg Phe		
65	70	75
Thr Lys Glu Thr Tyr Asp Phe Asp Ile Ala Val Leu Arg Leu Lys Thr		
85	90	95
Pro Ile Thr Phe Arg Met Asn Val Ala Pro Ala Cys Leu Pro Glu Arg		
100	105	110
Asp Trp Ala Glu Ser Thr Leu Met Thr Gln Lys Thr Gly Ile Val Ser		
115	120	125
Gly Phe Gly Arg Thr His Glu Lys Gly Arg Gln Ser Thr Arg Leu Lys		
130	135	140
Met Leu Glu Val Pro Tyr Val Asp Arg Asn Ser Cys Lys Leu Ser Ser		
145	150	155
Ser Phe Ile Ile Thr Gln Asn Met Phe Cys Ala Gly Tyr Asp Thr Lys		
165	170	175
Gln Glu Asp Ala Cys Gln Gly Asp Ala Gly Gly Pro His Val Thr Arg		
180	185	190
Phe Lys Asp Thr Tyr Phe Val Thr Gly Ile Val Ser Trp Gly Glu Gly		
195	200	205

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Cys Ala Arg Lys Gly Lys Tyr Gly Ile Tyr Thr Lys Val Thr Ala Phe
 210 215 220
 Leu Lys Trp Ile Asp Arg Ser Met Lys Thr Arg Gly Leu Pro Lys Ser
 225 230 235 240
 Val Ala Gln Ala Thr Ser Ser Ser Gly Glu Ala Pro Asp Ser Ile Thr
 245 250 255
 Trp Lys Pro Tyr Asp Ala Ala Asp Leu Asp Pro Thr Glu Asn Pro Phe
 260 265 270
 Asp Leu Leu Asp Phe Asn Gln Thr Gln Pro Glu Arg Gly Asp Asn Asn
 275 280 285
 Leu Thr Arg
 290

<210> SEQ ID NO 17
 <211> LENGTH: 322
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 17

Ile Val Gly Gly Gln Glu Cys Lys Asp Gly Glu Cys Pro Trp Gln Ala
 1 5 10 15
 Leu Leu Ile Asn Glu Glu Asn Glu Gly Phe Cys Gly Gly Thr Ile Leu
 20 25 30
 Ser Glu Phe Tyr Ile Leu Thr Ala Ala His Cys Leu Tyr Gln Ala Lys
 35 40 45
 Arg Phe Lys Val Arg Val Gly Asp Arg Asn Thr Glu Gln Glu Glu Gly
 50 55 60
 Gly Glu Ala Val His Glu Val Glu Val Val Ile Lys His Asn Arg Phe
 65 70 75 80
 Thr Lys Glu Thr Tyr Asp Phe Asp Ile Ala Val Leu Arg Leu Lys Thr
 85 90 95
 Pro Ile Thr Phe Arg Met Asn Val Ala Pro Ala Cys Leu Pro Glu Arg
 100 105 110
 Asp Trp Ala Glu Ser Thr Leu Met Thr Gln Lys Thr Gly Ile Val Ser
 115 120 125
 Gly Phe Gly Arg Thr His Glu Lys Gly Arg Gln Ser Thr Arg Leu Lys
 130 135 140
 Met Leu Glu Val Pro Tyr Val Asp Arg Asn Ser Cys Lys Leu Ser Ser
 145 150 155 160
 Ser Phe Ile Ile Thr Gln Asn Met Phe Cys Ala Gly Tyr Asp Thr Lys
 165 170 175
 Gln Glu Asp Ala Cys Gln Gly Asp Ala Gly Gly Pro His Val Thr Arg
 180 185 190
 Phe Lys Asp Thr Tyr Phe Val Thr Gly Ile Val Ser Trp Gly Glu Gly
 195 200 205
 Cys Ala Arg Lys Gly Lys Tyr Gly Ile Tyr Thr Lys Val Thr Ala Phe
 210 215 220
 Leu Lys Trp Ile Asp Arg Ser Met Lys Thr Arg Gly Leu Pro Lys Ser
 225 230 235 240
 Ser Gly Gly Ser Gly Gly Ser Gly Gly Ser Gly Gly Ser Gly Gly Ser
 245 250 255
 Gly Gly Ser Gly Gly Ser Gly Gly Ser Gly Gly Ser Gly Ser Val
 260 265 270

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Ala Gln Ala Thr Ser Ser Ser Gly Glu Ala Pro Asp Ser Ile Thr Trp
 275 280 285

Lys Pro Tyr Asp Ala Ala Asp Leu Asp Pro Thr Glu Asn Pro Phe Asp
 290 295 300

Leu Leu Asp Phe Asn Gln Thr Gln Pro Glu Arg Gly Asp Asn Asn Leu
 305 310 315 320

Thr Arg

<210> SEQ ID NO 18

<211> LENGTH: 345

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 18

Ala Asn Ser Phe Leu Phe Trp Asn Lys Tyr Lys Asp Gly Asp Gln Cys
 1 5 10 15

Glu Thr Ser Pro Cys Gln Asn Gln Gly Lys Cys Lys Asp Gly Leu Gly
 20 25 30

Glu Tyr Thr Cys Thr Cys Leu Glu Gly Phe Glu Gly Lys Asn Cys Glu
 35 40 45

Leu Phe Thr Arg Lys Leu Cys Ser Leu Asp Asn Gly Asp Cys Asp Gln
 50 55 60

Phe Cys His Glu Glu Gln Asn Ser Val Val Cys Ser Cys Ala Arg Gly
 65 70 75 80

Tyr Thr Leu Ala Asp Asn Gly Lys Ala Cys Ile Pro Thr Gly Pro Tyr
 85 90 95

Pro Cys Gly Lys Gln Thr Leu Glu Arg Arg Lys Arg Arg Lys Arg Ile
 100 105 110

Val Gly Gly Gln Glu Cys Lys Asp Gly Glu Cys Pro Trp Gln Ala Leu
 115 120 125

Leu Ile Asn Glu Glu Asn Glu Gly Phe Cys Gly Gly Thr Ile Leu Ser
 130 135 140

Glu Phe Tyr Ile Leu Thr Ala Ala His Cys Leu Tyr Gln Ala Lys Arg
 145 150 155 160

Phe Lys Val Arg Val Gly Asp Arg Asn Thr Glu Gln Glu Gly Gly
 165 170 175

Glu Ala Val His Glu Val Glu Val Ile Lys His Asn Arg Phe Thr
 180 185 190

Lys Glu Thr Tyr Asp Phe Asp Ile Ala Val Leu Arg Leu Lys Thr Pro
 195 200 205

Ile Thr Phe Arg Met Asn Val Ala Pro Ala Cys Leu Pro Glu Arg Asp
 210 215 220

Trp Ala Glu Ser Thr Leu Met Thr Gln Lys Thr Gly Ile Val Ser Gly
 225 230 235 240

Phe Gly Arg Thr His Glu Lys Gly Arg Gln Ser Thr Arg Leu Lys Met
 245 250 255

Leu Glu Val Pro Tyr Val Asp Arg Asn Ser Cys Lys Leu Ser Ser Ser
 260 265 270

Phe Ile Ile Thr Gln Asn Met Phe Cys Ala Gly Tyr Asp Thr Lys Gln
 275 280 285

Glu Asp Ala Cys Gln Gly Asp Ala Gly Gly Pro His Val Thr Arg Phe
 290 295 300

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Lys Asp Thr Tyr Phe Val Thr Gly Ile Val Ser Trp Gly Glu Gly Cys
305 310 315 320

Ala Arg Lys Gly Lys Tyr Gly Ile Tyr Thr Lys Val Thr Ala Phe Leu
325 330 335

Lys Trp Ile Asp Arg Ser Met Lys Thr
340 345

<210> SEQ ID NO 19

<211> LENGTH: 352

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 19

Ala Asn Ser Phe Leu Phe Trp Asn Lys Tyr Lys Asp Gly Asp Gln Cys
1 5 10 15

Glu Thr Ser Pro Cys Gln Asn Gln Gly Lys Cys Lys Asp Gly Leu Gly
20 25 30

Glu Tyr Thr Cys Thr Cys Leu Glu Gly Phe Glu Gly Lys Asn Cys Glu
35 40 45

Leu Phe Thr Arg Lys Leu Cys Ser Leu Asp Asn Gly Asp Cys Asp Gln
50 55 60

Phe Cys His Glu Glu Gln Asn Ser Val Val Cys Ser Cys Ala Arg Gly
65 70 75 80

Tyr Thr Leu Ala Asp Asn Gly Lys Ala Cys Ile Pro Thr Gly Pro Tyr
85 90 95

Pro Cys Gly Lys Gln Thr Leu Glu Arg Arg Lys Arg Arg Lys Arg Ile
100 105 110

Val Gly Gly Gln Glu Cys Lys Asp Gly Glu Cys Pro Trp Gln Ala Leu
115 120 125

Leu Ile Asn Glu Glu Asn Glu Gly Phe Cys Gly Gly Thr Ile Leu Ser
130 135 140

Glu Phe Tyr Ile Leu Thr Ala Ala His Cys Leu Tyr Gln Ala Lys Arg
145 150 155 160

Phe Lys Val Arg Val Gly Asp Arg Asn Thr Glu Gln Glu Glu Gly Gly
165 170 175

Glu Ala Val His Glu Val Glu Val Val Ile Lys His Asn Arg Phe Thr
180 185 190

Lys Glu Thr Tyr Asp Phe Asp Ile Ala Val Leu Arg Leu Lys Thr Pro
195 200 205

Ile Thr Phe Arg Met Asn Val Ala Pro Ala Cys Leu Pro Glu Arg Asp
210 215 220

Trp Ala Glu Ser Thr Leu Met Thr Gln Lys Thr Gly Ile Val Ser Gly
225 230 235 240

Phe Gly Arg Thr His Glu Lys Gly Arg Gln Ser Thr Arg Leu Lys Met
245 250 255

Leu Glu Val Pro Tyr Val Asp Arg Asn Ser Cys Lys Leu Ser Ser Ser
260 265 270

Phe Ile Ile Thr Gln Asn Met Phe Cys Ala Gly Tyr Asp Thr Lys Gln
275 280 285

Glu Asp Ala Cys Gln Gly Asp Ala Gly Gly Pro His Val Thr Arg Phe
290 295 300

Lys Asp Thr Tyr Phe Val Thr Gly Ile Val Ser Trp Gly Glu Gly Cys

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305              310              315              320
Ala Arg Lys Gly Lys Tyr Gly Ile Tyr Thr Lys Val Thr Ala Phe Leu
              325              330              335
Lys Trp Ile Asp Arg Ser Met Lys Thr Arg Gly Leu Pro Lys Ala Lys
              340              345              350

<210> SEQ ID NO 20
<211> LENGTH: 350
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

<400> SEQUENCE: 20
Ala Asn Ser Phe Leu Phe Trp Asn Lys Tyr Lys Asp Gly Asp Gln Cys
1              5              10              15
Glu Thr Ser Pro Cys Gln Asn Gln Gly Lys Cys Lys Asp Gly Leu Gly
      20              25              30
Glu Tyr Thr Cys Thr Cys Leu Glu Gly Phe Glu Gly Lys Asn Cys Glu
      35              40              45
Leu Phe Thr Arg Lys Leu Cys Ser Leu Asp Asn Gly Asp Cys Asp Gln
      50              55              60
Phe Cys His Glu Glu Gln Asn Ser Val Val Cys Ser Cys Ala Arg Gly
      65              70              75              80
Tyr Thr Leu Ala Asp Asn Gly Lys Ala Cys Ile Pro Thr Gly Pro Tyr
      85              90              95
Pro Cys Gly Lys Gln Thr Leu Glu Arg Arg Lys Arg Arg Lys Arg Ile
      100             105             110
Val Gly Gly Gln Glu Cys Lys Asp Gly Glu Cys Pro Trp Gln Ala Leu
      115             120             125
Leu Ile Asn Glu Glu Asn Glu Gly Phe Cys Gly Gly Thr Ile Leu Ser
      130             135             140
Glu Phe Tyr Ile Leu Thr Ala Ala His Cys Leu Tyr Gln Ala Lys Arg
      145             150             155             160
Phe Lys Val Arg Val Gly Asp Arg Asn Thr Glu Gln Glu Glu Gly Gly
      165             170             175
Glu Ala Val His Glu Val Glu Val Val Ile Lys His Asn Arg Phe Thr
      180             185             190
Lys Glu Thr Tyr Asp Phe Asp Ile Ala Val Leu Arg Leu Lys Thr Pro
      195             200             205
Ile Thr Phe Arg Met Asn Val Ala Pro Ala Cys Leu Pro Glu Arg Asp
      210             215             220
Trp Ala Glu Ser Thr Leu Met Thr Gln Lys Thr Gly Ile Val Ser Gly
      225             230             235             240
Phe Gly Arg Thr His Glu Lys Gly Arg Gln Ser Thr Arg Leu Lys Met
      245             250             255
Leu Glu Val Pro Tyr Val Asp Arg Asn Ser Cys Lys Leu Ser Ser Ser
      260             265             270
Phe Ile Ile Thr Gln Asn Met Phe Cys Ala Gly Tyr Asp Thr Lys Gln
      275             280             285
Glu Asp Ala Cys Gln Gly Asp Ala Gly Gly Pro His Val Thr Arg Phe
      290             295             300
Lys Asp Thr Tyr Phe Val Thr Gly Ile Val Ser Trp Gly Glu Gly Cys
305              310              315              320

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Ala Arg Lys Gly Lys Tyr Gly Ile Tyr Thr Lys Val Thr Ala Phe Leu
325 330 335

Lys Trp Ile Asp Arg Ser Met Lys Thr Arg Gly Leu Pro Lys
340 345 350

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<210> SEQ ID NO 21
<211> LENGTH: 405
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide
```

<400> SEQUENCE: 21

Ala Asn Ser Phe Leu Phe Trp Asn Lys Tyr Lys Asp Gly Asp Gln Cys
1 5 10 15

Glu Thr Ser Pro Cys Gln Asn Gln Gly Lys Cys Lys Asp Gly Leu Gly
20 25 30

Glu Tyr Thr Cys Thr Cys Leu Glu Gly Phe Glu Gly Lys Asn Cys Glu
35 40 45

Leu Phe Thr Arg Lys Leu Cys Ser Leu Asp Asn Gly Asp Cys Asp Gln
50 55 60

Phe Cys His Glu Glu Gln Asn Ser Val Val Cys Ser Cys Ala Arg Gly
65 70 75 80

Tyr Thr Leu Ala Asp Asn Gly Lys Ala Cys Ile Pro Thr Gly Pro Tyr
85 90 95

Pro Cys Gly Lys Gln Thr Leu Glu Arg Arg Lys Arg Ser Val Ala Gln
100 105 110

Ala Thr Ser Ser Ser Gly Glu Ala Pro Asp Ser Ile Thr Trp Lys Pro
115 120 125

Tyr Asp Ala Ala Asp Leu Asp Pro Thr Glu Asn Pro Phe Asp Leu Leu
130 135 140

Asp Phe Asn Gln Thr Gln Pro Glu Arg Gly Asp Asn Asn Leu Thr Arg
145 150 155 160

Arg Lys Arg Arg Lys Arg Ile Val Gly Gly Gln Glu Cys Lys Asp Gly
165 170 175

Glu Cys Pro Trp Gln Ala Leu Leu Ile Asn Glu Glu Asn Glu Gly Phe
180 185 190

Cys Gly Gly Thr Ile Leu Ser Glu Phe Tyr Ile Leu Thr Ala Ala His
195 200 205

Cys Leu Tyr Gln Ala Lys Arg Phe Lys Val Arg Val Gly Asp Arg Asn
210 215 220

Thr Glu Gln Glu Glu Gly Gly Glu Ala Val His Glu Val Glu Val Val
225 230 235 240

Ile Lys His Asn Arg Phe Thr Lys Glu Thr Tyr Asp Phe Asp Ile Ala
245 250 255

Val Leu Arg Leu Lys Thr Pro Ile Thr Phe Arg Met Asn Val Ala Pro
260 265 270

Ala Cys Leu Pro Glu Arg Asp Trp Ala Glu Ser Thr Leu Met Thr Gln
275 280 285

Lys Thr Gly Ile Val Ser Gly Phe Gly Arg Thr His Glu Lys Gly Arg
290 295 300

Gln Ser Thr Arg Leu Lys Met Leu Glu Val Pro Tyr Val Asp Arg Asn
305 310 315 320

Ser Cys Lys Leu Ser Ser Ser Phe Ile Ile Thr Gln Asn Met Phe Cys
325 330 335

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Ala Gly Tyr Asp Thr Lys Gln Glu Asp Ala Cys Gln Gly Asp Ala Gly
 340 345 350

Gly Pro His Val Thr Arg Phe Lys Asp Thr Tyr Phe Val Thr Gly Ile
 355 360 365

Val Ser Trp Gly Glu Gly Cys Ala Arg Lys Gly Lys Tyr Gly Ile Tyr
 370 375 380

Thr Lys Val Thr Ala Phe Leu Lys Trp Ile Asp Arg Ser Met Lys Thr
 385 390 395 400

Arg Gly Leu Pro Lys
 405

<210> SEQ ID NO 22
 <211> LENGTH: 409
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 22

Asp Gly Asp Gln Cys Glu Thr Ser Pro Cys Gln Asn Gln Gly Lys Cys
 1 5 10 15

Lys Asp Gly Leu Gly Glu Tyr Thr Cys Thr Cys Leu Glu Gly Phe Glu
 20 25 30

Gly Lys Asn Cys Glu Leu Phe Thr Arg Lys Leu Cys Ser Leu Asp Asn
 35 40 45

Gly Asp Cys Asp Gln Phe Cys His Glu Glu Gln Asn Ser Val Val Cys
 50 55 60

Ser Cys Ala Arg Gly Tyr Thr Leu Ala Asp Asn Gly Lys Ala Cys Ile
 65 70 75 80

Pro Thr Gly Pro Tyr Pro Cys Gly Lys Gln Thr Leu Glu Arg Arg Lys
 85 90 95

Arg Ser Val Ala Gln Ala Thr Ser Ser Ser Gly Glu Ala Pro Asp Ser
 100 105 110

Ile Thr Trp Lys Pro Tyr Asp Ala Ala Asp Leu Asp Pro Thr Glu Asn
 115 120 125

Pro Phe Asp Leu Leu Asp Phe Asn Gln Thr Gln Pro Glu Arg Gly Asp
 130 135 140

Asn Asn Leu Thr Arg Arg Lys Arg Arg Lys Arg Ile Val Gly Gly Gln
 145 150 155 160

Glu Cys Lys Asp Gly Glu Cys Pro Trp Gln Ala Leu Leu Ile Asn Glu
 165 170 175

Glu Asn Glu Gly Phe Cys Gly Gly Thr Ile Leu Ser Glu Phe Tyr Ile
 180 185 190

Leu Thr Ala Ala His Cys Leu Tyr Gln Ala Lys Arg Phe Lys Val Arg
 195 200 205

Val Gly Asp Arg Asn Thr Glu Gln Glu Glu Gly Gly Glu Ala Val His
 210 215 220

Glu Val Glu Val Val Ile Lys His Asn Arg Phe Thr Lys Glu Thr Tyr
 225 230 235 240

Asp Phe Asp Ile Ala Val Leu Arg Leu Lys Thr Pro Ile Thr Phe Arg
 245 250 255

Met Asn Val Ala Pro Ala Cys Leu Pro Glu Arg Asp Trp Ala Glu Ser
 260 265 270

Thr Leu Met Thr Gln Lys Thr Gly Ile Val Ser Gly Phe Gly Arg Thr

-continued

275	280	285
His Glu Lys Gly Arg Gln Ser Thr Arg Leu Lys Met Leu Glu Val Pro		
290	295	300
Tyr Val Asp Arg Asn Ser Cys Lys Leu Ser Ser Ser Phe Ile Ile Thr		
305	310	315
Gln Asn Met Phe Cys Ala Gly Tyr Asp Thr Lys Gln Glu Asp Ala Cys		
	325	330
Gln Gly Asp Ala Gly Gly Pro His Val Thr Arg Phe Lys Asp Thr Tyr		
	340	345
Phe Val Thr Gly Ile Val Ser Trp Gly Glu Gly Cys Ala Arg Lys Gly		
	355	360
Lys Tyr Gly Ile Tyr Thr Lys Val Thr Ala Phe Leu Lys Trp Ile Asp		
	370	375
Arg Ser Met Lys Thr Arg Gly Leu Pro Lys Ala Lys Ser His Ala Pro		
385	390	395
Glu Val Ile Thr Ser Ser Pro Leu Lys		
	405	

<210> SEQ ID NO 23
 <211> LENGTH: 394
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 23

Asp Gly Asp Gln Cys Glu Thr Ser Pro Cys Gln Asn Gln Gly Lys Cys
1 5 10 15
Lys Asp Gly Leu Gly Glu Tyr Thr Cys Thr Cys Leu Glu Gly Phe Glu
20 25 30
Gly Lys Asn Cys Glu Leu Phe Thr Arg Lys Leu Cys Ser Leu Asp Asn
35 40 45
Gly Asp Cys Asp Gln Phe Cys His Glu Glu Gln Asn Ser Val Val Cys
50 55 60
Ser Cys Ala Arg Gly Tyr Thr Leu Ala Asp Asn Gly Lys Ala Cys Ile
65 70 75 80
Pro Thr Gly Pro Tyr Pro Cys Gly Lys Gln Thr Leu Glu Arg Arg Lys
85 90 95
Arg Ser Val Ala Gln Ala Thr Ser Ser Ser Gly Glu Ala Pro Asp Ser
100 105 110
Ile Thr Trp Lys Pro Tyr Asp Ala Ala Asp Leu Asp Pro Thr Glu Asn
115 120 125
Pro Phe Asp Leu Leu Asp Phe Asn Gln Thr Gln Pro Glu Arg Gly Asp
130 135 140
Asn Asn Leu Thr Arg Arg Lys Arg Arg Lys Arg Ile Val Gly Gly Gln
145 150 155 160
Glu Cys Lys Asp Gly Glu Cys Pro Trp Gln Ala Leu Leu Ile Asn Glu
165 170 175
Glu Asn Glu Gly Phe Cys Gly Gly Thr Ile Leu Ser Glu Phe Tyr Ile
180 185 190
Leu Thr Ala Ala His Cys Leu Tyr Gln Ala Lys Arg Phe Lys Val Arg
195 200 205
Val Gly Asp Arg Asn Thr Glu Gln Glu Glu Gly Gly Glu Ala Val His
210 215 220

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Glu Val Glu Val Val Ile Lys His Asn Arg Phe Thr Lys Glu Thr Tyr
 225 230 235 240
 Asp Phe Asp Ile Ala Val Leu Arg Leu Lys Thr Pro Ile Thr Phe Arg
 245 250 255
 Met Asn Val Ala Pro Ala Cys Leu Pro Glu Arg Asp Trp Ala Glu Ser
 260 265 270
 Thr Leu Met Thr Gln Lys Thr Gly Ile Val Ser Gly Phe Gly Arg Thr
 275 280 285
 His Glu Lys Gly Arg Gln Ser Thr Arg Leu Lys Met Leu Glu Val Pro
 290 295 300
 Tyr Val Asp Arg Asn Ser Cys Lys Leu Ser Ser Ser Phe Ile Ile Thr
 305 310 315 320
 Gln Asn Met Phe Cys Ala Gly Tyr Asp Thr Lys Gln Glu Asp Ala Cys
 325 330 335
 Gln Gly Asp Ala Gly Gly Pro His Val Thr Arg Phe Lys Asp Thr Tyr
 340 345 350
 Phe Val Thr Gly Ile Val Ser Trp Gly Glu Gly Cys Ala Arg Lys Gly
 355 360 365
 Lys Tyr Gly Ile Tyr Thr Lys Val Thr Ala Phe Leu Lys Trp Ile Asp
 370 375 380
 Arg Ser Met Lys Thr Arg Gly Leu Pro Lys
 385 390

<210> SEQ ID NO 24
 <211> LENGTH: 391
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 24

Asp Gly Asp Gln Cys Glu Thr Ser Pro Cys Gln Asn Gln Gly Lys Cys
 1 5 10 15
 Lys Asp Gly Leu Gly Glu Tyr Thr Cys Thr Cys Leu Glu Gly Phe Glu
 20 25 30
 Gly Lys Asn Cys Glu Leu Phe Thr Arg Lys Leu Cys Ser Leu Asp Asn
 35 40 45
 Gly Asp Cys Asp Gln Phe Cys His Glu Glu Gln Asn Ser Val Val Cys
 50 55 60
 Ser Cys Ala Arg Gly Tyr Thr Leu Ala Asp Asn Gly Lys Ala Cys Ile
 65 70 75 80
 Pro Thr Gly Pro Tyr Pro Cys Gly Lys Gln Thr Leu Glu Arg Ser Val
 85 90 95
 Ala Gln Ala Thr Ser Ser Ser Gly Glu Ala Pro Asp Ser Ile Thr Trp
 100 105 110
 Lys Pro Tyr Asp Ala Ala Asp Leu Asp Pro Thr Glu Asn Pro Phe Asp
 115 120 125
 Leu Leu Asp Phe Asn Gln Thr Gln Pro Glu Arg Gly Asp Asn Asn Leu
 130 135 140
 Thr Arg Arg Lys Arg Arg Lys Arg Ile Val Gly Gly Gln Glu Cys Lys
 145 150 155 160
 Asp Gly Glu Cys Pro Trp Gln Ala Leu Leu Ile Asn Glu Glu Asn Glu
 165 170 175
 Gly Phe Cys Gly Gly Thr Ile Leu Ser Glu Phe Tyr Ile Leu Thr Ala
 180 185 190

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Ala His Cys Leu Tyr Gln Ala Lys Arg Phe Lys Val Arg Val Gly Asp
 195 200 205

Arg Asn Thr Glu Gln Glu Glu Gly Gly Glu Ala Val His Glu Val Glu
 210 215 220

Val Val Ile Lys His Asn Arg Phe Thr Lys Glu Thr Tyr Asp Phe Asp
 225 230 235 240

Ile Ala Val Leu Arg Leu Lys Thr Pro Ile Thr Phe Arg Met Asn Val
 245 250 255

Ala Pro Ala Cys Leu Pro Glu Arg Asp Trp Ala Glu Ser Thr Leu Met
 260 265 270

Thr Gln Lys Thr Gly Ile Val Ser Gly Phe Gly Arg Thr His Glu Lys
 275 280 285

Gly Arg Gln Ser Thr Arg Leu Lys Met Leu Glu Val Pro Tyr Val Asp
 290 295 300

Arg Asn Ser Cys Lys Leu Ser Ser Ser Phe Ile Ile Thr Gln Asn Met
 305 310 315 320

Phe Cys Ala Gly Tyr Asp Thr Lys Gln Glu Asp Ala Cys Gln Gly Asp
 325 330 335

Ala Gly Gly Pro His Val Thr Arg Phe Lys Asp Thr Tyr Phe Val Thr
 340 345 350

Gly Ile Val Ser Trp Gly Glu Gly Cys Ala Arg Lys Gly Lys Tyr Gly
 355 360 365

Ile Tyr Thr Lys Val Thr Ala Phe Leu Lys Trp Ile Asp Arg Ser Met
 370 375 380

Lys Thr Arg Gly Leu Pro Lys
 385 390

<210> SEQ ID NO 25
 <211> LENGTH: 391
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 25

Asp Gly Asp Gln Cys Glu Thr Ser Pro Cys Gln Asn Gln Gly Lys Cys
 1 5 10 15

Lys Asp Gly Leu Gly Glu Tyr Thr Cys Thr Cys Leu Glu Gly Phe Glu
 20 25 30

Gly Lys Asn Cys Glu Leu Phe Thr Arg Lys Leu Cys Ser Leu Asp Asn
 35 40 45

Gly Asp Cys Asp Gln Phe Cys His Glu Glu Gln Asn Ser Val Val Cys
 50 55 60

Ser Cys Ala Arg Gly Tyr Thr Leu Ala Asp Asn Gly Lys Ala Cys Ile
 65 70 75 80

Pro Thr Gly Pro Tyr Pro Cys Gly Lys Gln Thr Leu Glu Arg Arg Lys
 85 90 95

Arg Arg Lys Arg Ile Val Gly Gly Gln Glu Cys Lys Asp Gly Glu Cys
 100 105 110

Pro Trp Gln Ala Leu Leu Ile Asn Glu Glu Asn Glu Gly Phe Cys Gly
 115 120 125

Gly Thr Ile Leu Ser Glu Phe Tyr Ile Leu Thr Ala Ala His Cys Leu
 130 135 140

Tyr Gln Ala Lys Arg Phe Lys Val Arg Val Gly Asp Arg Asn Thr Glu

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145	150	155	160
Gln Glu Glu Gly Gly Glu Ala Val His Glu Val Glu Val Val Ile Lys			
	165	170	175
His Asn Arg Phe Thr Lys Glu Thr Tyr Asp Phe Asp Ile Ala Val Leu			
	180	185	190
Arg Leu Lys Thr Pro Ile Thr Phe Arg Met Asn Val Ala Pro Ala Cys			
	195	200	205
Leu Pro Glu Arg Asp Trp Ala Glu Ser Thr Leu Met Thr Gln Lys Thr			
	210	215	220
Gly Ile Val Ser Gly Phe Gly Arg Thr His Glu Lys Gly Arg Gln Ser			
225	230	235	240
Thr Arg Leu Lys Met Leu Glu Val Pro Tyr Val Asp Arg Asn Ser Cys			
	245	250	255
Lys Leu Ser Ser Ser Phe Ile Ile Thr Gln Asn Met Phe Cys Ala Gly			
	260	265	270
Tyr Asp Thr Lys Gln Glu Asp Ala Cys Gln Gly Asp Ala Gly Gly Pro			
	275	280	285
His Val Thr Arg Phe Lys Asp Thr Tyr Phe Val Thr Gly Ile Val Ser			
	290	295	300
Trp Gly Glu Gly Cys Ala Arg Lys Gly Lys Tyr Gly Ile Tyr Thr Lys			
305	310	315	320
Val Thr Ala Phe Leu Lys Trp Ile Asp Arg Ser Met Lys Thr Arg Gly			
	325	330	335
Leu Pro Lys Ser Val Ala Gln Ala Thr Ser Ser Ser Gly Glu Ala Pro			
	340	345	350
Asp Ser Ile Thr Trp Lys Pro Tyr Asp Ala Ala Asp Leu Asp Pro Thr			
	355	360	365
Glu Asn Pro Phe Asp Leu Leu Asp Phe Asn Gln Thr Gln Pro Glu Arg			
	370	375	380
Gly Asp Asn Asn Leu Thr Arg			
385	390		

<210> SEQ ID NO 26

<211> LENGTH: 422

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 26

Asp Gly Asp Gln Cys Glu Thr Ser Pro Cys Gln Asn Gln Gly Lys Cys			
1	5	10	15
Lys Asp Gly Leu Gly Glu Tyr Thr Cys Thr Cys Leu Glu Gly Phe Glu			
	20	25	30
Gly Lys Asn Cys Glu Leu Phe Thr Arg Lys Leu Cys Ser Leu Asp Asn			
	35	40	45
Gly Asp Cys Asp Gln Phe Cys His Glu Glu Gln Asn Ser Val Val Cys			
	50	55	60
Ser Cys Ala Arg Gly Tyr Thr Leu Ala Asp Asn Gly Lys Ala Cys Ile			
65	70	75	80
Pro Thr Gly Pro Tyr Pro Cys Gly Lys Gln Thr Leu Glu Arg Arg Lys			
	85	90	95
Arg Arg Lys Arg Ile Val Gly Gly Gln Glu Cys Lys Asp Gly Glu Cys			
	100	105	110

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Pro Trp Gln Ala Leu Leu Ile Asn Glu Glu Asn Glu Gly Phe Cys Gly
   115                               120               125

Gly Thr Ile Leu Ser Glu Phe Tyr Ile Leu Thr Ala Ala His Cys Leu
   130                               135               140

Tyr Gln Ala Lys Arg Phe Lys Val Arg Val Gly Asp Arg Asn Thr Glu
   145                               150               155               160

Gln Glu Glu Gly Gly Glu Ala Val His Glu Val Glu Val Val Ile Lys
   165                               170               175

His Asn Arg Phe Thr Lys Glu Thr Tyr Asp Phe Asp Ile Ala Val Leu
   180                               185               190

Arg Leu Lys Thr Pro Ile Thr Phe Arg Met Asn Val Ala Pro Ala Cys
   195                               200               205

Leu Pro Glu Arg Asp Trp Ala Glu Ser Thr Leu Met Thr Gln Lys Thr
   210                               215               220

Gly Ile Val Ser Gly Phe Gly Arg Thr His Glu Lys Gly Arg Gln Ser
   225                               230               235               240

Thr Arg Leu Lys Met Leu Glu Val Pro Tyr Val Asp Arg Asn Ser Cys
   245                               250               255

Lys Leu Ser Ser Ser Phe Ile Ile Thr Gln Asn Met Phe Cys Ala Gly
   260                               265               270

Tyr Asp Thr Lys Gln Glu Asp Ala Cys Gln Gly Asp Ala Gly Gly Pro
   275                               280               285

His Val Thr Arg Phe Lys Asp Thr Tyr Phe Val Thr Gly Ile Val Ser
   290                               295               300

Trp Gly Glu Gly Cys Ala Arg Lys Gly Lys Tyr Gly Ile Tyr Thr Lys
   305                               310               315               320

Val Thr Ala Phe Leu Lys Trp Ile Asp Arg Ser Met Lys Thr Arg Gly
   325                               330               335

Leu Pro Lys Ser Ser Gly Gly Ser Gly Gly Ser Gly Gly Ser Gly Gly
   340                               345               350

Ser Gly Gly Ser Gly Gly Ser Gly Gly Ser Gly Gly Ser Gly Gly Ser
   355                               360               365

Gly Ser Ser Val Ala Gln Ala Thr Ser Ser Ser Gly Glu Ala Pro Asp
   370                               375               380

Ser Ile Thr Trp Lys Pro Tyr Asp Ala Ala Asp Leu Asp Pro Thr Glu
   385                               390               395               400

Asn Pro Phe Asp Leu Leu Asp Phe Asn Gln Thr Gln Pro Glu Arg Gly
   405                               410               415

Asp Asn Asn Leu Thr Arg
   420

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<210> SEQ ID NO 27
<211> LENGTH: 950
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

<400> SEQUENCE: 27

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Asp Ala His Lys Ser Glu Val Ala His Arg Phe Lys Asp Leu Gly Glu
 1                               5               10               15

Glu Asn Phe Lys Ala Leu Val Leu Ile Ala Phe Ala Gln Tyr Leu Gln
 20                               25               30

Gln Ser Pro Phe Glu Asp His Val Lys Leu Val Asn Glu Val Thr Glu
 35                               40               45

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Phe	Ala	Lys	Thr	Cys	Val	Ala	Asp	Glu	Ser	Ala	Glu	Asn	Cys	Asp	Lys
50						55					60				
Ser	Leu	His	Thr	Leu	Phe	Gly	Asp	Lys	Leu	Cys	Thr	Val	Ala	Thr	Leu
65					70					75					80
Arg	Glu	Thr	Tyr	Gly	Glu	Met	Ala	Asp	Cys	Cys	Ala	Lys	Gln	Glu	Pro
				85					90					95	
Glu	Arg	Asn	Glu	Cys	Phe	Leu	Gln	His	Lys	Asp	Asp	Asn	Pro	Asn	Leu
			100					105					110		
Pro	Arg	Leu	Val	Arg	Pro	Glu	Val	Asp	Val	Met	Cys	Thr	Ala	Phe	His
		115					120					125			
Asp	Asn	Glu	Glu	Thr	Phe	Leu	Lys	Lys	Tyr	Leu	Tyr	Glu	Ile	Ala	Arg
	130					135					140				
Arg	His	Pro	Tyr	Phe	Tyr	Ala	Pro	Glu	Leu	Leu	Phe	Phe	Ala	Lys	Arg
145					150					155					160
Tyr	Lys	Ala	Ala	Phe	Thr	Glu	Cys	Cys	Gln	Ala	Ala	Asp	Lys	Ala	Ala
				165					170					175	
Cys	Leu	Leu	Pro	Lys	Leu	Asp	Glu	Leu	Arg	Asp	Glu	Gly	Lys	Ala	Ser
			180					185					190		
Ser	Ala	Lys	Gln	Arg	Leu	Lys	Cys	Ala	Ser	Leu	Gln	Lys	Phe	Gly	Glu
		195					200					205			
Arg	Ala	Phe	Lys	Ala	Trp	Ala	Val	Ala	Arg	Leu	Ser	Gln	Arg	Phe	Pro
	210					215					220				
Lys	Ala	Glu	Phe	Ala	Glu	Val	Ser	Lys	Leu	Val	Thr	Asp	Leu	Thr	Lys
225					230					235					240
Val	His	Thr	Glu	Cys	Cys	His	Gly	Asp	Leu	Leu	Glu	Cys	Ala	Asp	Asp
			245					250						255	
Arg	Ala	Asp	Leu	Ala	Lys	Tyr	Ile	Cys	Glu	Asn	Gln	Asp	Ser	Ile	Ser
		260						265					270		
Ser	Lys	Leu	Lys	Glu	Cys	Cys	Glu	Lys	Pro	Leu	Leu	Glu	Lys	Ser	His
		275					280					285			
Cys	Ile	Ala	Glu	Val	Glu	Asn	Asp	Glu	Met	Pro	Ala	Asp	Leu	Pro	Ser
	290					295					300				
Leu	Ala	Ala	Asp	Phe	Val	Glu	Ser	Lys	Asp	Val	Cys	Lys	Asn	Tyr	Ala
305					310					315					320
Glu	Ala	Lys	Asp	Val	Phe	Leu	Gly	Met	Phe	Leu	Tyr	Glu	Tyr	Ala	Arg
			325					330						335	
Arg	His	Pro	Asp	Tyr	Ser	Val	Val	Leu	Leu	Leu	Arg	Leu	Ala	Lys	Thr
			340					345					350		
Tyr	Glu	Thr	Thr	Leu	Glu	Lys	Cys	Cys	Ala	Ala	Ala	Asp	Pro	His	Glu
		355					360					365			
Cys	Tyr	Ala	Lys	Val	Phe	Asp	Glu	Phe	Lys	Pro	Leu	Val	Glu	Glu	Pro
	370					375					380				
Gln	Asn	Leu	Ile	Lys	Gln	Asn	Cys	Glu	Leu	Phe	Glu	Gln	Leu	Gly	Glu
385					390					395					400
Tyr	Lys	Phe	Gln	Asn	Ala	Leu	Leu	Val	Arg	Tyr	Thr	Lys	Lys	Val	Pro
			405						410					415	
Gln	Val	Ser	Thr	Pro	Thr	Leu	Val	Glu	Val	Ser	Arg	Asn	Leu	Gly	Lys
			420					425					430		
Val	Gly	Ser	Lys	Cys	Cys	Lys	His	Pro	Glu	Ala	Lys	Arg	Met	Pro	Cys
	435						440					445			
Ala	Glu	Asp	Tyr	Leu	Ser	Val	Val	Leu	Asn	Gln	Leu	Cys	Val	Leu	His
	450					455					460				

-continued

Glu	Lys	Thr	Pro	Val	Ser	Asp	Arg	Val	Thr	Lys	Cys	Cys	Thr	Glu	Ser	465	470	475	480
Leu	Val	Asn	Arg	Arg	Pro	Cys	Phe	Ser	Ala	Leu	Glu	Val	Asp	Glu	Thr	485	490	495	
Tyr	Val	Pro	Lys	Glu	Phe	Asn	Ala	Glu	Thr	Phe	Thr	Phe	His	Ala	Asp	500	505	510	
Ile	Cys	Thr	Leu	Ser	Glu	Lys	Glu	Arg	Gln	Ile	Lys	Lys	Gln	Thr	Ala	515	520	525	
Leu	Val	Glu	Leu	Val	Lys	His	Lys	Pro	Lys	Ala	Thr	Lys	Glu	Gln	Leu	530	535	540	
Lys	Ala	Val	Met	Asp	Asp	Phe	Ala	Ala	Phe	Val	Glu	Lys	Cys	Cys	Lys	545	550	555	560
Ala	Asp	Asp	Lys	Glu	Thr	Cys	Phe	Ala	Glu	Glu	Gly	Lys	Lys	Leu	Val	565	570	575	
Ala	Ala	Ser	Gln	Ala	Ala	Leu	Gly	Leu	Ala	Asn	Ser	Phe	Leu	Phe	Trp	580	585	590	
Asn	Lys	Tyr	Lys	Asp	Gly	Asp	Gln	Cys	Glu	Thr	Ser	Pro	Cys	Gln	Asn	595	600	605	
Gln	Gly	Lys	Cys	Lys	Asp	Gly	Leu	Gly	Glu	Tyr	Thr	Cys	Thr	Cys	Leu	610	615	620	
Glu	Gly	Phe	Glu	Gly	Lys	Asn	Cys	Glu	Leu	Phe	Thr	Arg	Lys	Leu	Cys	625	630	635	640
Ser	Leu	Asp	Asn	Gly	Asp	Cys	Asp	Gln	Phe	Cys	His	Glu	Glu	Gln	Asn	645	650	655	
Ser	Val	Val	Cys	Ser	Cys	Ala	Arg	Gly	Tyr	Thr	Leu	Ala	Asp	Asn	Gly	660	665	670	
Lys	Ala	Cys	Ile	Pro	Thr	Gly	Pro	Tyr	Pro	Cys	Gly	Lys	Gln	Thr	Leu	675	680	685	
Glu	Arg	Arg	Lys	Arg	Arg	Lys	Arg	Ile	Val	Gly	Gly	Gln	Glu	Cys	Lys	690	695	700	
Asp	Gly	Glu	Cys	Pro	Trp	Gln	Ala	Leu	Leu	Ile	Asn	Glu	Glu	Asn	Glu	705	710	715	720
Gly	Phe	Cys	Gly	Gly	Thr	Ile	Leu	Ser	Glu	Phe	Tyr	Ile	Leu	Thr	Ala	725	730	735	
Ala	His	Cys	Leu	Tyr	Gln	Ala	Lys	Arg	Phe	Lys	Val	Arg	Val	Gly	Asp	740	745	750	
Arg	Asn	Thr	Glu	Gln	Glu	Glu	Gly	Gly	Glu	Ala	Val	His	Glu	Val	Glu	755	760	765	
Val	Val	Ile	Lys	His	Asn	Arg	Phe	Thr	Lys	Glu	Thr	Tyr	Asp	Phe	Asp	770	775	780	
Ile	Ala	Val	Leu	Arg	Leu	Lys	Thr	Pro	Ile	Thr	Phe	Arg	Met	Asn	Val	785	790	795	800
Ala	Pro	Ala	Cys	Leu	Pro	Glu	Arg	Asp	Trp	Ala	Glu	Ser	Thr	Leu	Met	805	810	815	
Thr	Gln	Lys	Thr	Gly	Ile	Val	Ser	Gly	Phe	Gly	Arg	Thr	His	Glu	Lys	820	825	830	
Gly	Arg	Gln	Ser	Thr	Arg	Leu	Lys	Met	Leu	Glu	Val	Pro	Tyr	Val	Asp	835	840	845	
Arg	Asn	Ser	Cys	Lys	Leu	Ser	Ser	Ser	Phe	Ile	Ile	Thr	Gln	Asn	Met	850	855	860	
Phe	Cys	Ala	Gly	Tyr	Asp	Thr	Lys	Gln	Glu	Asp	Ala	Cys	Gln	Gly	Asp	865	870	875	880
Ala	Gly	Gly	Pro	His	Val	Thr	Arg	Phe	Lys	Asp	Thr	Tyr	Phe	Val	Thr				

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885	890	895
Gly Ile Val Ser Trp Gly Glu Gly Cys Ala Arg Lys Gly Lys Tyr Gly		
900	905	910
Ile Tyr Thr Lys Val Thr Ala Phe Leu Lys Trp Ile Asp Arg Ser Met		
915	920	925
Lys Thr Arg Gly Leu Pro Lys Ala Lys Ser His Ala Pro Glu Val Ile		
930	935	940
Thr Ser Ser Pro Leu Lys		
945	950	

<210> SEQ ID NO 28
 <211> LENGTH: 935
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 28

Ala Asn Ser Phe Leu Phe Trp Asn Lys Tyr Lys Asp Gly Asp Gln Cys		
1	5	10 15
Glu Thr Ser Pro Cys Gln Asn Gln Gly Lys Cys Lys Asp Gly Leu Gly		
20	25	30
Glu Tyr Thr Cys Thr Cys Leu Glu Gly Phe Glu Gly Lys Asn Cys Glu		
35	40	45
Leu Phe Thr Arg Lys Leu Cys Ser Leu Asp Asn Gly Asp Cys Asp Gln		
50	55	60
Phe Cys His Glu Glu Gln Asn Ser Val Val Cys Ser Cys Ala Arg Gly		
65	70	75 80
Tyr Thr Leu Ala Asp Asn Gly Lys Ala Cys Ile Pro Thr Gly Pro Tyr		
85	90	95
Pro Cys Gly Lys Gln Thr Leu Glu Arg Arg Lys Arg Arg Lys Arg Ile		
100	105	110
Val Gly Gly Gln Glu Cys Lys Asp Gly Glu Cys Pro Trp Gln Ala Leu		
115	120	125
Leu Ile Asn Glu Glu Asn Glu Gly Phe Cys Gly Gly Thr Ile Leu Ser		
130	135	140
Glu Phe Tyr Ile Leu Thr Ala Ala His Cys Leu Tyr Gln Ala Lys Arg		
145	150	155 160
Phe Lys Val Arg Val Gly Asp Arg Asn Thr Glu Gln Glu Glu Gly Gly		
165	170	175
Glu Ala Val His Glu Val Glu Val Val Ile Lys His Asn Arg Phe Thr		
180	185	190
Lys Glu Thr Tyr Asp Phe Asp Ile Ala Val Leu Arg Leu Lys Thr Pro		
195	200	205
Ile Thr Phe Arg Met Asn Val Ala Pro Ala Cys Leu Pro Glu Arg Asp		
210	215	220
Trp Ala Glu Ser Thr Leu Met Thr Gln Lys Thr Gly Ile Val Ser Gly		
225	230	235 240
Phe Gly Arg Thr His Glu Lys Gly Arg Gln Ser Thr Arg Leu Lys Met		
245	250	255
Leu Glu Val Pro Tyr Val Asp Arg Asn Ser Cys Lys Leu Ser Ser Ser		
260	265	270
Phe Ile Ile Thr Gln Asn Met Phe Cys Ala Gly Tyr Asp Thr Lys Gln		
275	280	285

Glu 290	Ala	Cys	Gln	Gly	Asp 295	Ala	Gly	Gly	Pro	His 300	Val	Thr	Arg	Phe
Lys 305	Asp	Thr	Tyr	Phe	Val 310	Thr	Gly	Ile	Val	Ser 315	Trp	Gly	Glu	Cys 320
Ala	Arg	Lys	Gly	Lys 325	Tyr	Gly	Ile	Tyr	Thr 330	Lys	Val	Thr	Ala	Leu 335
Lys	Trp	Ile	Asp	Arg	Ser	Met	Lys	Thr	Arg	Gly	Leu	Pro	Lys	Ala 350
His	Lys	Ser	Glu	Val	Ala	His	Arg	Phe	Lys	Asp	Leu	Gly	Glu	Asn 365
Phe	Lys	Ala	Leu	Val	Leu	Ile	Ala	Phe	Ala	Gln	Tyr	Leu	Gln	Ser 380
Pro 385	Phe	Glu	Asp	His	Val 390	Lys	Leu	Val	Asn	Glu	Val	Thr	Glu	Ala 400
Lys	Thr	Cys	Val	Ala	Asp	Glu	Ser	Ala	Glu	Asn	Cys	Asp	Lys	Leu 415
His	Thr	Leu	Phe	Gly	Asp	Lys	Leu	Cys	Thr	Val	Ala	Thr	Leu	Glu 430
Thr	Tyr	Gly	Glu	Met	Ala	Asp	Cys	Cys	Ala	Lys	Gln	Glu	Pro	Arg 445
Asn 450	Glu	Cys	Phe	Leu	Gln	His	Lys	Asp	Asp	Asn	Pro	Asn	Leu	Arg 460
Leu 465	Val	Arg	Pro	Glu	Val	Asp	Val	Met	Cys	Thr	Ala	Phe	His	Asn 480
Glu	Glu	Thr	Phe	Leu	Lys	Lys	Tyr	Leu	Tyr	Glu	Ile	Ala	Arg	His 495
Pro	Tyr	Phe	Tyr	Ala	Pro	Glu	Leu	Leu	Phe	Phe	Ala	Lys	Arg	Lys 510
Ala	Ala	Phe	Thr	Glu	Cys	Cys	Gln	Ala	Ala	Asp	Lys	Ala	Ala	Leu 525
Leu 530	Pro	Lys	Leu	Asp	Glu	Leu	Arg	Asp	Glu	Gly	Lys	Ala	Ser	Ala 540
Lys 545	Gln	Arg	Leu	Lys	Cys	Ala	Ser	Leu	Gln	Lys	Phe	Gly	Glu	Ala 555
Phe	Lys	Ala	Trp	Ala	Val	Ala	Arg	Leu	Ser	Gln	Arg	Phe	Pro	Ala 570
Glu	Phe	Ala	Glu	Val	Ser	Lys	Leu	Val	Thr	Asp	Leu	Thr	Lys	His 585
Thr	Glu	Cys	Cys	His	Gly	Asp	Leu	Leu	Glu	Cys	Ala	Asp	Asp	Ala 600
Asp 610	Leu	Ala	Lys	Tyr	Ile	Cys	Glu	Asn	Gln	Asp	Ser	Ile	Ser	Lys 615
Leu 625	Lys	Glu	Cys	Cys	Glu	Lys	Pro	Leu	Leu	Glu	Lys	Ser	His	Ile 630
Ala	Glu	Val	Glu	Asn	Asp	Glu	Met	Pro	Ala	Asp	Leu	Pro	Ser	Ala 645
Ala	Asp	Phe	Val	Glu	Ser	Lys	Asp	Val	Cys	Lys	Asn	Tyr	Ala	Ala 660
Lys	Asp	Val	Phe	Leu	Gly	Met	Phe	Leu	Tyr	Glu	Tyr	Ala	Arg	His 675
Pro 690	Asp	Tyr	Ser	Val	Val	Leu	Leu	Leu	Arg	Leu	Ala	Lys	Thr	Glu 695
Thr	Thr	Leu	Glu	Lys	Cys	Cys	Ala	Ala	Ala	Asp	Pro	His	Glu	Tyr

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705	710	715	720
Ala Lys Val Phe Asp Glu Phe Lys Pro Leu Val Glu Glu Pro Gln Asn			
	725	730	735
Leu Ile Lys Gln Asn Cys Glu Leu Phe Glu Gln Leu Gly Glu Tyr Lys			
	740	745	750
Phe Gln Asn Ala Leu Leu Val Arg Tyr Thr Lys Lys Val Pro Gln Val			
	755	760	765
Ser Thr Pro Thr Leu Val Glu Val Ser Arg Asn Leu Gly Lys Val Gly			
	770	775	780
Ser Lys Cys Cys Lys His Pro Glu Ala Lys Arg Met Pro Cys Ala Glu			
	785	790	800
Asp Tyr Leu Ser Val Val Leu Asn Gln Leu Cys Val Leu His Glu Lys			
	805	810	815
Thr Pro Val Ser Asp Arg Val Thr Lys Cys Cys Thr Glu Ser Leu Val			
	820	825	830
Asn Arg Arg Pro Cys Phe Ser Ala Leu Glu Val Asp Glu Thr Tyr Val			
	835	840	845
Pro Lys Glu Phe Asn Ala Glu Thr Phe Thr Phe His Ala Asp Ile Cys			
	850	855	860
Thr Leu Ser Glu Lys Glu Arg Gln Ile Lys Lys Gln Thr Ala Leu Val			
	865	870	880
Glu Leu Val Lys His Lys Pro Lys Ala Thr Lys Glu Gln Leu Lys Ala			
	885	890	895
Val Met Asp Asp Phe Ala Ala Phe Val Glu Lys Cys Cys Lys Ala Asp			
	900	905	910
Asp Lys Glu Thr Cys Phe Ala Glu Glu Gly Lys Lys Leu Val Ala Ala			
	915	920	925
Ser Gln Ala Ala Leu Gly Leu			
	930	935	

<210> SEQ ID NO 29

<211> LENGTH: 385

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 29

Asp Gly Asp Gln Cys Glu Thr Ser Pro Cys Gln Asn Gln Gly Lys Cys			
1	5	10	15
Lys Asp Gly Leu Gly Glu Tyr Thr Cys Thr Cys Leu Glu Gly Phe Glu			
	20	25	30
Gly Lys Asn Cys Glu Leu Phe Thr Arg Lys Leu Cys Ser Leu Asp Asn			
	35	40	45
Gly Asp Cys Asp Gln Phe Cys His Glu Glu Gln Asn Ser Val Val Cys			
	50	55	60
Ser Cys Ala Arg Gly Tyr Thr Leu Ala Asp Asn Gly Lys Ala Cys Ile			
	65	70	75
Pro Thr Gly Pro Tyr Pro Cys Gly Lys Gln Thr Leu Glu Arg Ile Val			
	85	90	95
Gly Gly Gln Glu Cys Lys Asp Gly Glu Cys Pro Trp Gln Ala Leu Leu			
	100	105	110
Ile Asn Glu Glu Asn Glu Gly Phe Cys Gly Gly Thr Ile Leu Ser Glu			
	115	120	125

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Phe Tyr Ile Leu Thr Ala Ala His Cys Leu Tyr Gln Ala Lys Arg Phe
 130          135          140

Lys Val Arg Val Gly Asp Arg Asn Thr Glu Gln Glu Gly Gly Glu
 145          150          155          160

Ala Val His Glu Val Glu Val Val Ile Lys His Asn Arg Phe Thr Lys
          165          170          175

Glu Thr Tyr Asp Phe Asp Ile Ala Val Leu Arg Leu Lys Thr Pro Ile
          180          185          190

Thr Phe Arg Met Asn Val Ala Pro Ala Cys Leu Pro Glu Arg Asp Trp
          195          200          205

Ala Glu Ser Thr Leu Met Thr Gln Lys Thr Gly Ile Val Ser Gly Phe
 210          215          220

Gly Arg Thr His Glu Lys Gly Arg Gln Ser Thr Arg Leu Lys Met Leu
 225          230          235          240

Glu Val Pro Tyr Val Asp Arg Asn Ser Cys Lys Leu Ser Ser Ser Phe
          245          250          255

Ile Ile Thr Gln Asn Met Phe Cys Ala Gly Tyr Asp Thr Lys Gln Glu
          260          265          270

Asp Ala Cys Gln Gly Asp Ala Gly Gly Pro His Val Thr Arg Phe Lys
          275          280          285

Asp Thr Tyr Phe Val Thr Gly Ile Val Ser Trp Gly Glu Gly Cys Ala
          290          295          300

Arg Lys Gly Lys Tyr Gly Ile Tyr Thr Lys Val Thr Ala Phe Leu Lys
 305          310          315          320

Trp Ile Asp Arg Ser Met Lys Thr Arg Gly Leu Pro Lys Ser Val Ala
          325          330          335

Gln Ala Thr Ser Ser Ser Gly Glu Ala Pro Asp Ser Ile Thr Trp Lys
          340          345          350

Pro Tyr Asp Ala Ala Asp Leu Asp Pro Thr Glu Asn Pro Phe Asp Leu
          355          360          365

Leu Asp Phe Asn Gln Thr Gln Pro Glu Arg Gly Asp Asn Asn Leu Thr
          370          375          380

Arg
385

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<210> SEQ ID NO 30
<211> LENGTH: 416
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

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<400> SEQUENCE: 30

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Asp Gly Asp Gln Cys Glu Thr Ser Pro Cys Gln Asn Gln Gly Lys Cys
 1          5          10          15

Lys Asp Gly Leu Gly Glu Tyr Thr Cys Thr Cys Leu Glu Gly Phe Glu
          20          25          30

Gly Lys Asn Cys Glu Leu Phe Thr Arg Lys Leu Cys Ser Leu Asp Asn
          35          40          45

Gly Asp Cys Asp Gln Phe Cys His Glu Glu Gln Asn Ser Val Val Cys
          50          55          60

Ser Cys Ala Arg Gly Tyr Thr Leu Ala Asp Asn Gly Lys Ala Cys Ile
          65          70          75          80

Pro Thr Gly Pro Tyr Pro Cys Gly Lys Gln Thr Leu Glu Arg Ile Val
          85          90          95

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Gly Gly Gln Glu Cys Lys Asp Gly Glu Cys Pro Trp Gln Ala Leu Leu
 100 105 110
 Ile Asn Glu Glu Asn Glu Gly Phe Cys Gly Gly Thr Ile Leu Ser Glu
 115 120 125
 Phe Tyr Ile Leu Thr Ala Ala His Cys Leu Tyr Gln Ala Lys Arg Phe
 130 135 140
 Lys Val Arg Val Gly Asp Arg Asn Thr Glu Gln Glu Glu Gly Gly Glu
 145 150 155 160
 Ala Val His Glu Val Glu Val Val Ile Lys His Asn Arg Phe Thr Lys
 165 170 175
 Glu Thr Tyr Asp Phe Asp Ile Ala Val Leu Arg Leu Lys Thr Pro Ile
 180 185 190
 Thr Phe Arg Met Asn Val Ala Pro Ala Cys Leu Pro Glu Arg Asp Trp
 195 200 205
 Ala Glu Ser Thr Leu Met Thr Gln Lys Thr Gly Ile Val Ser Gly Phe
 210 215 220
 Gly Arg Thr His Glu Lys Gly Arg Gln Ser Thr Arg Leu Lys Met Leu
 225 230 235 240
 Glu Val Pro Tyr Val Asp Arg Asn Ser Cys Lys Leu Ser Ser Ser Phe
 245 250 255
 Ile Ile Thr Gln Asn Met Phe Cys Ala Gly Tyr Asp Thr Lys Gln Glu
 260 265 270
 Asp Ala Cys Gln Gly Asp Ala Gly Gly Pro His Val Thr Arg Phe Lys
 275 280 285
 Asp Thr Tyr Phe Val Thr Gly Ile Val Ser Trp Gly Glu Gly Cys Ala
 290 295 300
 Arg Lys Gly Lys Tyr Gly Ile Tyr Thr Lys Val Thr Ala Phe Leu Lys
 305 310 315 320
 Trp Ile Asp Arg Ser Met Lys Thr Arg Gly Leu Pro Lys Ser Ser Gly
 325 330 335
 Gly Ser Gly Gly Ser Gly Gly Ser Gly Gly Ser Gly Gly Ser Gly Gly
 340 345 350
 Ser Gly Gly Ser Gly Gly Ser Gly Gly Ser Gly Ser Ser Val Ala Gln
 355 360 365
 Ala Thr Ser Ser Ser Gly Glu Ala Pro Asp Ser Ile Thr Trp Lys Pro
 370 375 380
 Tyr Asp Ala Ala Asp Leu Asp Pro Thr Glu Asn Pro Phe Asp Leu Leu
 385 390 395 400
 Asp Phe Asn Gln Thr Gln Pro Glu Arg Gly Asp Asn Asn Leu Thr Arg
 405 410 415

<210> SEQ ID NO 31

<211> LENGTH: 35

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 31

Ala Glu Thr Val Phe Pro Asp Val Asp Tyr Val Asn Ser Thr Glu Ala
 1 5 10 15
 Glu Thr Ile Leu Asp Asn Ile Thr Gln Ser Thr Gln Ser Phe Asn Asp
 20 25 30
 Phe Thr Arg

-continued

35

<210> SEQ ID NO 32
<211> LENGTH: 38
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 32

Met Ser Glu Thr Ser Arg Thr Ala Phe Gly Gly Arg Arg Ala Val Pro
1 5 10 15

Pro Asn Asn Ser Asn Ala Ala Glu Asp Asp Leu Pro Thr Val Glu Leu
20 25 30

Gln Gly Val Val Pro Arg
35

<210> SEQ ID NO 33
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 33

Asp Thr Glu Asp Gln Glu Asp Gln Val Asp Pro Arg
1 5 10

<210> SEQ ID NO 34
<211> LENGTH: 40
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 34

Met Gly Arg Pro Leu His Leu Val Leu Leu Ser Ala Ser Leu Ala Gly
1 5 10 15

Leu Leu Leu Leu Gly Glu Ser Leu Phe Ile Arg Arg Glu Gln Ala Asn
20 25 30

Asn Ile Leu Ala Arg Val Thr Arg
35 40

<210> SEQ ID NO 35
<211> LENGTH: 24
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 35

Met Ala His Val Arg Gly Leu Gln Leu Pro Gly Cys Leu Ala Leu Ala
1 5 10 15

Ala Leu Cys Ser Leu Val His Ser
20

<210> SEQ ID NO 36
<211> LENGTH: 19
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 36

Met Arg Leu Ala Val Gly Ala Leu Leu Val Cys Ala Val Leu Gly Leu
1 5 10 15

Cys Leu Ala

-continued

<210> SEQ ID NO 37
 <211> LENGTH: 43
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 37

Met Ala His Val Arg Gly Leu Gln Leu Pro Gly Cys Leu Ala Leu Ala
 1 5 10 15

Ala Leu Cys Ser Leu Val His Ser Gln His Val Phe Leu Ala Pro Gln
 20 25 30

Gln Ala Arg Ser Leu Leu Gln Arg Val Arg Arg
 35 40

<210> SEQ ID NO 38
 <211> LENGTH: 20
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 38

Arg Gly Leu Pro Lys Ala Lys Ser His Ala Pro Glu Val Ile Thr Ser
 1 5 10 15

Ser Pro Leu Lys
 20

<210> SEQ ID NO 39
 <211> LENGTH: 155
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 39

Ala Asn Thr Phe Leu Glu Glu Val Arg Lys Gly Asn Leu Glu Arg Glu
 1 5 10 15

Cys Val Glu Glu Thr Cys Ser Tyr Glu Glu Ala Phe Glu Ala Leu Glu
 20 25 30

Ser Ser Thr Ala Thr Asp Val Phe Trp Ala Lys Tyr Thr Ala Cys Glu
 35 40 45

Thr Ala Arg Thr Pro Arg Asp Lys Leu Ala Ala Cys Leu Glu Gly Asn
 50 55 60

Cys Ala Glu Gly Leu Gly Thr Asn Tyr Arg Gly His Val Asn Ile Thr
 65 70 75 80

Arg Ser Gly Ile Glu Cys Gln Leu Trp Arg Ser Arg Tyr Pro His Lys
 85 90 95

Pro Glu Ile Asn Ser Thr Thr His Pro Gly Ala Asp Leu Gln Glu Asn
 100 105 110

Phe Cys Arg Asn Pro Asp Ser Ser Thr Thr Gly Pro Trp Cys Tyr Thr
 115 120 125

Thr Asp Pro Thr Val Arg Arg Gln Glu Cys Ser Ile Pro Val Cys Gly
 130 135 140

Gln Asp Gln Val Thr Val Ala Met Thr Pro Arg
 145 150 155

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<210> SEQ ID NO 40
 <211> LENGTH: 129
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 40

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Ser Glu Gly Ser Ser Val Asn Leu Ser Pro Pro Leu Glu Gln Cys Val
1          5          10          15
Pro Asp Arg Gly Gln Gln Tyr Gln Gly Arg Leu Ala Val Thr Thr His
20          25          30
Gly Leu Pro Cys Leu Ala Trp Ala Ser Ala Gln Ala Lys Ala Leu Ser
35          40          45
Lys His Gln Asp Phe Asn Ser Ala Val Gln Leu Val Glu Asn Phe Cys
50          55          60
Arg Asn Pro Asp Gly Asp Glu Glu Gly Val Trp Cys Tyr Val Ala Gly
65          70          75          80
Lys Pro Gly Asp Phe Gly Tyr Cys Asp Leu Asn Tyr Cys Glu Glu Ala
85          90          95
Val Glu Glu Glu Thr Gly Asp Gly Leu Asp Glu Asp Ser Asp Arg Ala
100         105         110
Ile Glu Gly Arg Thr Ala Thr Ser Glu Tyr Gln Thr Phe Phe Asn Pro
115        120        125

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Arg

<210> SEQ ID NO 41
 <211> LENGTH: 471
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 41

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Ala Asn Ser Phe Leu Phe Trp Asn Lys Tyr Lys Asp Gly Asp Gln Cys
1          5          10          15
Glu Thr Ser Pro Cys Gln Asn Gln Gly Lys Cys Lys Asp Gly Leu Gly
20          25          30
Glu Tyr Thr Cys Thr Cys Leu Glu Gly Phe Glu Gly Lys Asn Cys Glu
35          40          45
Leu Phe Thr Arg Lys Leu Cys Ser Leu Asp Asn Gly Asp Cys Asp Gln
50          55          60
Phe Cys His Glu Glu Gln Asn Ser Val Val Cys Ser Cys Ala Arg Gly
65          70          75          80
Tyr Thr Leu Ala Asp Asn Gly Lys Ala Cys Ile Pro Thr Gly Pro Tyr
85          90          95
Pro Cys Gly Lys Gln Thr Leu Glu Arg Arg Lys Arg Ser Val Ala Gln
100         105         110
Ala Thr Ser Ser Ser Gly Glu Ala Pro Asp Ser Ile Thr Trp Lys Pro
115        120        125
Tyr Asp Ala Ala Asp Leu Asp Pro Thr Glu Asn Pro Phe Asp Leu Leu
130        135        140
Asp Phe Asn Gln Thr Gln Pro Glu Arg Gly Asp Asn Asn Leu Thr Arg
145        150        155        160
Arg Lys Arg Arg Lys Arg Ile Val Gly Gly Gln Glu Cys Lys Asp Gly

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165					170					175					
Glu	Cys	Pro	Trp	Gln	Ala	Leu	Leu	Ile	Asn	Glu	Glu	Asn	Glu	Gly	Phe
			180					185					190		
Cys	Gly	Gly	Thr	Ile	Leu	Ser	Glu	Phe	Tyr	Ile	Leu	Thr	Ala	Ala	His
		195					200					205			
Cys	Leu	Tyr	Gln	Ala	Lys	Arg	Phe	Lys	Val	Arg	Val	Gly	Asp	Arg	Asn
		210				215					220				
Thr	Glu	Gln	Glu	Glu	Gly	Gly	Glu	Ala	Val	His	Glu	Val	Glu	Val	Val
225					230					235					240
Ile	Lys	His	Asn	Arg	Phe	Thr	Lys	Glu	Thr	Tyr	Asp	Phe	Asp	Ile	Ala
			245						250					255	
Val	Leu	Arg	Leu	Lys	Thr	Pro	Ile	Thr	Phe	Arg	Met	Asn	Val	Ala	Pro
			260					265					270		
Ala	Cys	Leu	Pro	Glu	Arg	Asp	Trp	Ala	Glu	Ser	Thr	Leu	Met	Thr	Gln
		275					280						285		
Lys	Thr	Gly	Ile	Val	Ser	Gly	Phe	Gly	Arg	Thr	His	Glu	Lys	Gly	Arg
		290				295					300				
Gln	Ser	Thr	Arg	Leu	Lys	Met	Leu	Glu	Val	Pro	Tyr	Val	Asp	Arg	Asn
305					310					315					320
Ser	Cys	Lys	Leu	Ser	Ser	Ser	Phe	Ile	Ile	Thr	Gln	Asn	Met	Phe	Cys
			325						330					335	
Ala	Gly	Tyr	Asp	Thr	Lys	Gln	Glu	Asp	Ala	Cys	Gln	Gly	Asp	Ala	Gly
			340					345					350		
Gly	Pro	His	Val	Thr	Arg	Phe	Lys	Asp	Thr	Tyr	Phe	Val	Thr	Gly	Ile
		355					360					365			
Val	Ser	Trp	Gly	Glu	Gly	Cys	Ala	Arg	Lys	Gly	Lys	Tyr	Gly	Ile	Tyr
		370				375					380				
Thr	Lys	Val	Thr	Ala	Phe	Leu	Lys	Trp	Ile	Asp	Arg	Ser	Met	Lys	Thr
385					390					395					400
Arg	Gly	Leu	Pro	Lys	Ala	Lys	Ser	His	Ala	Pro	Glu	Val	Ile	Thr	Ser
			405						410					415	
Ser	Pro	Leu	Lys	Val	Ala	Gln	Ala	Thr	Ser	Ser	Ser	Gly	Glu	Ala	Pro
			420					425					430		
Asp	Ser	Ile	Thr	Trp	Lys	Pro	Tyr	Asp	Ala	Ala	Asp	Leu	Asp	Pro	Thr
		435					440					445			
Glu	Asn	Pro	Phe	Asp	Leu	Leu	Asp	Phe	Asn	Gln	Thr	Gln	Pro	Glu	Arg
		450				455					460				
Gly	Asp	Asn	Asn	Leu	Thr	Arg									
465					470										

<210> SEQ ID NO 42

<211> LENGTH: 939

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 42

Asp	Ala	His	Lys	Ser	Glu	Val	Ala	His	Arg	Phe	Lys	Asp	Leu	Gly	Glu
1				5					10					15	

Glu	Asn	Phe	Lys	Ala	Leu	Val	Leu	Ile	Ala	Phe	Ala	Gln	Tyr	Leu	Gln
			20					25					30		

Gln	Ser	Pro	Phe	Glu	Asp	His	Val	Lys	Leu	Val	Asn	Glu	Val	Thr	Glu
		35					40					45			

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Phe 50	Ala	Lys	Thr	Cys	Val	Ala 55	Asp	Glu	Ser	Ala	Glu 60	Asn	Cys	Asp	Lys
Ser 65	Leu	His	Thr	Leu	Phe 70	Gly	Asp	Lys	Leu	Cys 75	Thr	Val	Ala	Thr	Leu 80
Arg	Glu	Thr	Tyr	Gly 85	Glu	Met	Ala	Asp	Cys 90	Cys	Ala	Lys	Gln	Glu 95	Pro
Glu	Arg	Asn	Glu	Cys 100	Phe	Leu	Gln	His 105	Lys	Asp	Asp	Asn	Pro	Asn	Leu
Pro	Arg	Leu	Val	Arg 115	Pro	Glu	Val	Asp 120	Val	Met	Cys	Thr	Ala	Phe	His
Asp	Asn	Glu	Glu	Thr 130	Phe	Leu	Lys	Lys 135	Tyr	Leu	Tyr	Glu	Ile	Ala	Arg
Arg	His	Pro	Tyr	Phe 145	Tyr	Ala	Pro	Glu 150	Leu	Leu	Phe	Phe	Ala	Lys	Arg 160
Tyr	Lys	Ala	Ala	Phe 165	Thr	Glu	Cys	Cys 170	Gln	Ala	Ala	Asp	Lys	Ala	Ala
Cys	Leu	Leu	Pro	Lys 180	Leu	Asp	Glu	Leu 185	Arg	Asp	Glu	Gly	Lys	Ala	Ser
Ser	Ala	Lys	Gln	Arg 195	Leu	Lys	Cys	Ala 200	Ser	Leu	Gln	Lys	Phe	Gly	Glu
Arg	Ala	Phe	Lys	Ala 210	Trp	Ala	Val	Ala 215	Arg	Leu	Ser	Gln	Arg	Phe	Pro
Lys	Ala	Glu	Phe	Ala 225	Glu	Val	Ser	Lys 230	Leu	Val	Thr	Asp	Leu	Thr	Lys 240
Val	His	Thr	Glu	Cys 245	Cys	His	Gly	Asp 250	Leu	Leu	Glu	Cys	Ala	Asp	Asp
Arg	Ala	Asp	Leu	Ala 260	Lys	Tyr	Ile	Cys 265	Glu	Asn	Gln	Asp	Ser	Ile	Ser
Ser	Lys	Leu	Lys	Glu 275	Cys	Cys	Glu	Lys 280	Pro	Leu	Leu	Glu	Lys	Ser	His
Cys	Ile	Ala	Glu	Val 290	Glu	Asn	Asp	Glu 295	Met	Pro	Ala	Asp	Leu	Pro	Ser
Leu	Ala	Ala	Asp	Phe 305	Val	Glu	Ser	Lys 310	Asp	Val	Cys	Lys	Asn	Tyr	Ala 320
Glu	Ala	Lys	Asp	Val 325	Phe	Leu	Gly	Met 330	Phe	Leu	Tyr	Glu	Tyr	Ala	Arg 335
Arg	His	Pro	Asp	Tyr 340	Ser	Val	Val	Leu 345	Leu	Leu	Arg	Leu	Ala	Lys	Thr
Tyr	Glu	Thr	Thr	Leu 355	Glu	Lys	Cys	Cys 360	Ala	Ala	Ala	Asp	Pro	His	Glu
Cys	Tyr	Ala	Lys	Val 370	Phe	Asp	Glu	Phe 375	Lys	Pro	Leu	Val	Glu	Glu	Pro
Gln	Asn	Leu	Ile	Lys 385	Gln	Asn	Cys	Glu 390	Leu	Phe	Glu	Gln	Leu	Gly	Glu 400
Tyr	Lys	Phe	Gln	Asn 405	Ala	Leu	Leu	Val 410	Arg	Tyr	Thr	Lys	Lys	Val	Pro
Gln	Val	Ser	Thr	Pro 420	Thr	Leu	Val	Glu 425	Val	Ser	Arg	Asn	Leu	Gly	Lys
Val	Gly	Ser	Lys	Cys 435	Cys	Lys	His	Pro 440	Glu	Ala	Lys	Arg	Met	Pro	Cys
Ala	Glu	Asp	Tyr	Leu 450	Ser	Val	Val	Leu 455	Asn	Gln	Leu	Cys	Val	Leu	His
Glu	Lys	Thr	Pro	Val	Ser	Asp	Arg	Val	Thr	Lys	Cys	Cys	Thr	Glu	Ser

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465	470	475	480
Leu Val Asn Arg Arg Pro Cys Phe Ser Ala Leu Glu Val Asp Glu Thr			
	485	490	495
Tyr Val Pro Lys Glu Phe Asn Ala Glu Thr Phe Thr Phe His Ala Asp			
	500	505	510
Ile Cys Thr Leu Ser Glu Lys Glu Arg Gln Ile Lys Lys Gln Thr Ala			
	515	520	525
Leu Val Glu Leu Val Lys His Lys Pro Lys Ala Thr Lys Glu Gln Leu			
	530	535	540
Lys Ala Val Met Asp Asp Phe Ala Ala Phe Val Glu Lys Cys Cys Lys			
	545	550	555
Ala Asp Asp Lys Glu Thr Cys Phe Ala Glu Glu Gly Lys Lys Leu Val			
	565	570	575
Ala Ala Ser Gln Ala Ala Leu Gly Leu Asp Gly Asp Gln Cys Glu Thr			
	580	585	590
Ser Pro Cys Gln Asn Gln Gly Lys Cys Lys Asp Gly Leu Gly Glu Tyr			
	595	600	605
Thr Cys Thr Cys Leu Glu Gly Phe Glu Gly Lys Asn Cys Glu Leu Phe			
	610	615	620
Thr Arg Lys Leu Cys Ser Leu Asp Asn Gly Asp Cys Asp Gln Phe Cys			
	625	630	635
His Glu Glu Gln Asn Ser Val Val Cys Ser Cys Ala Arg Gly Tyr Thr			
	645	650	655
Leu Ala Asp Asn Gly Lys Ala Cys Ile Pro Thr Gly Pro Tyr Pro Cys			
	660	665	670
Gly Lys Gln Thr Leu Glu Arg Arg Lys Arg Arg Lys Arg Ile Val Gly			
	675	680	685
Gly Gln Glu Cys Lys Asp Gly Glu Cys Pro Trp Gln Ala Leu Leu Ile			
	690	695	700
Asn Glu Glu Asn Glu Gly Phe Cys Gly Gly Thr Ile Leu Ser Glu Phe			
	705	710	715
Tyr Ile Leu Thr Ala Ala His Cys Leu Tyr Gln Ala Lys Arg Phe Lys			
	725	730	735
Val Arg Val Gly Asp Arg Asn Thr Glu Gln Glu Glu Gly Gly Glu Ala			
	740	745	750
Val His Glu Val Glu Val Val Ile Lys His Asn Arg Phe Thr Lys Glu			
	755	760	765
Thr Tyr Asp Phe Asp Ile Ala Val Leu Arg Leu Lys Thr Pro Ile Thr			
	770	775	780
Phe Arg Met Asn Val Ala Pro Ala Cys Leu Pro Glu Arg Asp Trp Ala			
	785	790	795
Glu Ser Thr Leu Met Thr Gln Lys Thr Gly Ile Val Ser Gly Phe Gly			
	805	810	815
Arg Thr His Glu Lys Gly Arg Gln Ser Thr Arg Leu Lys Met Leu Glu			
	820	825	830
Val Pro Tyr Val Asp Arg Asn Ser Cys Lys Leu Ser Ser Ser Phe Ile			
	835	840	845
Ile Thr Gln Asn Met Phe Cys Ala Gly Tyr Asp Thr Lys Gln Glu Asp			
	850	855	860
Ala Cys Gln Gly Asp Ala Gly Gly Pro His Val Thr Arg Phe Lys Asp			
	865	870	875
Thr Tyr Phe Val Thr Gly Ile Val Ser Trp Gly Glu Gly Cys Ala Arg			
	885	890	895

-continued

Lys Gly Lys Tyr Gly Ile Tyr Thr Lys Val Thr Ala Phe Leu Lys Trp
 900 905 910

Ile Asp Arg Ser Met Lys Thr Arg Gly Leu Pro Lys Ala Lys Ser His
 915 920 925

Ala Pro Glu Val Ile Thr Ser Ser Pro Leu Lys
 930 935

<210> SEQ ID NO 43

<211> LENGTH: 976

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 43

Asp Ala His Lys Ser Glu Val Ala His Arg Phe Lys Asp Leu Gly Glu
 1 5 10 15

Glu Asn Phe Lys Ala Leu Val Leu Ile Ala Phe Ala Gln Tyr Leu Gln
 20 25 30

Gln Ser Pro Phe Glu Asp His Val Lys Leu Val Asn Glu Val Thr Glu
 35 40 45

Phe Ala Lys Thr Cys Val Ala Asp Glu Ser Ala Glu Asn Cys Asp Lys
 50 55 60

Ser Leu His Thr Leu Phe Gly Asp Lys Leu Cys Thr Val Ala Thr Leu
 65 70 75 80

Arg Glu Thr Tyr Gly Glu Met Ala Asp Cys Cys Ala Lys Gln Glu Pro
 85 90 95

Glu Arg Asn Glu Cys Phe Leu Gln His Lys Asp Asp Asn Pro Asn Leu
 100 105 110

Pro Arg Leu Val Arg Pro Glu Val Asp Val Met Cys Thr Ala Phe His
 115 120 125

Asp Asn Glu Glu Thr Phe Leu Lys Lys Tyr Leu Tyr Glu Ile Ala Arg
 130 135 140

Arg His Pro Tyr Phe Tyr Ala Pro Glu Leu Leu Phe Phe Ala Lys Arg
 145 150 155 160

Tyr Lys Ala Ala Phe Thr Glu Cys Cys Gln Ala Ala Asp Lys Ala Ala
 165 170 175

Cys Leu Leu Pro Lys Leu Asp Glu Leu Arg Asp Glu Gly Lys Ala Ser
 180 185 190

Ser Ala Lys Gln Arg Leu Lys Cys Ala Ser Leu Gln Lys Phe Gly Glu
 195 200 205

Arg Ala Phe Lys Ala Trp Ala Val Ala Arg Leu Ser Gln Arg Phe Pro
 210 215 220

Lys Ala Glu Phe Ala Glu Val Ser Lys Leu Val Thr Asp Leu Thr Lys
 225 230 235 240

Val His Thr Glu Cys Cys His Gly Asp Leu Leu Glu Cys Ala Asp Asp
 245 250 255

Arg Ala Asp Leu Ala Lys Tyr Ile Cys Glu Asn Gln Asp Ser Ile Ser
 260 265 270

Ser Lys Leu Lys Glu Cys Cys Glu Lys Pro Leu Leu Glu Lys Ser His
 275 280 285

Cys Ile Ala Glu Val Glu Asn Asp Glu Met Pro Ala Asp Leu Pro Ser
 290 295 300

Leu Ala Ala Asp Phe Val Glu Ser Lys Asp Val Cys Lys Asn Tyr Ala

-continued

305	310	315	320
Glu Ala Lys Asp Val Phe Leu Gly Met Phe Leu Tyr Glu Tyr Ala Arg	325	330	335
Arg His Pro Asp Tyr Ser Val Val Leu Leu Leu Arg Leu Ala Lys Thr	340	345	350
Tyr Glu Thr Thr Leu Glu Lys Cys Cys Ala Ala Ala Asp Pro His Glu	355	360	365
Cys Tyr Ala Lys Val Phe Asp Glu Phe Lys Pro Leu Val Glu Glu Pro	370	375	380
Gln Asn Leu Ile Lys Gln Asn Cys Glu Leu Phe Glu Gln Leu Gly Glu	385	390	395
Tyr Lys Phe Gln Asn Ala Leu Leu Val Arg Tyr Thr Lys Lys Val Pro	405	410	415
Gln Val Ser Thr Pro Thr Leu Val Glu Val Ser Arg Asn Leu Gly Lys	420	425	430
Val Gly Ser Lys Cys Cys Lys His Pro Glu Ala Lys Arg Met Pro Cys	435	440	445
Ala Glu Asp Tyr Leu Ser Val Val Leu Asn Gln Leu Cys Val Leu His	450	455	460
Glu Lys Thr Pro Val Ser Asp Arg Val Thr Lys Cys Cys Thr Glu Ser	465	470	475
Leu Val Asn Arg Arg Pro Cys Phe Ser Ala Leu Glu Val Asp Glu Thr	485	490	495
Tyr Val Pro Lys Glu Phe Asn Ala Glu Thr Phe Thr Phe His Ala Asp	500	505	510
Ile Cys Thr Leu Ser Glu Lys Glu Arg Gln Ile Lys Lys Gln Thr Ala	515	520	525
Leu Val Glu Leu Val Lys His Lys Pro Lys Ala Thr Lys Glu Gln Leu	530	535	540
Lys Ala Val Met Asp Asp Phe Ala Ala Phe Val Glu Lys Cys Cys Lys	545	550	555
Ala Asp Asp Lys Glu Thr Cys Phe Ala Glu Glu Gly Lys Lys Leu Val	565	570	575
Ala Ala Ser Gln Ala Ala Leu Gly Leu Asp Gly Asp Gln Cys Glu Thr	580	585	590
Ser Pro Cys Gln Asn Gln Gly Lys Cys Lys Asp Gly Leu Gly Glu Tyr	595	600	605
Thr Cys Thr Cys Leu Glu Gly Phe Glu Gly Lys Asn Cys Glu Leu Phe	610	615	620
Thr Arg Lys Leu Cys Ser Leu Asp Asn Gly Asp Cys Asp Gln Phe Cys	625	630	635
His Glu Glu Gln Asn Ser Val Val Cys Ser Cys Ala Arg Gly Tyr Thr	645	650	655
Leu Ala Asp Asn Gly Lys Ala Cys Ile Pro Thr Gly Pro Tyr Pro Cys	660	665	670
Gly Lys Gln Thr Leu Glu Arg Arg Lys Arg Arg Lys Arg Ile Val Gly	675	680	685
Gly Gln Glu Cys Lys Asp Gly Glu Cys Pro Trp Gln Ala Leu Leu Ile	690	695	700
Asn Glu Glu Asn Glu Gly Phe Cys Gly Gly Thr Ile Leu Ser Glu Phe	705	710	715
Tyr Ile Leu Thr Ala Ala His Cys Leu Tyr Gln Ala Lys Arg Phe Lys	725	730	735

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Val Arg Val Gly Asp Arg Asn Thr Glu Gln Glu Glu Gly Gly Glu Ala
 740 745 750
 Val His Glu Val Glu Val Val Ile Lys His Asn Arg Phe Thr Lys Glu
 755 760 765
 Thr Tyr Asp Phe Asp Ile Ala Val Leu Arg Leu Lys Thr Pro Ile Thr
 770 775 780
 Phe Arg Met Asn Val Ala Pro Ala Cys Leu Pro Glu Arg Asp Trp Ala
 785 790 795 800
 Glu Ser Thr Leu Met Thr Gln Lys Thr Gly Ile Val Ser Gly Phe Gly
 805 810 815
 Arg Thr His Glu Lys Gly Arg Gln Ser Thr Arg Leu Lys Met Leu Glu
 820 825 830
 Val Pro Tyr Val Asp Arg Asn Ser Cys Lys Leu Ser Ser Ser Phe Ile
 835 840 845
 Ile Thr Gln Asn Met Phe Cys Ala Gly Tyr Asp Thr Lys Gln Glu Asp
 850 855 860
 Ala Cys Gln Gly Asp Ala Gly Gly Pro His Val Thr Arg Phe Lys Asp
 865 870 875 880
 Thr Tyr Phe Val Thr Gly Ile Val Ser Trp Gly Glu Gly Cys Ala Arg
 885 890 895
 Lys Gly Lys Tyr Gly Ile Tyr Thr Lys Val Thr Ala Phe Leu Lys Trp
 900 905 910
 Ile Asp Arg Ser Met Lys Thr Arg Gly Leu Pro Lys Ser Val Ala Gln
 915 920 925
 Ala Thr Ser Ser Ser Gly Glu Ala Pro Asp Ser Ile Thr Trp Lys Pro
 930 935 940
 Tyr Asp Ala Ala Asp Leu Asp Pro Thr Glu Asn Pro Phe Asp Leu Leu
 945 950 955 960
 Asp Phe Asn Gln Thr Gln Pro Glu Arg Gly Asp Asn Asn Leu Thr Arg
 965 970 975

<210> SEQ ID NO 44
 <211> LENGTH: 51
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 44

Val Ala Gln Ala Thr Ser Ser Ser Gly Glu Ala Pro Asp Ser Ile Thr
 1 5 10 15
 Trp Lys Pro Tyr Asp Ala Ala Asp Leu Asp Pro Thr Glu Asn Pro Phe
 20 25 30
 Asp Leu Leu Asp Phe Asn Gln Thr Gln Pro Glu Arg Gly Asp Asn Asn
 35 40 45
 Leu Thr Arg
 50

<210> SEQ ID NO 45
 <211> LENGTH: 20
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 45

Met Glu Thr Asp Thr Leu Leu Leu Trp Val Leu Leu Leu Trp Val Pro
 1 5 10 15

-continued

Gly Ser Thr Gly
20

<210> SEQ ID NO 46
<211> LENGTH: 330
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 46

Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Lys
1 5 10 15
Ser Thr Ser Gly Gly Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr
20 25 30
Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser
35 40 45
Gly Val His Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser
50 55 60
Leu Ser Ser Val Val Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr
65 70 75 80
Tyr Ile Cys Asn Val Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys
85 90 95
Lys Val Glu Pro Lys Ser Cys Asp Lys Thr His Thr Cys Pro Pro Cys
100 105 110
Pro Ala Pro Glu Leu Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro
115 120 125
Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys
130 135 140
Val Val Val Asp Val Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp
145 150 155 160
Tyr Val Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu
165 170 175
Glu Gln Tyr Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu
180 185 190
His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn
195 200 205
Lys Ala Leu Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly
210 215 220
Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Asp Glu
225 230 235 240
Leu Thr Lys Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr
245 250 255
Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn
260 265 270
Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe
275 280 285
Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn
290 295 300
Val Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr
305 310 315 320
Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys
325 330

<210> SEQ ID NO 47
<211> LENGTH: 735
<212> TYPE: PRT

-continued

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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
    polypeptide

<400> SEQUENCE: 47

Met Gly Arg Pro Leu His Leu Val Leu Leu Ser Ala Ser Leu Ala Gly
 1          5          10          15
Leu Leu Leu Leu Gly Glu Ser Leu Phe Ile Arg Arg Glu Gln Ala Asn
 20          25          30
Asn Ile Leu Ala Arg Val Thr Arg Ala Asn Ser Phe Leu Phe Trp Asn
 35          40          45
Lys Tyr Lys Asp Gly Asp Gln Cys Glu Thr Ser Pro Cys Gln Asn Gln
 50          55          60
Gly Lys Cys Lys Asp Gly Leu Gly Glu Tyr Thr Cys Thr Cys Leu Glu
 65          70          75          80
Gly Phe Glu Gly Lys Asn Cys Glu Leu Phe Thr Arg Lys Leu Cys Ser
 85          90          95
Leu Asp Asn Gly Asp Cys Asp Gln Phe Cys His Glu Glu Gln Asn Ser
100          105          110
Val Val Cys Ser Cys Ala Arg Gly Tyr Thr Leu Ala Asp Asn Gly Lys
115          120          125
Ala Cys Ile Pro Thr Gly Pro Tyr Pro Cys Gly Lys Gln Thr Leu Glu
130          135          140
Arg Arg Lys Arg Arg Lys Arg Ile Val Gly Gly Gln Glu Cys Lys Asp
145          150          155          160
Gly Glu Cys Pro Trp Gln Ala Leu Leu Ile Asn Glu Glu Asn Glu Gly
165          170          175
Phe Cys Gly Gly Thr Ile Leu Ser Glu Phe Tyr Ile Leu Thr Ala Ala
180          185          190
His Cys Leu Tyr Gln Ala Lys Arg Phe Lys Val Arg Val Gly Asp Arg
195          200          205
Asn Thr Glu Gln Glu Glu Gly Gly Glu Ala Val His Glu Val Glu Val
210          215          220
Val Ile Lys His Asn Arg Phe Thr Lys Glu Thr Tyr Asp Phe Asp Ile
225          230          235          240
Ala Val Leu Arg Leu Lys Thr Pro Ile Thr Phe Arg Met Asn Val Ala
245          250          255
Pro Ala Cys Leu Pro Glu Arg Asp Trp Ala Glu Ser Thr Leu Met Thr
260          265          270
Gln Lys Thr Gly Ile Val Ser Gly Phe Gly Arg Thr His Glu Lys Gly
275          280          285
Arg Gln Ser Thr Arg Leu Lys Met Leu Glu Val Pro Tyr Val Asp Arg
290          295          300
Asn Ser Cys Lys Leu Ser Ser Ser Phe Ile Ile Thr Gln Asn Met Phe
305          310          315          320
Cys Ala Gly Tyr Asp Thr Lys Gln Glu Asp Ala Cys Gln Gly Asp Ala
325          330          335
Gly Gly Pro His Val Thr Arg Phe Lys Asp Thr Tyr Phe Val Thr Gly
340          345          350
Ile Val Ser Trp Gly Glu Gly Cys Ala Arg Lys Gly Lys Tyr Gly Ile
355          360          365
Tyr Thr Lys Val Thr Ala Phe Leu Lys Trp Ile Asp Arg Ser Met Lys
370          375          380

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Thr Arg Gly Leu Pro Lys Ala Lys Ser His Ala Pro Glu Val Ile Thr
385                               390                               395                               400

Ser Ser Pro Leu Lys Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu
                               405                               410                               415

Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr Ala Ala Leu Gly Cys
                               420                               425                               430

Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser
                               435                               440                               445

Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val Leu Gln Ser
                               450                               455                               460

Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro Ser Ser Ser
465                               470                               475                               480

Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val Asn His Lys Pro Ser Asn
                               485                               490                               495

Thr Lys Val Asp Lys Lys Val Glu Pro Lys Ser Cys Asp Lys Thr His
                               500                               505                               510

Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly Pro Ser Val
                               515                               520                               525

Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr
530                               535                               540

Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu Asp Pro Glu
545                               550                               555                               560

Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala Lys
                               565                               570                               575

Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg Val Val Ser
                               580                               585                               590

Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys
595                               600                               605

Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu Lys Thr Ile
610                               615                               620

Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro
625                               630                               635                               640

Pro Ser Arg Asp Glu Leu Thr Lys Asn Gln Val Ser Leu Thr Cys Leu
                               645                               650                               655

Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn
660                               665                               670

Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser
675                               680                               685

Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg
690                               695                               700

Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu
705                               710                               715                               720

His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys
725                               730                               735

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<210> SEQ ID NO 48

<211> LENGTH: 715

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 48

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Met Glu Thr Asp Thr Leu Leu Leu Trp Val Leu Leu Leu Trp Val Pro
1           5           10           15

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-continued

Gly Ser Thr Gly Ala Asn Ser Phe Leu Phe Trp Asn Lys Tyr Lys Asp
 20 25 30
 Gly Asp Gln Cys Glu Thr Ser Pro Cys Gln Asn Gln Gly Lys Cys Lys
 35 40 45
 Asp Gly Leu Gly Glu Tyr Thr Cys Thr Cys Leu Glu Gly Phe Glu Gly
 50 55 60
 Lys Asn Cys Glu Leu Phe Thr Arg Lys Leu Cys Ser Leu Asp Asn Gly
 65 70 75 80
 Asp Cys Asp Gln Phe Cys His Glu Glu Gln Asn Ser Val Val Cys Ser
 85 90 95
 Cys Ala Arg Gly Tyr Thr Leu Ala Asp Asn Gly Lys Ala Cys Ile Pro
 100 105 110
 Thr Gly Pro Tyr Pro Cys Gly Lys Gln Thr Leu Glu Arg Arg Lys Arg
 115 120 125
 Arg Lys Arg Ile Val Gly Gly Gln Glu Cys Lys Asp Gly Glu Cys Pro
 130 135 140
 Trp Gln Ala Leu Leu Ile Asn Glu Glu Asn Glu Gly Phe Cys Gly Gly
 145 150 155 160
 Thr Ile Leu Ser Glu Phe Tyr Ile Leu Thr Ala Ala His Cys Leu Tyr
 165 170 175
 Gln Ala Lys Arg Phe Lys Val Arg Val Gly Asp Arg Asn Thr Glu Gln
 180 185 190
 Glu Glu Gly Gly Glu Ala Val His Glu Val Glu Val Val Ile Lys His
 195 200 205
 Asn Arg Phe Thr Lys Glu Thr Tyr Asp Phe Asp Ile Ala Val Leu Arg
 210 215 220
 Leu Lys Thr Pro Ile Thr Phe Arg Met Asn Val Ala Pro Ala Cys Leu
 225 230 235 240
 Pro Glu Arg Asp Trp Ala Glu Ser Thr Leu Met Thr Gln Lys Thr Gly
 245 250 255
 Ile Val Ser Gly Phe Gly Arg Thr His Glu Lys Gly Arg Gln Ser Thr
 260 265 270
 Arg Leu Lys Met Leu Glu Val Pro Tyr Val Asp Arg Asn Ser Cys Lys
 275 280 285
 Leu Ser Ser Ser Phe Ile Ile Thr Gln Asn Met Phe Cys Ala Gly Tyr
 290 295 300
 Asp Thr Lys Gln Glu Asp Ala Cys Gln Gly Asp Ala Gly Gly Pro His
 305 310 315 320
 Val Thr Arg Phe Lys Asp Thr Tyr Phe Val Thr Gly Ile Val Ser Trp
 325 330 335
 Gly Glu Gly Cys Ala Arg Lys Gly Lys Tyr Gly Ile Tyr Thr Lys Val
 340 345 350
 Thr Ala Phe Leu Lys Trp Ile Asp Arg Ser Met Lys Thr Arg Gly Leu
 355 360 365
 Pro Lys Ala Lys Ser His Ala Pro Glu Val Ile Thr Ser Ser Pro Leu
 370 375 380
 Lys Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Ser Ser
 385 390 395 400
 Lys Ser Thr Ser Gly Gly Thr Ala Ala Leu Gly Cys Leu Val Lys Asp
 405 410 415
 Tyr Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr
 420 425 430

-continued

Ser	Gly	Val	His	Thr	Phe	Pro	Ala	Val	Leu	Gln	Ser	Ser	Gly	Leu	Tyr
	435						440					445			
Ser	Leu	Ser	Ser	Val	Val	Thr	Val	Pro	Ser	Ser	Ser	Leu	Gly	Thr	Gln
	450						455				460				
Thr	Tyr	Ile	Cys	Asn	Val	Asn	His	Lys	Pro	Ser	Asn	Thr	Lys	Val	Asp
	465				470					475					480
Lys	Lys	Val	Glu	Pro	Lys	Ser	Cys	Asp	Lys	Thr	His	Thr	Cys	Pro	Pro
				485					490					495	
Cys	Pro	Ala	Pro	Glu	Leu	Leu	Gly	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro
			500					505					510		
Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr
		515					520					525			
Cys	Val	Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	Val	Lys	Phe	Asn
	530					535					540				
Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg
	545				550					555					560
Glu	Glu	Gln	Tyr	Asn	Ser	Thr	Tyr	Arg	Val	Val	Ser	Val	Leu	Thr	Val
				565					570					575	
Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser
			580					585					590		
Asn	Lys	Ala	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys
		595					600					605			
Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Asp
	610					615					620				
Glu	Leu	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu	Val	Lys	Gly	Phe
	625				630					635					640
Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu
				645					650					655	
Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu	Asp	Ser	Asp	Gly	Ser	Phe
		660						665					670		
Phe	Leu	Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly
		675					680					685			
Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu	His	Asn	His	Tyr
	690					695					700				
Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys					
	705					710				715					

<210> SEQ ID NO 49
 <211> LENGTH: 33
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 49

Lys	Ser	Ser	Gly	Gly	Ser	Gly	Gly	Ser	Gly	Gly	Ser	Gly	Gly	Ser	Gly
1			5				10					15			
Gly	Ser	Gly	Gly	Ser	Gly	Gly	Ser	Gly	Gly	Ser	Gly	Gly	Ser	Gly	Ser
		20					25					30			

Ser

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What is claimed is:

1. A protein comprising the amino acid sequence of formula (I):

LC-L1-HC-L2-AP

(I),

wherein:

LC comprises the amino acid sequence that is at least 85% identical to the amino acid sequence of SEQ ID NO:13,

L1 is a peptide linker comprising a protease recognition site,

HC comprises the amino acid sequence that is at least 85% identical to the amino acid sequence of SEQ ID NO:11,

L2 is absent or is a peptide linker that cannot be processed by the protease recognizing the protease recognition site in L1, and

AP is an activation peptide comprising the amino acid sequence that is at least 90% identical to the amino acid sequence of SEQ ID NO: 12; and

wherein, when the L1 is processed by the protease, a two-chain polypeptide comprising the LC and the HC on separate chains is produced, and the two-chain polypeptide binds to a factor Xa (fXa) inhibitor.

2. The protein of claim 1, wherein the protease is furin.

3. The protein of claim 1, wherein the L1 comprises the amino acid sequence of RKR or RKRRKR (SEQ ID NO:7).

4. The protein of claim 1, wherein the L2 is 0-50 amino acid residues in length.

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5. The protein of claim 4, wherein at least 50% of the amino acid residues of L2 is Gly or Ser.

6. The protein of claim 1, wherein the activation peptide comprises a glycosylation site.

7. The protein of claim 1, wherein the activation peptide comprises the amino acid sequence of SEQ ID NO: 12.

8. The protein of claim 1, wherein the protein further comprises human serum albumin (HSA) at the N-terminus of the formula (I).

9. The protein of claim 8, wherein the LC does not comprise amino acid residues 1-45 of SEQ ID NO:2.

10. The protein of claim 1, wherein the produced two-chain polypeptide has reduced capacity to assemble into a prothrombin complex as compared to a wild-type fXa.

11. The protein of claim 1, wherein the LC comprises the amino acid sequence that is at least 90% identical to the amino acid sequence of SEQ ID NO:6.

12. The protein of claim 1, wherein the produced two-chain polypeptide has reduced catalytic activity as compared to a wild-type fXa.

13. The protein of claim 1, wherein the HC comprises the amino acid sequence that is at least 90% identical to the amino acid sequence of SEQ ID NO:10.

14. The protein of claim 1, wherein the HC comprises the amino acid sequence of SEQ ID NO:11.

* * * * *